PROJECT RECORD

NASA and ESA Data Projects

Learning and exploring with open data from NASA and ESA

Asteroid Tracker · Image Explorer · Mars Data

EDITION
November 12, 2026 · repository f98086c · 142 recorded changes

DEVELOPER
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PREPARED WITH
Claude (Anthropic), from the project repository

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Document control

Title	NASA and ESA Data Projects: Project Record
Edition	November 12, 2026, built from repository commit `f98086c`
Owner	Malcolm Syme · mcsyme11@gmail.com
Source	The project repository on GitHub (mcsyme11-byte/NASA-ESA-Data-and-API-Projects), main branch
How it is kept	Rebuilt automatically on GitHub after every change to the repository, except the daily data updates (which change only data files, none of them in this record), and saved to the repository straight after, so each edition matches the repository at the commit it names
Status	Public project record (the repository is public)

Latest changes

The twelve most recent changes in this edition, not counting the record's own rebuilds and the daily data updates, with long descriptions shortened; the full list is in Part 15, Repository History.

Date	Commit	Change
2026-11-12	`f98086c`	Rule: each project's main page opens on a Home tab (mission, data and agencies, how to use, links to ...
2026-11-12	`b8e908c`	Image Explorer home mock-up: no back button, as on the Mars Data home page
2026-11-12	`8a9bf70`	Mars Data: home page mock-up for approval
2026-11-12	`acbcd3e`	Image Explorer: mock-up of the project's own home page, for approval
2026-11-12	`ffa0c2d`	Mars Data: no back button on the project's home page
2026-11-12	`30a4ca2`	Asteroid Tracker tabs wrap at every width instead of scrolling off the edge
2026-11-12	`f879fa6`	Mars Data: the approved Two craters chart (Gale and Jezero) on the page
2026-11-12	`7011690`	Shorter footer on every page and a Legal page for Privacy, Terms, Credits and the licence; detailed ...
2026-11-12	`dccb2d2`	Mars Data: Perseverance MEDA data (sols 0-1859) and a two-craters mock-up for approval
2026-11-12	`0d29781`	Map page: remove the Back button; the tracker's tabs are right below the title
2026-11-12	`abb918f`	Layout fixes: spacecraft filters, card close button, Sky Finder place panel, tracker tabs
2026-11-12	`6f594a2`	Mars Data: a distinct colour for each of the eight Mars Years

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PART 1

Introduction

About this record

This is the complete record of **NASA and ESA Data Projects**, Malcolm Syme's three learning-and-exploring projects built on the open data of NASA and ESA (the European Space Agency): what each project is and how it works, where everything stands, what is planned, every change so far, the data sources and their terms, the design, the decisions and working rules, the site's own pages, the licence and credits, and the repository's full history.

It is built from the project's public GitHub repository (mcsyme11-byte/NASA-ESA-Data-and-API-Projects) and rebuilt automatically after every change, so it matches the repository at the commit named on its cover and on the Document control page. The one exception is the daily data updates: automatic commits that only refresh data files (the asteroid data, the astronaut photos and similar). None of those files are in this record, so they don't start a rebuild, and the Repository History counts them instead of listing them. Each chapter names the repository file it comes from; links lead to that file on GitHub.

Developer	Malcolm Syme · mcsyme11@gmail.com
Built with	Claude (Anthropic)
Repository	github.com/mcsyme11-byte/NASA-ESA-Data-and-API-Projects (public)
The site	mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects
Licence	MIT License (the project's own code and documents); outside data and libraries keep their own terms
Edition	see the cover and Document control

The projects at a glance

Project	What it does	Data	Where it stands
Asteroid Tracker	The near-Earth objects passing Earth this week and every year since 2000, sorted by size and distance (also in Moon distances), a page and orbit diagram for each, a search, and a map of all of them with the planets and spacecraft; impact-risk lists from NASA/JPL and ESA side by side	NASA NeoWs; NASA/JPL close-approach data, Small-Body Database, Horizons and Sentry; ESA NEO Coordination Centre	Live since November 8, 2026; its data is refreshed daily
Image Explorer	The Earth from orbit on any day since 2000, a globe inside the real sky with natural events and astronaut photos, and the picture libraries of NASA, ESA/Webb and ESA/Hubble, searchable by subject, mission and date	NASA GIBS, EONET, Image and Video Library and astronaut photography; ESA/Webb and ESA/Hubble	Live since November 9, 2026
Mars Data	Weather measured on the surface of Mars, cleaned, organised by sol, season and time of day, with statistics and charts that show the patterns of a Martian year	NASA Planetary Data System (Curiosity's REMS first), ESA where it helps	Planned; started in its own chat on November 11, 2026

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Where things stand

- **Live:** the Asteroid Tracker (with its "Visible from here" tab), the Image Explorer, Mars Data (Curiosity's weather, since November 12, 2026), Sky Finder (a phone app for finding passing asteroids, since November 12, 2026) and the spacecraft map (since November 12, 2026), on the site, with the site's information pages (About, Contact, and a Legal page gathering Privacy, Terms of use, Credits and the licence), linked from every page's footer.
- **NASA and ESA:** since November 9, 2026 ESA's data is added alongside NASA's where it helps; NASA stays the main source. Every page credits its sources and says the project isn't endorsed by NASA or ESA.
- **Keys:** visitors need no key. Since November 12, 2026 the Asteroid Tracker's daily data job copies the NASA data that needs one, using the developer's key kept as a GitHub repository secret (`NASA_API_KEY`), never in the code.
- **Waiting:** the astronaut photos on the globe wait for a key from NASA's astronaut photography service, kept as a GitHub repository secret, never in the code.
- **Next:** the developer's testing of the November 12, 2026 additions, and Mars Data's next steps, as listed in the roadmap.
- **The repository:** public, with `main` protected against force pushes and deletion; GitHub checks every push for wording, personal details and keys, and rebuilds this record.

How the work is done

- **The developer decides:** what is built and when. Questions are answered without building anything until the developer explicitly says go; new pages start as mock-ups the developer approves.
- **Real data, checked first:** every source is checked against NASA's or ESA's current documentation before anything is built on it, recorded in the data-source notes, and credited wherever its data appears.
- **Free to run:** static pages on GitHub Pages, free NASA and ESA services, no paid services, no keys in the code.
- **Tested before it goes live:** every page is checked at phone, tablet and computer widths, with its empty, loading and error states.
- **Every change is recorded:** each finished change is pushed with its roadmap and change-log entries, and this record is rebuilt.
- **Public-ready:** no personal details anywhere, except the developer credit: © 2026 Malcolm Syme · mcsyme11@gmail.com.

The projects in brief

Source: `README.md` · *last changed 2026-11-12*

Three ways to learn and explore with real data from NASA and ESA (the European Space Agency) and their APIs. Designed by **Malcolm Syme** and built with Claude.

The site: https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/

The projects

Project	What it is	Data	Status
Asteroid Tracker	The near-Earth objects passing Earth this week and every year since 2000, a page and orbit diagram for each, a search, and a turnable map of all 43,000 with the planets and spacecraft	NASA NeoWs; NASA/JPL close-approach data, Small-Body Database, Horizons and Sentry; ESA NEO Coordination Centre (risk lists, close approaches)	**Live**
Mars Data	Weather measured on Mars, cleaned and organised by time and place, with statistics and charts that show its patterns	NASA Planetary Data System (Curiosity's REMS)	Planned
Image Explorer	The Earth from orbit on any day since 2000, a globe inside the real sky with natural events and astronaut photos, and the picture libraries of NASA and ESA, searchable by subject, mission and date	NASA GIBS, EONET and Image and Video Library; ESA/Webb and ESA/Hubble picture archives	**Live**

Alongside them (since November 12, 2026):

Utility	What it is	Data
Sky Finder	An installable phone web app: hold the phone up to the sky to find tonight's passing asteroid, with the Moon, planets, bright stars and the space stations to guide you	NASA/JPL close approaches and Horizons; CelesTrak; d3-celestial; GeoNames
Spacecraft map	The active spacecraft of NASA, ESA and others, from low Earth orbit to the Voyagers, with what each does and links to its mission pages; previewed on the home page	NASA/JPL Horizons; CelesTrak

Order of work: Asteroid Tracker (live), then Image Explorer (live), then Mars Data (next); each builds skills the next one uses.

How it fits together

```
NASA-ESA-Data-and-API-Projects/
├── index.html          home page linking the three projects
├── about/ legal/ privacy/ terms/ credits/ contact/   the site's information pages
├── shared/             the look (theme.css, emblem), the NASA API helper (nasa.js), the footer links
(footer.js),
│                       the sky maths (sky.js, passes.js, sats.js, place.js) and satellite.js (vendor/)
├── asteroid-tracker/   Project 1
├── image-explorer/     Project 2
├── mars-data/          Project 3 (Python analysis + a charts page)
├── sky-finder/         Sky Finder, the phone web app
├── spacecraft/         the spacecraft map and its data job
├── docs/               design guide, data sources, notes, roadmap, history, and the Project Record
(record/ builds it)
└── .github/workflows/  checks after every push, the daily data updates, and the Project Record rebuild
```

- **Pages run in the browser,** with no build step: plain HTML, CSS and JavaScript, hosted free on GitHub Pages.
- **One look everywhere:** the Flight Deck Suite's "old money" style (navy, brass and cream, serif type) with a night-sky touch. See docs/DESIGN.md.
- **API keys are never committed.** Visitors need no key: the GitHub data jobs copy the data that needs one, using keys kept as repository secrets, and the pages use NASA's shared `DEMO_KEY` for anything else. Python scripts read `NASA_API_KEY` from a local `.env` file, which Git ignores.

Getting started (developers)

1. Get a free NASA API key at **api.nasa.gov** (issued by email).
2. For the Python analysis: copy `.env.example` to `.env` and paste the key in.
3. To try the pages on your own computer, serve the folder (modules don't load from `file://`):

```
python3 -m http.server 8000
```

then open `http://localhost:8000`.

Documentation

Document	What's in it
docs/DESIGN.md	Colours, type, components and chart colours
docs/DATA-SOURCES.md	Every NASA and ESA source used: endpoints, keys, limits, terms and credit lines
docs/PROJECT-NOTES.md	How it's built, decisions, how to run and publish
docs/ROADMAP.md	What's next and every pending to-do
docs/HISTORY.md	How the project came about, in order
Project Record (PDF)	The whole project as one PDF, rebuilt automatically after every change (how)
CHANGELOG.md	Every change, newest first
LICENSE.md	The MIT License (the project's own code and documents)
CREDITS.md	Every outside data source and library, with its terms

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Data courtesy of NASA and ESA. Imagery from NASA's Global Imagery Browse Services (GIBS), part of NASA's Earth Science Data and Information System (ESDIS). An independent project, not endorsed by NASA or ESA.

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How this record is made

Source: `docs/record/README.md` · *last changed 2026-11-12*

`docs/NASA and ESA Data Projects - Project Record.pdf` is the project's complete record as one PDF: what each project is and how it works, where everything stands, the roadmap, the history, the change log, the data sources, the design, the decisions and working rules, the site's own pages, the licence and credits, the full repository history and a glossary.

It is built from this repository, never edited by hand, and **rebuilt automatically on GitHub** (since Nov 11, 2026) after every push to main: `.github/workflows/project-record.yml` runs the build below and commits the record on its own ("Project Record rebuilt (automatic)"). Each edition matches the repository at the commit its cover and Document control name. Chats don't build or commit it themselves; they only check that the automatic commit arrived after their push. It can also be run by hand: GitHub › Actions › Project Record › Run workflow.

Daily data updates are skipped (decided Nov 11, 2026). Pushes that change only data files (`asteroid-tracker/data/`, `image-explorer/data/`, `mars-data/data/`), such as the daily asteroid and astronaut-photo updates, don't start a rebuild: none of those files are in the record, so each rebuild would make the same PDF again. A push that also changes anything else starts one as usual. The Repository History chapter counts the data updates instead of listing them.

Build

The automatic build runs exactly this. To build by hand (only if the automatic run failed), use a clone with full history: a shallow clone cuts the Repository History chapter short.

```
sh docs/record/build.sh          # writes docs/NASA and ESA Data Projects - Project Record.pdf
```

It is laid out as a Word document in a temporary folder first, then made into the PDF; only the PDF is kept. Needs python3 (with pypdf and fontTools), node with the npm packages `docx` and `marked`, and LibreOffice (`soffice`).

File	What it does
`build.sh`	Runs everything below, then checks the record for personal details (a personal folder path, any e-mail address but the developer's and NASA's and ESA's own, at nasa.gov and esa.int) and stops if it finds one
`build_record.js`	Lays out the document: cover, Document control, Contents, the Parts. The list of repository documents in each Part is `PARTS` at the top
`export_extras.py`	The parts that aren't Markdown files: the repository history (without the daily data updates and the record's own rebuilds, which are counted), the site's own pages (About, Privacy, Terms of use, Credits, Contact, from their HTML) and the working rules (from CLAUDE.md)
`page_map.py`	Measures which page each Part and chapter starts on (a first build laid out by LibreOffice), for the page numbers in the Contents
`fonts.sh`	Build machine only: installs Gelasio, which has Georgia's exact letter widths, so the measured page numbers match Word
`front.md`	About this record, the projects at a glance, where things stand, how the work is done (keep "Where things stand" current)
`glossary.md`	The glossary

The build is adapted from the Flight Deck Suite's and Hot Mic's Project Records, which work the same way.

Everything goes in, every time

Every Markdown and text document in the repository is in the record, except the few left out on purpose below. A document listed in `PARTS` (`build_record.js`) sits in its own Part. **Any other document goes in by itself**, in full, in a Part called "Newly added documents", and the build names it in its output ("Newly added documents (not yet in PARTS): ..."). So nothing can be missed, even if a chat forgets the list.

When adding a document: add it to the right Part in `PARTS` in the same change, so it doesn't sit in "Newly added documents".

Left out on purpose (`LEFT_OUT` in `build_record.js`): `docs/handoffs/` (working notes between chats), `docs/CLAUDE-PROJECT-SETUP.md` (the Claude project's set-up), `CLAUDE.md` (its standing rules are in Part Decisions and Working Rules), the outside libraries' own licence texts and copies (`vendor/`), data files (`data/` folders), test data and GitHub's settings.

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PART 2

Status and Roadmap

What is agreed next and every pending to-do.

Roadmap

Source: `docs/ROADMAP.md` · *last changed 2026-11-12*

What's next, as of November 8, 2026. Every pending to-do lives here; items are done at the developer's own pace. Everything finished moves to CHANGELOG.md.

0. Repository (set up Nov 8, 2026)

1. Asteroid Tracker (version 1 live Nov 8, 2026)

- ☐ **Home tab on each project's main page** (rule decided Nov 12, 2026, in CLAUDE.md), each from a mock-up the developer approves: Asteroid Tracker (mock-up awaiting approval, Nov 12, 2026), Mars Data, Image Explorer.
- ☐ **Developer testing of the Nov 12, 2026 additions** (built and published Nov 12, 2026):
  - Asteroid Tracker, "Visible from here" tab: "Use my location" and a typed town; the passes list, the night chart, "Find it ›" opening Sky Finder; "Forget this place".
  - Sky Finder (`sky-finder/`), on a phone outdoors at night: add to the home screen (iPhone and Android); the safety notice; camera, location and motion prompts; calibration on the Moon or a planet; the markers lining up with the real Moon, planets and bright stars; an asteroid's diamond and path; red light; the ISS or Tiangong passes; the how-to page.
  - Spacecraft map (`spacecraft/`) and the home page's new panel: the five views, search, agency and kind filters, the card for several craft, Follow, the path, Share, time controls; the asteroid map link under the panel.
- ☐ Read the first runs of the new "Spacecraft data" workflow and the "Fetch sky positions" step of "Asteroid data" (Actions logs): every craft should be found; any "not found" line (Horizons name or CelesTrak name) needs its `src` corrected in `spacecraft/catalogue.json`. Links reported broken in `spacecraft/data/links.json` need new addresses.
- ☐ Spacecraft map, from the first data run (Nov 12, 2026): add TESS, SPHEREx and WSF-M1, which CelesTrak did not return by name (TESS not even by its catalogue number 43435), by catalogue numbers (`#nnnnn` in `src.tle`, checked on celestrak.org); find Tianwen-1 and Tianwen-2 in Horizons if they are added there; replace the links the check reported broken (Chandra, Einstein Probe and SOHO at ESA, Mars Odyssey, Curiosity, LRO, KASA) with their current addresses.
- ☐ Review `spacecraft/catalogue.json` every few months: take missions that have ended off the list, add new ones.
- ☐ Watch the scheduled "Asteroid data" runs (06:17 and the 14:47 backup, added Nov 9, 2026): at least one should run each day.
- ☐ Confirm the monthly refresh of `data/famous.json` on the 1st of the month (the "Asteroid data" log shows "famous.json written").

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- ☐ Watch the "Fetch impact-risk lists and ESA close approaches" step in the "Asteroid data" runs: ESA calls its automated access experimental, so a change of format stops the step (the previous files are kept) and needs a fix in `fetch_risk.py`.
- ☐ Optional: if ESA's credit form or terms for the NEOCC figures are in doubt, ask the NEOCC by email (`neocc@esa.int`) and record the answer in `docs/DATA-SOURCES.md`.

2. Image Explorer (version 1 live Nov 9, 2026; additions Nov 9)

- ☐ Astronaut photos: the free API key (requested by email from jsc-earthweb@mail.nasa.gov; no answer by Nov 11,
  a. would add cataloged photos with their centre points. Without it the job uses the public Latest ISS Imagery set (since Nov 11, 2026). If the key arrives: repository secret `EOL_API_KEY`, then Actions › Astronaut photos › Run workflow.
- ☐ **Developer testing of the Nov 11, 2026 additions:** tapping the Earth map in a desert, at sea and near a town (country, feature, distance); the gaps between satellite passes filled; Map lines and names (borders, coastlines, rivers, names, grid) in all three views; the globe's earthquakes, volcanoes, launch sites, natural wonders and astronaut photos.
- ☐ **Developer testing of the Nov 10, 2026 additions,** on a computer and a phone: the Earth map's colour keys and the note under the map when a layer shows another day; place search for smaller places (for example Moraine Lake, Mount Rundle); the globe zoomed in close (sharper Earth) and the Day and night button; ESA videos with "Pictures and videos" or "Videos".
- ☐ Picture library: tune the ranking further as the developer reviews searches (the weights are in `scoreOf` in `image-explorer/app.js`).
- ☐ Polar views: a sea ice layer that works through the polar night. GIBS's microwave sea-ice records ended (SSMIS Aug 2021, AMSR2 Sep 2025); the current one is MODIS (a camera, daylight only). Ice surface temperature (`MODIS_Terra_Ice_Surface_Temp_Night`, thermal) is an option to consider.
- ☐ Further rounds of changes to the globe, as the developer reviews it.
- ☐ Picture library: subtitles for videos (NASA's files cannot be attached from another site; would need a copy).
- ☐ Shared parts used by both projects (tabs, "How to read this", the big search box, the tooltip) moved from `asteroid-tracker/tracker.css` into `shared/theme.css`; the page still loads `tracker.css` for them.
- ☐ Globe: more sky objects (fainter galaxies), if wanted.
- ☐ Keyboard use of the globe's markers (today they are reached through Find). With reduced motion set, the globe does not turn by itself and moves without animation; to be rechecked on a phone.

3. Mars Data (third)

Plan confirmed Nov 11, 2026 (see PROJECT-NOTES.md): three missions, updated automatically by a GitHub Actions job.

Data and analysis

- ☑ REMS, the whole record (sols 1–4843, August 7, 2012 to March 22, 2026; 4,703 sols with data), cleaned with rules 1.1 in chunks; every cleaning report checked (Nov 12, 2026).
- ☑ Corrected UV (sols 1–3423, 35 files of about 1.2 GB): all done and checked (row counts match NASA's labels).
- ☐ Next: the scheduled "Mars data" job on main (twice a week): pick up new REMS releases (sols after 4843), process only new or replaced files, check, commit, and open a GitHub issue (emailed) when new

data arrives or a run fails. It replaces the branch job; then delete the one-off probe (`mars-data/probe/`, `mars-probe.yml`) and the `mars-probe` branch.

- ☐ Review the values set aside (`mars-data/analysis/exclusions.csv`, CLEANING.md rule 10): air temperature on 26 sols (2803, 2844, 4087, 4089, 4092, 4094, 4174, 4179, 4186, 4208, 4221, 4222, 4263, 4274, 4282, 4298, 4432, 4775, 4780, 4781, 4792, 4808, 4809, 4817, 4821, 4841). Each has one hour at about 18:00–20:00 local reading about −115 to −120 °C between hours of −35 to −55 °C: a recurring evening sensor glitch, more frequent since sol 4087. Consider an automatic rule for a single hour far out of line with the hours on both sides (25 K or more), applied to the minute data before the daily statistics.
- ☐ Optional: ask ESA (data.licences@esa.int) to confirm the spelling of the Mars Express MaRS credit line ("M. Plätzold, K. Pieter" on ESA's terms page; the archive names Martin Pätzold).
- ✅ Perseverance MEDA (Jezero Crater), sols 0–1859, cleaned with MEDA rules 1.0 (`CLEANING-MEDA.md`); every report checked (Nov 12, 2026). Reviewed, real weather: sol 1078 (+20 Pa step in the dusty season) and sol 1113 (11 hours measured).
- ✅ Two-craters chart (Gale and Jezero pressure): approved and on the page (Nov 12, 2026).
- ☐ Mars Express MaRS profiles (ESA): waiting for ESA's archive (401 during maintenance, Nov 12, 2026); then run `run_mex.py` and add a profiles chart (mock-up first). Alternative to check: NASA's PDS Geosciences Node copy.

Automatic updates

- ☐ GitHub Actions job "Mars data", twice a week to start (to be reduced to weekly once stable): check each mission's archive for new or replaced files, download only those, clean and summarise with the same code, run the checks, and commit the summaries if they changed.
- ☐ Manifest of processed files (name, size, checksum), committed, so nothing is downloaded twice and replaced files are reprocessed.
- ☐ Checks before saving: expected columns, values within physical limits, sol count never decreasing, no duplicate sols. On any failure the previous summaries are kept and the job stops with an error.
- ☐ Email notice: when a run finds new data, the job opens a GitHub issue ("New Mars data: …"), which GitHub emails to the repository owner; a failed run is reported the same way.
- ☐ The page shows, for each mission, "Data up to sol X · checked [date]".

Page

- ✅ Page live (Nov 12, 2026) with the whole record, sols 1–4843.
- ☐ A short write-up on the page of what the patterns show, once the whole record is in.
- ☐ `CREDITS.md`, the Credits page and the Privacy page updated for the new sources in the same change.

PART 3

Project History

History

Source: `docs/HISTORY.md` · *last changed 2026-11-12*

How NASA and ESA Data Projects came about, in order.

November 8, 2026 — the idea and the repository

- The developer planned three learning projects built on NASA's open data and APIs, in one Claude project named "NASA and ESA Data Projects": an **asteroid tracker** (near-Earth objects sorted by size and distance, with an interactive dashboard), a **Mars data** study (cleaning, organising, statistics and charts of mission measurements) and a **NASA image explorer** (searching NASA imagery by date, location and mission, starting from GIBS).
- Notes from the first discussion: NeoWs's feed covers at most 7 days per request; some older Mars APIs (such as InSight's weather feed) ended with their missions, so archived PDS data is the better source; GIBS is imagery of Earth without missions to filter by, so it is paired with the NASA Image and Video Library.
- Decided: one repository, a folder per project; public, for free GitHub Pages hosting and as a portfolio, with no API key ever committed.
- The repository was set up with the shared look of the developer's other projects, a NASA API helper, a home page, placeholder pages and the documents in `docs/`.

November 8, 2026 — Asteroid Tracker, version 1

- The plan was confirmed in the project's own chat: hand-drawn SVG charts, the year view from JPL data in version 1, and a mock-up first. The mock-up (with made-up data) was approved the same day.
- Checking the live services showed two limits: the JPL close-approach API cannot be called from a web page, and NeoWs's rate-limit headers cannot be read by one. The year view therefore reads a file refreshed daily by GitHub Actions, and the key settings show the key's limits instead of the requests left.
- The page went live on GitHub Pages with the week view, filters, three graphs, the year view and key settings.

November 8, 2026 — Asteroid Tracker: each asteroid's page, the orbit diagram, past years

- After version 1 the charts gained "How to read this" notes, and three additions were built the same evening: a page for each asteroid, past years in the year view, and an orbit diagram, the one singled out for particular care.
- The orbit maths was checked before the diagram was drawn: Earth's distance from the Sun at perihelion and aphelion, and an asteroid's distance from Earth on the day of a close approach, matched NASA's figures.

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November 9, 2026 — famous close passes and going back

- The developer reported choppy frames when zoomed in close, a poorly laid-out spacecraft list, identical full-screen and exit icons, and the lack of a way back to the exact spot on the previous page; he also asked whether the map shows Earth where it is right now. All were addressed, together with three approved ideas: links for every name in the "By year" tables, famous close passes, and a background thread for the orbit maths.
- The zoomed-in slowness came from lines that ran far off screen: the browser still worked out every dash of them. Cutting each line to the visible area removed it.
- Two-body orbits cannot show a famous pass correctly, because Earth's pull bends the asteroid's path during the pass. Precise positions from NASA/JPL Horizons are used for those passes instead.

November 8–9, 2026 — the solar system map

- The developer asked for a map of all the asteroids around the Sun that could be turned smoothly and seen from different vantage points, with spacecraft such as the Voyagers, plus a search with results after every letter. A mock-up with made-up asteroids was approved; the developer then asked for a guide to asteroid names, year filters and a fix to the full-screen button, built directly.
- Smoothness was measured throughout. Two changes made playing reach 60 frames a second: each object's orbit is solved starting from the previous frame, and the dots are drawn at 1.5× resolution rather than 2×.
- The real data showed that NASA's list of near-Earth objects includes about 200 comets (Halley among them), which became their own colour on the map.

November 9, 2026 — Image Explorer, version 1

- The plan was confirmed in the project's own chat (OpenLayers, eight GIBS layers, tabs, place search), and the live services were checked first. The fire layers turned out to work only in the `epsg4326` projection, which set the map's projection.
- The first mock-up showed the daily satellite map. The developer found it disappointing (the gaps between satellite passes, no detail down to houses) and asked for a globe like the solar system map, with pictures pinned to places on Earth and to galaxies in the sky; the daily map was kept for following fires and storms.
- NASA's event feed holds over 7,000 "open" events, most with no report in months; the globe shows those reported in the last 30 days, about 35.
- The globe mock-up was approved the same night and published with about 1,350 sky objects.

November 9, 2026 — Image Explorer: five additions overnight

- The developer asked for five additions and went to bed. NASA, SIMBAD and the astronaut photo site cannot be reached from the build session, so a temporary GitHub Actions job on a separate branch checked them and committed what it found; Natural Earth was downloaded directly.
- The check of the hand-listed sky positions found one real mistake: the Rho Ophiuchi marker sat on the star, not on the cloud complex named after it.
- The astronaut photo site turned out to have a documented API that needs a free key, requested by email. The daily job was built to start working as soon as the key is stored as a GitHub secret.

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November 9, 2026 — NASA and ESA

- The project became **NASA and ESA Data Projects**: ESA (the European Space Agency) data is added alongside NASA's where it helps, with NASA as the main source. The repository was renamed NASA-ESA-Data-and-API-Projects, and the site moved to https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/.
- Project-wide rules for ESA sources were added (check ESA's documentation first, credit ESA wherever its data appears, never ESA's logo, free to run), and every page now says the project isn't endorsed by NASA or ESA.

November 9, 2026 — ESA's data in the Asteroid Tracker

- The Asteroid Tracker became the first project to show ESA's data beside NASA's: the impact-risk lists of ESA's NEO Coordination Centre and NASA/JPL's Sentry, side by side in a new Impact risk tab, on each asteroid's page and on the map. Neither centre's server answers web pages directly, so the twice-daily data job copies both lists.
- The first run showed a speed given as "nan", which made the file unreadable in a browser; the job now keeps only finite numbers.

November 11, 2026 — Project Record

- The project gained a Project Record: the whole project as one PDF, rebuilt by GitHub after every change except the daily data updates, in the same way as the developer's other projects.

November 12, 2026 — No key needed for visitors

- The Asteroid Tracker's "This week" list and asteroid pages had relied on each visitor's NASA key, or NASA's shared demo key (30 requests an hour and 50 a day per connection). The daily data job now copies that data with the developer's key, kept as a GitHub secret, so the site works for every visitor without a key.

November 11–12, 2026 — Mars Data, version 1

- Mars Data was planned around Curiosity's weather station (REMS), with Perseverance (MEDA) and ESA's Mars Express to follow, and the whole record from sol 1, cleaned slowly and in chunks. The archives turned out to be reachable only from GitHub's servers, so the work ran there, on a separate branch, with a cleaning report for every chunk.
- Careful checking caught several traps: a column named "solar longitude" that is the Sun's direction from the rover, not the season; NASA's index stopping at sol 4133 while the data reaches sol 4843; two tables in one ancillary file; quality flags whose first character is not a quality check at all.
- The developer approved the mock-up the same night, and the page went live with Mars Years 31–33 (sols 1–1293).

November 12, 2026 — Sky Finder and the spacecraft map

- The developer proposed a phone app that points the camera at a passing asteroid and a page of the passes visible from the visitor's place, then a map of everything humans have sent into space that is still working. Mock-ups were approved with changes (more realistic icons, a plainer banner, facts checked line by line, the phone experience first) and both were built and published together: the "Visible from here" tab, **Sky Finder** and the **spacecraft map**, which took over the home page's large panel.

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- Over the night of November 11–12 the rest of the record was processed in batches of about five chunks, each checked before the next, on a schedule of checks every two hours. By midday on November 12 the page held the whole released Curiosity record: sols 1–4843, Mars Years 31–38, with the 2018 global dust storm and the rover's 990 m climb up Mount Sharp visible in the charts. A recurring evening sensor glitch (one hour at about −120 °C) was found on 26 sols and set aside for review rather than published.

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PART 4

Change Log

Every change to the site and the projects, newest first.

Source: `CHANGELOG.md` · *last changed 2026-11-12*

November 12, 2026 — Mars Data: Two craters

- The approved "Two craters" chart is on the Mars Data page: air pressure at Gale Crater (Curiosity) and Jezero Crater (Perseverance) through the same seasons. The page's data credit names the MEDA team; the mock-up file is removed.
- The Mars Data page has no back button: it is the project's own home page, reached from the site's menu.

November 12, 2026 — a shorter footer and a Legal page

- **Footer:** one row with About, Legal, Contact and Source code ↗ and the copyright line, then one short credit line ("Data courtesy of NASA and ESA. An independent project, not endorsed by NASA or ESA."), on every page.
- **Legal page** (`legal/`): Privacy, Terms of use, Credits and the MIT licence, one card each with a one-line summary. Their own addresses are unchanged; opened directly, they go back to Legal.
- **Data credits:** each page's detailed credits (sources, ESA's credit lines, the GIBS acknowledgement) moved out of the footer to small lines just above it, with the content they cover.

November 12, 2026 — Mars Data: Perseverance (Jezero Crater) and a two-craters mock-up

- Perseverance's weather data (MEDA, NASA PDS), sols 0–1859 (February 2021 to May 2026), cleaned with the MEDA rules (`mars-data/analysis/CLEANING-MEDA.md`, `run_meda.py`); a cleaning report for every chunk (`mars-data/data/meda/reports/`). Page file `mars-data/data/meda.json`.
- Mock-up for approval: a "Two craters" chart comparing air pressure at Gale and Jezero (`mars-data/mockup.html`).
- MEDA added to `CREDITS.md`, the Credits page and `docs/DATA-SOURCES.md`. Mars Express (ESA) waits for ESA's archive, down for maintenance.

November 12, 2026 — Mars Data: the whole Curiosity record

- Curiosity REMS sols 4258–4843 (Mars Years 37–38) added; the page now covers the whole released record: sols 1–4843 (August 7, 2012 to March 22, 2026), 4,703 sols with data, Mars Years 31–38. Every cleaning report checked.
- The recurring evening air-temperature glitch was found on 26 sols in all; their air temperature is set aside for review (`mars-data/analysis/exclusions.csv`).
- The charts use eight colours, one for each Mars Year (two added to the theme's six for this page).

November 12, 2026 — Mars Data: sols 3645–4257 added; evening glitch set aside

- Curiosity REMS sols 3645–4257 (Mars Year 37); every cleaning report checked. The page now covers sols 1–4257.

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- A recurring air-temperature glitch (one hour at about 19:00–20:00 local reading about −120 °C) was found on 12 sols across the record; their air temperature is set aside for review (`mars-data/analysis/exclusions.csv`).

November 12, 2026 — Mars Data: sols 3069–3644 added; corrected UV complete

- Curiosity REMS sols 3069–3644 (Mars Year 36) and the last corrected UV files (to sol 3423, the end of NASA's corrected UV set); every cleaning report checked, nothing flagged. The page now covers sols 1–3644.

November 12, 2026 — Mars Data: sols 2483–3068 added

- Curiosity REMS sols 2483–3068 (Mars Years 35–36) and corrected UV for sols 2501–3100; every cleaning report checked, nothing flagged by the automatic review.
- New cleaning rule 10: a value that passes every rule but is plainly not weather is listed in `mars-data/analysis/exclusions.csv` and left out of the page until reviewed. First entry: sol 2844 air temperature (one hour at −119.6 °C between hours of −38 and −49 °C). The page now covers sols 1–3068.

November 12, 2026 — Mars Data: sols 1870–2482 added

- Curiosity REMS sols 1870–2482 (Mars Years 34–35) and corrected UV for sols 1901–2500; every cleaning report checked, nothing flagged. Gaps with no data released: sols 2172–2199 and 2321–2346. The 2018 global dust storm (around sols 2085–2120) shows as very low midday UV and daily pressure swings of up to 140 Pa. The page now covers sols 1–2482.

November 12, 2026 — Mars Data: sols 1294–1869 added

- Curiosity REMS sols 1294–1869 (Mars Years 33–34) and corrected UV for sols 1301–1900, cleaned with rules 1.1; every cleaning report checked (no problems, nothing flagged for review; UV row counts match NASA's labels). The page now covers sols 1–1869.

November 12, 2026 — Sky Finder, "Visible from here" and the spacecraft map

- **Layout fixes (same day):** the spacecraft map's agency and kind filters were squeezed to a sliver on desktop and are now full height; the craft card keeps its title and close button in view while it scrolls; a "group" of one craft is drawn as the craft itself; Sky Finder's place panel has a close button (and Escape); the Asteroid Tracker's tabs wrap to a second line at every width (phones included) instead of running off the edge, and the map and asteroid pages list the "Visible from here" tab too. The map page no longer has a "Back" button above its title: the tracker's tabs are right below it.
- **Asteroid Tracker, "Visible from here" tab** (`#visible`): choose a place ("Use my location" or a town from the GeoNames list); the close passes of the next 30 days that rise at least 10° above that horizon in full darkness (the Sun 12° down), each with its best moment, highest point, brightness and what is needed to see it (naked eye up to magnitude 6, binoculars 6–9, small telescope 9–12, large telescope 12–15, too faint above 15); a night chart of its height hour by hour; "Find it ›" opens Sky Finder. Data: `data/sky.json`, written daily by `fetch_sky.py` (NASA/JPL close approaches and Horizons). The place stays in the browser.
- **Sky Finder** (`sky-finder/`): an installable phone web app. Tonight's list (asteroid passes, the Moon, the planets and the ISS and Tiangong passes you can see); a camera view with a marker and the hour-by-hour path of the chosen asteroid, the Moon, planets, bright stars, the stations, the horizon and the compass points, an arrow to anything off the screen and a line telling which way to turn; calibration on

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the Moon, a planet or a bright star; drag-to-look where there are no motion sensors; red light; a safety notice on first use; works offline once visited; a how-to page. Linked from the tracker's new tab and the Image Explorer's globe notes. Sky maths in `shared/sky.js`, checked against Astropy.

- **Spacecraft map** (`spacecraft/`): 82 active spacecraft and stations of NASA, ESA, JAXA, ISRO, CNSA and others, from low Earth orbit to the Voyagers, on the asteroid map's controls, with five views (Solar system, Near Earth, Earth orbit, Moon, Mars), search, agency and kind filters, a card for each craft (figures, launch, description, timeline, links to the agencies' pages), Follow, its path, Share and time controls. Small drawings show what each kind of craft looks like. Data: NASA/JPL Horizons and CelesTrak, copied twice a day by the new "Spacecraft data" workflow; satellites worked out in the browser with satellite.js (MIT).
- **Home page:** the large panel now previews the spacecraft map (the asteroid map's preview takes its place if the spacecraft data cannot be loaded); the asteroid map is a link under it; a Sky Finder banner.
- **Privacy, Terms and Credits** updated first: the place, Sky Finder's settings and offline copy; camera, location and motion use; safety at night and never looking at the Sun; CelesTrak and satellite.js credits.

November 12, 2026 — Mars Data page live

- **Mars Data page** (`mars-data/index.html`), built from the approved mock-up: pressure through the Martian year (with a correction for the rover's climb), the daily pressure swing, humidity at night (preliminary), midday ultraviolet, temperature by season and hour, any sol hour by hour, daily highs and lows, and the rover's climb. Loading and error states; charts drawn at the width they are shown.
- **Data:** Curiosity REMS sols 1–1293 (Mars Years 31–33), cleaned with rules 1.1, with the cleaning reports (`mars-data/data/rems/reports/`); page files built by `mars-data/analysis/make_page_data.py`.
- The mock-up files are removed; the home page lists Mars Data as live.

November 12, 2026 — site-wide polish

A full check of every page at phone, tablet and computer sizes, and these fixes:

- **Feel:** quick hover and press feedback on every button, chip, tab and link; a brass glow on focused fields; a solid header with a soft shadow; dark date pickers and dropdowns.
- **Phones:** tab bars and chip rows that don't fit scroll sideways with a soft edge fade, and the selected tab scrolls into view (Asteroid Tracker tabs and search chips, the Globe's and Picture library's chip rows); tap targets raised to 40 px (the site name, "Its page ›", "Clear", sliders, the "Asteroid Tracker" link above titles); search placeholders shortened to fit.
- **Number tiles** shrink their figures slightly on narrow tiles, so values such as "0.75–1.10 au" and "11 Nov 2026" are never cut off at the tile's edge.
- "**How to read this**" is easier to read (12 px).
- **Asteroid Tracker:** one tab bar on every page (the map and asteroid pages gained Impact risk); the map page always has its Back button; space between the Impact risk intro and its tiles; labels on the Solar System Map and the orbit diagram outlined so they read clearly over dots and orbits; the orbit diagram's date no longer covers its labels on phones (a map-only rule had applied to it); "Asteroid not found" as the title for an unknown number; the asteroid page's tags and links in one consistent style.
- **Image Explorer:** fewer place names on a phone-sized Earth map, so they don't crowd; the Picture library's source filter is now "Library" (it shared "From" with the year), and its labels match the other rows; its Clear button appears once something is typed.
- The rules behind these are in `docs/DESIGN.md` (8a. Feel) and CLAUDE.md, so new pages follow them.

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November 12, 2026 — Asteroid Tracker: no key needed for visitors

- The daily data job copies NASA's NeoWs data for the current week, the two weeks before and after it, and every asteroid in them (`fetch_neows.py`, `data/neows/`), using the developer's key from the repository secret `NASA_API_KEY`. The "This week" list and the asteroid pages read that copy first, so visitors need no key and use none of their own allowance; other weeks and asteroids are still asked of NASA directly.
- The Key settings box was removed: with the daily copy no visitor needs a key. A key saved by the earlier version is deleted from the browser on the next visit; the Privacy and Terms pages say so.
- The secret `NASA_API_KEY` was added the same day; the first run copied 5 weeks and 199 asteroid pages, and the live site was confirmed to open the week and asteroid pages from the copy without asking NASA.

November 11, 2026 — Image Explorer: map tap card, map lines and names, gaps filled, more on the globe

- **Tapping the Earth map** says where the spot is: the country (from Natural Earth outlines), a natural feature on or near it (for example "Hamada de Tinrhert · desert · Algeria"), and the distance and direction from the nearest town ("347 km south of Hassi Messaoud"); at sea, the sea or ocean. "Open water or remote land" is gone. Rivers' points (their mouths) are left out of "near".
- **Gaps between satellite passes** show a cloud-free Blue Marble picture instead of black (not under night lights).
- **Map lines and names:** coastlines, land borders, rivers, a latitude and longitude grid, and names of countries, capitals and cities (more as the map zooms in; bigger places win where names crowd), each with a button; in all three views. Built from Natural Earth 1:50m by `tools/build_shapes.py`.
- **Globe:** earthquakes of magnitude 4.5 or more in the past week (USGS, live; circles sized by magnitude), 276 volcanoes, 24 launch sites and 711 natural wonders (national parks, waterfalls, glaciers, gorges, caves, reefs; off at first), each with its own button, card, "See it on the Earth map", pictures, and in Find (`data/globe-spots.json`, `tools/build_globe_spots.py`).
- **Astronaut photos without the key:** the daily job reads NASA's public Latest ISS Imagery set when there is no `EOL_API_KEY`; photos not yet cataloged are pinned below the Station and say so.
- Privacy and Credits list the US Geological Survey; Credits list the new Natural Earth and GeoNames uses.

November 12, 2026 — Project Record builds again

- **Fixed:** the Project Record's automatic rebuild stopped on its personal-details check from Nov 11, 2026 (commit 01c5a37 onward), because the Mars Data notes quote ESA's licensing contact address. Every push since then failed the rebuild and sent the developer an e-mail; no commit was refused, and only the PDF stayed at its earlier edition. The check now allows any address at nasa.gov or esa.int (official agency contacts); any other address still stops the build.
- `mars-data/analysis/CLEANING.md` has its place in the record's Mars Data Part.

November 12, 2026 — Mars Data: sources checked, cleaning rules and a mock-up

- **Sources checked** from GitHub's servers and recorded in `docs/DATA-SOURCES.md`: Curiosity REMS (NASA PDS Atmospheres Node, sols 1–4843), Perseverance MEDA (NASA PDS), Mars Express MaRS profiles (ESA Planetary Science Archive, CC BY-NC 3.0 IGO), with their citations and ESA's credit line.
- **Cleaning rules 1.1** (`mars-data/analysis/CLEANING.md`): NASA's quality flags read character by character, physical limits, coverage of each sol, daily statistics, humidity (labelled preliminary), corrected UV. Code in `mars-data/analysis/`; Ls and Mars Year calculated from the date (`mars_time.py`), checked against NASA's values.

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- **Mock-up** of the charts page for approval: `mars-data/mockup.html`, drawn from the cleaned data for sols 1–1293, including humidity (preliminary) and corrected UV. A midday UV peak counts only between 10:30 and 13:30 local time on a sol with at least 30 minutes of UV. Approved by the developer on November 11, 2026.

November 11, 2026 — Project Record

- **Project Record:** `docs/NASA and ESA Data Projects - Project Record.pdf`, the whole project as one PDF (cover, Document control, contents, 14 Parts, glossary), built from the repository by `docs/record/`, adapted from the Flight Deck Suite's and Hot Mic's records.
- **Rebuilt automatically** by `.github/workflows/project-record.yml` after every push to main, except pushes that change only data files (the daily data updates); the Repository History counts those instead of listing them.
- The build stops if a personal detail would end up in the record; the agencies' public contact addresses quoted in the data-source notes are allowed.
- **Wording check:** data files are checked for keys only, since they hold real place and object names.

November 10, 2026 — Image Explorer: ESA's videos; smaller places in place search

- **ESA's videos:** with "Pictures and videos" or "Videos", searches also ask ESA/Webb and ESA/Hubble's video archives. An ESA video plays the small file from ESA's servers and links the full HD file; the viewer shows its credit line (with line breaks, including the music credit), length and release date, and notes that the music has its own terms. The page waits at most 12 s for each ESA archive; a slower one is left out of that search, and the results line says it did not answer in time.
- **Smaller places in place search:** about 85,000 lakes, rivers, glaciers, peaks, passes, islands, parks and other natural features from GeoNames that have an English Wikipedia article (for example Lake Louise, Moraine Lake, the Bow River, Mount Rundle, the Athabasca Glacier), with their English names. The list loads after the main place list; a name typed in full is shown first, then the most widely named places (Antarctica's many small named features last). It is built by the manual job "Build smaller places" (`.github/workflows/build-places.yml`, `tools/build_geonames_features.py`).
- The map's place card offers "Pictures of …" (NASA's and ESA's libraries) instead of "NASA pictures of …".

November 10, 2026 — Asteroid Tracker: risk colour on the map, ESA and NASA compared, share links

- **Orbit diagram:** the tilt turns all the way round with a mouse, as on the map; "seen from below" labels the view from underneath.
- **Map:** asteroids on ESA's or NASA's impact-risk list are drawn in rose (2 × 2 pixels) and, zoomed in closer than about 0.35 au, ringed with a small diamond; "Show in their orbit colours" under Orbit types turns this off (kept on Back and reload).
- **Asteroid page:** "ESA and NASA compared", a chart of both centres' chances (the most likely impact and all possible impacts together, on a log scale) and the years in which each finds an impact possible, with the most likely year marked. ESA's special risk list gives no years or total chance; the chart says "not given" there.
- **Share links:** "Share this view" on the map (zoom, angle, centre, selection, date, filters, or the famous pass in progress) and "Share this week" on the tracker (the week, filters and open row). On a phone the share sheet opens; elsewhere the link is copied, or shown ready to copy. The link's settings are removed from the address once applied.

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November 11, 2026 — Projects numbered in build order; Mars Data uses the whole Curiosity record

- Image Explorer is now Project 2 and Mars Data Project 3, on the home page (in that order), the project pages and the READMEs.
- Mars Data plan: the whole Curiosity record is used from the start, downloaded and cleaned in chunks, with a cleaning report for each chunk.

November 11, 2026 — Mars Data: plan for automatic updates

- The plan for Mars Data is set in `docs/ROADMAP.md` and `docs/PROJECT-NOTES.md`: three missions (Curiosity REMS, Perseverance MEDA, Mars Express MaRS), a GitHub Actions job twice a week that picks up new archive releases with checks before anything is saved, and an email notice through a GitHub issue when new data arrives or a run fails. Planning only; nothing is built yet.

November 10, 2026 — Image Explorer: sharper globe close up; day and night

- Zooming in on the globe lays finer Blue Marble pictures over the part of the Earth in view, about 4, 2, 1 and 0.5 km a pixel as the view comes closer (about 8 km before); each level starts where the one before would show fewer picture pixels than screen pixels.
- **Day and night:** the half of the Earth facing away from the Sun is shaded, with a soft edge for twilight, from the Sun's position at the current time; the Earth keeps turning with the time while the page is open. A "Day and night" button turns the shading off.

November 10, 2026 — Image Explorer: colour keys in the project's look

- The Earth map's colour keys are drawn on the page from GIBS's colour maps: the colour bar, up to seven labelled values, the two ends, and the measurement and its units (cloud-top temperature in °C instead of kelvin); a layer's categories (for example snow's) as small swatches. GIBS's own legend picture stands in while a colour map loads or if it cannot be read.

November 10, 2026 — Image Explorer: each Earth map layer's days read from GIBS

- Each layer's days are read from GIBS (its time domain) instead of a fixed lag: a layer shows the chosen day, or its nearest earlier day when it has none (MUR sea temperature and sea ice miss a few days), and the note under the map says which ("newest day …" or "no picture that day; nearest earlier day …"). The fixed lag stays only as a fallback while the days load or if GIBS cannot be reached.
- The old test branches `image-explorer-probe` and `map-data-test` were deleted.

November 10, 2026 — Image Explorer: code libraries served from the site

- OpenLayers 10.11.0, proj4js 2.22.0 and three.js 0.187.0 are kept in `image-explorer/vendor/` (unchanged, each with its licence) and loaded from the site instead of the jsDelivr CDN, so the page no longer depends on jsDelivr. jsDelivr is removed from the Privacy page; the wording check skips the vendor folder.
- The developer's testing of the Image Explorer (Nov 10, 2026) passed, including ESA's pictures on the live site.

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November 10, 2026 — main branch protected

- A branch ruleset, "Protect main", blocks force pushes to `main` and its deletion. Ordinary pushes (the chats and the daily data jobs) work as before. Mistakes are undone with a new commit (`git revert`).

November 10, 2026 — Asteroid Tracker: the map tilts all the way round

- The map's tilt is no longer stopped at "straight down" (0°) and "edge-on" (90°): dragging up or down, or the ↑ ↓ keys, carry on past edge-on to underneath and round again. The Top, Tilted and Side buttons turn the short way back.

November 9, 2026 — Image Explorer: searching message names the libraries

- While a search runs, the Picture library says which libraries it is asking, following From: "Searching NASA's and ESA's libraries…", "Searching NASA's library…" or "Searching ESA's archives…" (it said NASA's library in every case).

November 9, 2026 — README brought up to date

- The README links the live site, gives the projects' status in the order of work, names NASA and ESA sources in the documentation table, and lists the GitHub workflows (checks and daily data updates).

November 9, 2026 — ESA check across the site

- Every ESA source now on the site was checked against the project rules: ESA's NEO Coordination Centre (risk list, special risk list, close approaches; Asteroid Tracker) and the ESA/Webb and ESA/Hubble picture archives (Image Explorer) are in CREDITS.md, on the Credits page and in DATA-SOURCES, and their credit lines appear in the footers of the pages that show them. NASA/JPL's Sentry impact monitoring, added at the same time, is listed too.
- **Privacy:** the ESA/Hubble privacy policy is linked; the list of sites the pages link to includes ESA's NEO Coordination Centre (the Asteroid Tracker links to it but the browser never contacts it; its data is fetched by the daily data job on GitHub).
- **Terms:** ESA/Webb and ESA/Hubble pictures are released under CC BY 4.0, each with its own credit line.
- **About, the home page, the README and CLAUDE.md** describe the ESA data in the Asteroid Tracker and Image Explorer; the README shows the Image Explorer as live.

November 9, 2026 — Image Explorer: ESA's Webb and Hubble pictures in the Picture library

- **Each search also asks ESA/Webb and ESA/Hubble's picture archives** and mixes their pictures with NASA's, sorted by the same score and taking turns between libraries. A tag on each picture says whose it is (NASA, ESA/Webb or ESA/Hubble); **From** picks NASA and ESA, NASA, or ESA; the results line counts each library.
- **Viewer for ESA pictures:** the credit line in full (with its links), the licence (CC BY 4.0), the caption, the release date, type, constellation and category, read from ESA's page for the picture; a link to that page.
- **Mission buttons:** James Webb and Hubble also list ESA's archive for that telescope.
- **Globe:** the selected object's pictures come from NASA's and ESA's libraries in turn, with a count for each.

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- If one library is out of reach the others still show, and the results line says which is out of reach.
- Credits page, CREDITS.md and the Privacy page list ESA/Webb and ESA/Hubble; the footer reads "Data courtesy of NASA and ESA". The mock-up page was removed.

November 9, 2026 — Asteroid Tracker: ESA's impact-risk list and close approaches

- **Impact risk tab:** every object on ESA's risk list (NEO Coordination Centre, with its special risk list) or NASA/JPL's Sentry list, side by side: each centre's chance, Palermo and Torino values and most likely impact date, with filters (which list, size, name), sorting, row details linking to both centres' pages, and the rules for reading the chance, Torino, Palermo and why the two lists differ.
- **Asteroid page:** an "Impact risk" section with ESA's and NASA's figures, or "Not listed"; the selected approach shows ESA's distance when ESA lists the same pass.
- **Week details:** ESA's miss distance and the impact chance, when ESA has them.
- **Map:** an "On a risk list" filter, and arrow buttons for panning that appear with ✥ (tap to move, hold to glide).
- **Data job:** `fetch_risk.py` writes `data/risk.json` and `data/esa_close.json` twice a day; a change in ESA's format keeps the previous file instead of writing wrong figures.
- ESA's credit ("ESA NEO Coordination Centre (©ESA)") on every page that shows its figures; ESA NEOCC and NASA/JPL Sentry in `CREDITS.md`, on the Credits page and in `docs/DATA-SOURCES.md`.

November 9, 2026 — Image Explorer: ESA picture archives checked; mock-up

- ESA/Webb and ESA/Hubble picture archives checked (search, paging, fields, credit lines, licence, browser access); findings in `docs/DATA-SOURCES.md`.
- Mock-up of the Picture library with ESA's pictures beside NASA's: `image-explorer/mockup-esa.html`.

November 9, 2026 — ESA alongside NASA

- **Decided:** ESA (the European Space Agency) data is added alongside NASA's where it helps; NASA stays the main source. Each project's chat adds ESA to its own project.
- **Rules for ESA sources** in CLAUDE.md: checked against ESA's current documentation first, recorded in DATA-SOURCES, ESA's credit wherever its data appears, listed in CREDITS.md and on the Credits page, no ESA logo, free to run, and any ESA service the browser contacts added to the Privacy page first.
- **Every page** says the project isn't endorsed by NASA or ESA; the site's own pages credit "NASA and ESA (the European Space Agency)"; the home page and About describe data from NASA and ESA.
- **Credits** (page and CREDITS.md), **Terms**, **LICENSE.md**, **DATA-SOURCES** and **DESIGN** cover ESA: its CC BY-SA 3.0 IGO licence and credit form, its terms for other material, and that its logo is not used.
- The Claude project's new name, description and instructions are in `docs/CLAUDE-PROJECT-SETUP.md`.

November 9, 2026 — renamed NASA and ESA Data Projects

- **New name** on every page, document and code comment: NASA and ESA Data Projects.
- **New repository address:** github.com/mcsyme11-byte/NASA-ESA-Data-and-API-Projects; the site is now at https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/. Every full link (the footer's licence and source links, About, Terms, Credits, the README, the docs, the data jobs' contact line) points to the new address.
- Saved browser data (the API key, places in pages) keeps its old storage names, so nothing saved is lost.

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November 9, 2026 — Asteroid Tracker: panning on the map; Back goes one page

- **Map panning:** drag with the right or middle mouse button, or hold Shift while dragging; on a touch screen, drag with two fingers; or press the new ✥ button beside the zoom buttons, after which a plain drag pans. "Re-centre" appears once the map is panned. Choosing Sun, Earth or Selected, an "Out to" view or a famous pass also re-centres. Alt and the arrow keys pan from the keyboard. The pan is kept when returning to the map.
- The mouse wheel, a double click and a pinch now zoom towards the pointer rather than the middle of the map.
- **Fix:** the "‹ Back" buttons on an asteroid's page and on the map went back two pages, because the site-wide Back code in `shared/footer.js` and the tracker's own code both handled the click. The tracker now leaves naming and clicking to the footer, and only shows the map's button when the map was opened from another page.

November 9, 2026 — Asteroid Tracker: clearer close approach in the orbit diagram; slower playing

- **Orbit diagram (each asteroid's page):** the selected close approach is marked by two rings joined by a line, where Earth (blue) and the asteroid (brass) are at the moment they are closest, labelled "Closest to Earth" with the date and NASA's distance. Before the approach the side panel counts down ("Closest approach in 40 days"). Play stops exactly at the approach and the rings turn solid; pressing Play again carries on. The "Near Earth" view marks the same point on the asteroid's path past Earth.
- **Speeds:** Slow 1 day a second, Normal 5 days a second (was 60), Fast 20, Fastest 90; each named in the menu.
- **Map:** pressing a highlighted famous-pass button again ends the tour and returns to today.
- In the "Near Earth" view, the "towards the Sun" label no longer overlaps the time marks.

November 9, 2026 — shared files always load fresh

- **Fixed:** browsers that had saved an older `shared/theme.css` kept showing the old layout (narrow text, an unstyled Back button) after an update. Every page now links `theme.css` and `footer.js` with a version tag (`?v=2026110902`), raised whenever either file changes, so the new copy is fetched at once.

November 9, 2026 — site pages: full width and Back buttons

- **Full width:** About, Privacy, Terms and Credits now use the page's full content width, like the other pages.
- **Back buttons:** About, Privacy, Terms, Credits, Contact and the Mars Data page have a "‹ Back" button above the heading. It is named for the page it returns to ("‹ Back to the Asteroid Tracker") and returns there the way the browser's Back does; opened directly, it reads "‹ Home".

November 9, 2026 — Asteroid Tracker: closest pass on the home page, size comparisons, three more famous passes

- **Home page:** a line under the live map shows the closest asteroid pass of the next 7 days, in the visitor's time zone, with its distance, size comparison and a link to its page (new daily file `data/soon.json`).
- **Size comparisons** on each asteroid's page, in the tracker's row details and on the map's cards: "about the length of a school bus (12 m)", "about 2 times the length of a Canadian football field (137 m)". One rule set (`compare.js`) replaces the earlier phrase.

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- **Famous close passes:** three more, Toutatis (2004), 2012 TC4 (2017) and 2023 BU (2023), now eight in date order. Times, distances and speeds now come from NASA/JPL close-approach data stored with each pass, so they follow NASA's latest figures (2020 QG: 9,317 km; Duende: 34,053 km; 2019 OK: 71,355 km). The pass path is smoothed between samples; the Moon's-orbit label shows only when the ring is large enough to label clearly.
- **Data job:** a backup daily run at 14:47 UTC, and a push that rebases onto other work and retries.

November 9, 2026 — site pages: About, Privacy, Terms, Credits, Contact

- **Five new pages:** About (what the site is, the projects, how it works, who made it), Privacy (nothing collected; what stays in the browser; the services the browser contacts), Terms of use (free and educational, as is, not for safety decisions, NASA, the API key, no warranty), Credits (every data source and library with its terms, the GIBS acknowledgement, the MIT License) and Contact (the address, Write an email, Copy address, and three ready-made subjects: bug report, the data, ideas).
- **Site links in every footer** (`shared/footer.js`): About · Privacy · Terms · Credits · Contact · Licence (MIT) · Source code, on every page of the site.
- Checked at 1280 and 390 pixels wide; no page errors.

November 9, 2026 — Image Explorer: picture library search and viewer

- **Viewer:** Close sits in its own bar above the picture and stays in view while the text scrolls; the white scrollbar is gone (dark scrollbars throughout); videos and pictures are labelled.
- **Space pictures first:** each result is scored from what NASA records with it (Photojournal and Space Telescope Science Institute pictures, Apollo and Space Station frames and words such as nebula or crater count up; launches, people, events and artists' illustrations count down). Launches, people, events and illustrations move under "Behind the scenes", closed by default.
- **Mission buttons** search several phrasings at once and merge them (for example James Webb: nebulae, galaxies, NIRCam images and "james webb").
- **Results as you type** from 3 letters (after a pause of 0.35 s); a search already made is not asked again.
- **Spelling:** suggestions as you type allow for small mistakes, from about 150 missions, 1,350 sky objects, countries, natural features and towns. A search with fewer than 3 results is repeated with the nearest known spelling ("hubbel" shows Hubble, with a link to search the words as typed).
- **"See it in the sky":** a picture whose title or keywords name a galaxy, nebula or star cluster on the globe opens that object on the globe.
- Escape closes the suggestions without clearing the words.

November 9, 2026 — MIT License and credits

- **Licence:** the project's own code and documents are now under the MIT License (`LICENSE.md`), replacing "all rights reserved".
- **Credits:** `CREDITS.md` lists every outside data source and library with its terms (NASA and JPL sources, GIBS with its acknowledgement, d3-celestial, GeoNames, Natural Earth, OpenLayers, proj4js, three.js).

November 9, 2026 — Image Explorer: date changes reload the pictures

- **Fixed:** changing the date (date box, slider, day buttons, Play) did not reload the satellite pictures already on the map, so an earlier day could show the newest day's picture. Every layer now reloads for the new date. Found while checking the polar views with live imagery.

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- Polar views checked with live NASA imagery: the Arctic (Nov 8: polar night over the pole; Aug 15: the whole Arctic in daylight with summer fires) and the Antarctic (Nov 8: the continent in spring sunlight, snow layer).
- The polar sea ice layer's colour key is a single swatch ("Sea ice seen that day, daylight only") in place of GIBS's large classification legend; the page explains the polar night.

November 9, 2026 — Asteroid Tracker: data check

- Checked the "Asteroid data" job: the 06:17 UTC scheduled run did not start, so the job was started by hand at 12:01 UTC. It succeeded: close approaches for 2026 updated (2027 unchanged), the list of near-Earth objects and the spacecraft positions rewritten, and the site republished. Parker Solar Probe now has 13,493 positions (every 6 hours), and its path curves smoothly around the Sun on the map.

November 9, 2026 — Image Explorer: polar views, natural features, videos, astronaut photos

- **Arctic and Antarctic views** on the Earth map, looking straight down on the poles (GIBS's polar maps, with proj4js). Layers a view lacks are greyed out ("World view only"): sea temperature and smoke and dust in both, fires around Antarctica; sea ice there comes from the daily MODIS sea-ice layer. Choosing a place far from the pole goes back to the world view.
- **Natural features in place search:** 3,978 mountains (with heights), ranges, lakes, rivers, deserts, islands, seas, gulfs and bays from Natural Earth (public domain); choosing a lake, range, river or sea shows all of it.
- **Playable videos** in the picture library: Pictures / Pictures and videos / Videos; a play mark on video cards; the viewer plays the smallest version NASA offers and links the full-size file.
- **Astronaut photos on the globe** (blue squares): a daily job (`astronaut-photos.yml`, `fetch_astronaut_photos.py`) fetches the newest cataloged photos of Earth with their positions from NASA's Gateway to Astronaut Photography of Earth; each opens with the photo, its date, what it shows, a link to NASA's page and NASA's credit line. Switched off until the free API key is stored as the secret `EOL_API_KEY`.
- **Sky positions checked:** Westerlund 2 moved 0.02°; the Rho Ophiuchi marker moved 1.2°, from the star onto the cloud complex named after it.

November 9, 2026 — Image Explorer version 1

- **Live** at `image-explorer/`, from the approved mock-up; the mock-up files are removed. The home page lists the project as live.
- **Globe:** about 1,350 galaxies, nebulae and star clusters (every Messier object, every object with a common name and every galaxy to magnitude 12), the 88 constellations drawn as faint lines with their names, 5,044 stars, all from files kept in the repository (`image-explorer/data/`, built by `tools/build_sky.py`; d3-celestial, BSD licence). Fainter objects are drawn smaller; famous ones are named.
- **Place search:** 32,117 towns of 10,000 people or more and every capital (GeoNames), and countries by name, with suggestions after every letter; tapping the map names the nearest town within about 100 km.
- **Picture library:** live from the start, with an empty search box and a prompt.
- Checked at 320–1440 px: no sideways scrolling, no script errors, and the "out of reach" messages when NASA cannot be reached.

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November 9, 2026 — Asteroid Tracker: smooth spacecraft paths

- Spacecraft paths on the map are drawn through every sample as smooth curves (centripetal Catmull-Rom) instead of a thinned set of straight lines, which looked jagged near the Sun; each spacecraft's dot moves along the same curve.
- Parker Solar Probe and Juno, which repeat the same loop many times, show their last year of flight; selecting one shows its whole path, drawn brighter.
- Parker Solar Probe is sampled every 6 hours instead of daily (`data/spacecraft.json`), so its turn around the Sun is captured: near the Sun it covers about 16 million km a day while passing about 7 million km from it.

November 9, 2026 — Image Explorer: the Globe tab (mock-up)

- **Globe tab** in the mock-up: NASA's Blue Marble Earth, turned to the current sidereal time, inside the real sky (5,044 stars). Pins on the Earth for the natural events NASA EONET reported in the last 30 days; markers in the sky for the 110 Messier objects and famous Hubble and Webb targets (galaxies as rings, nebulae as diamonds, star clusters as dots). Drag to turn, scroll or pinch to zoom, turn slowly, find by name, show or hide each kind. Selecting a sky object shows its facts and NASA's pictures of it; selecting an event opens the Earth map on that place and day.
- **Picture library** in the mock-up now searches NASA live, with the centre, year and type filters and Show more.

November 9, 2026 — Asteroid Tracker: famous close passes, back buttons, smoother zoom

- **Famous close passes** on the map: Apophis (2029), Duende (2013), 2020 QG (2020), 2019 OK (2019) and 2024 YR4 (2032, with its pass by the Moon). Each flies the map to Earth, plays the pass slowly and pauses at the moment of closest approach, with a card of NASA's figures and a short account. Earth, the Moon and the asteroid are placed from NASA/JPL Horizons (new monthly file `data/famous.json`); the distances shown match NASA's close-approach data to within 16 km.
- **Back buttons:** each asteroid's page, and the map when opened from another page, has a "‹ Back" button naming where it returns to. Returning (with that button or the browser's) puts the tracker back on the same week or year, filters, sort, open row, search words and scroll position, and the map back on the same view, selection, date and filters.
- NASA's answers are kept for an hour in the browser tab, so going back or reloading spends no extra requests.
- **"By year" tables:** every name now opens that asteroid's own page (ids from `data/neos.json`, loaded in the background); comets open JPL's page.
- **Map:** zooming in goes down to Earth's true size; Earth, the Sun and the planets are drawn at true size when large enough; the Moon's orbit appears near Earth; the scale bar switches to km close in. A **live clock**: on today's date the map follows the present moment.
- **Smoother:** the orbit maths runs in a background thread (Web Worker), and lines are cut to the visible area, so zooming in close and playing stays at 60 frames a second (slow frames when playing zoomed in fell from about a third to under 2%).
- **Layout:** full-screen and exit-full-screen buttons now look different (with an "Exit" label); the spacecraft list is a full-width card under the map instead of a tall side column; the "Found by" and "Close to Earth in" controls stack on tablets instead of overlapping; a selected asteroid's label no longer touches its ring; starting a famous pass scrolls the whole map into view.

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November 9, 2026 — Image Explorer: data checks and a mock-up for review

- GIBS and the NASA Image and Video Library checked with live requests from the published site; findings recorded in `docs/DATA-SOURCES.md` (the fire layers work only in the `epsg4326` projection; the library allows 10,000 results per search; dates and missing picture sizes explained).
- Plan recorded in `docs/PROJECT-NOTES.md`: OpenLayers, eight layers, two tabs, place search, Arctic and Antarctic views.
- **Mock-up** (`image-explorer/mockup.html`): a live map of NASA imagery with the eight layers, date controls, a day-by-day slider and Play, sample place search, tapping the map, and the picture library with a fixed sample of real results and the picture viewer.

November 9, 2026 — Asteroid Tracker: solar system map, search, home preview, names explained

- **Solar System Map** (`asteroid-tracker/map.html`): all 43,186 known near-Earth objects (asteroids and 209 comets) coloured by orbit type, the eight planets and eleven spacecraft (from Parker Solar Probe to the Voyagers, with the paths they have flown), at any date from 1900 to 2050. Drag to turn, scroll or pinch to zoom, keyboard controls; Top, Tilted and Side views; zoom presets from Mars to the Voyagers; centre on the Sun, Earth or a selection; Play at four speeds; Find; filters (potentially hazardous, over 140 m, over 1 km); this week's visitors ringed; full screen, with a separate exit button and a fallback for phones that cannot make part of a page full screen.
- **Found by the end of:** a discovery timeline from 1960 to today, with "Watch the discoveries"; the objects found in the year shown shine white. **Close to Earth in:** only the asteroids that passed within 10 LD in a chosen year.
- **Search tab:** instant results with every letter over every near-Earth object, a "Found in" filter (a year or a decade, biggest first), cards with an orbit sketch, size, orbit type and a link to the map; typed codes are decoded as they are typed.
- **How asteroid names work:** a guide in the Search tab, and each name decoded on search cards, the map and each asteroid's page (for example "2004 MN4: found June 16–30, 2004, the 113th new small body given a code in that half-month").
- **Home page:** a live preview of the map (the 2,552 potentially hazardous asteroids and the inner planets), pausing when off screen and still for people who prefer reduced motion; it opens the map.
- **Tabs** across the Asteroid Tracker (This week · By year · Map · Search), also on each asteroid's page, which gains "Show on the solar system map".
- New daily data: `fetch_map_data.py` writes `data/neos.json`, `data/preview.json` and `data/spacecraft.json`.
- Layout checked on every page at 320–1440 px; fixes at small widths.

November 8, 2026 — Asteroid Tracker: tidier notes and layout

- "How to read this" notes are now closed by default and open with a tap.
- Layout fixes found in a check of every page at 320 to 1440 px: the "Open its page" button no longer breaks onto two lines; the week bar keeps its dates on one or two neat lines on phones (and reads "Nov 8–14, 2026"); the orbit diagram's time buttons fit on one line at 320 px; the legend lines up on small screens.
- Turning the orbit diagram redraws at most once per screen refresh, so dragging stays smooth.

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November 8, 2026 — Asteroid Tracker: each asteroid's page, the orbit diagram, past years

- **A page for each asteroid** (`object.html?id=…`, 1 NASA request): size, length of its year, distance from the Sun, orbit tilt and the closest its orbit comes to Earth's; every close approach to Earth from 1900 to 2200 on a chart and in a list; facts about how its orbit was worked out. Opened from a row's details in the week view, and from numbered asteroids in the year tables.
- **The orbit diagram:** the Sun, Mercury to Mars (and Jupiter for far-reaching orbits) and the asteroid, drawn to scale and placed at any date from 1900 to 2050. Top, Tilted and Side views, drag to turn, zoom, and "Near Earth" (the path past Earth over ±120 days, with the Moon's orbit for scale). Play, step by day or month, a two-year slider, today and the next close approach. Readings for the date: distance from Earth and the Sun, speed around the Sun, height above Earth's orbital plane; NASA's official figures beside the diagram's at a close approach, and a plain statement of how exact the drawing is for the date shown.
- **Past years:** the year view ("By year") covers 2000 to next year, with a year picker, links such as `#year-2012`, tiles suited to past, current and next year, and a chart of the approaches found each year.
- The daily data job now writes one file per year plus an index (`data/years.json`).
- Checked with real data after publishing: for Apophis's December 2027 pass the diagram gives 114.5 LD against NASA's 114.22 LD. In the "Near Earth" view the time ticks follow the asteroid's speed (hourly for a fast close pass such as Apophis's on April 13, 2029, every few days for a slow distant one). For passes closer than the Moon the page notes that Earth's own pull, left out of the diagram, bends the path.

November 8, 2026 — Asteroid Tracker: charts explained

- Each chart now has a short "How to read this" note with an example.
- The size-against-distance chart shades the area closer than the Moon and labels its directions ("closer than the Moon", "bigger ↑ · farther →") above the plot, where they never cover data.

November 8, 2026 — Asteroid Tracker, version 1

- **This week** (live from NASA NeoWs, 1 request per week shown): tiles for the number passing Earth, the closest, the biggest, the fastest and those flagged potentially hazardous; previous and next weeks.
- **Filters:** potentially hazardous only, size, distance and day. **Sorting** by any column (a "Sort by" menu on phones). Each row opens its details with sizes, distances and speeds people can picture.
- **Three graphs,** hand-drawn SVG in the project's look: size against miss distance (each size a min–max range), approaches per day, and the week's biggest. Tooltips on hover or tap.
- **This year** (NASA/JPL close-approach data): every approach within 1, 5 or 10 Moon distances on a timeline, the closest of the year and the ones coming up. The data file is refreshed daily by a new GitHub Actions workflow, because the JPL API cannot be called from a web page.
- **Key settings:** add or remove a personal NASA key (kept in the browser only); the page shows which key is in use.
- Loading, empty and error states (NASA out of reach, demo key used up, data file missing), checked at 320 to 1440 px.
- Docs: the checked facts about NeoWs and the JPL API, and the size-from-brightness rule, in DATA-SOURCES.

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November 8, 2026 — documentation tone

- **New rule:** all documentation is written in a professional tone, as project documentation, never in the tone of an AI assistant talking to a person (CLAUDE.md, PROJECT-NOTES, the Claude project's instructions).
- The documents were reviewed against it and the remaining informal lines reworded.
- `dev/check_wording.py` now also flags common chat phrases (first-person promises, greetings and sign-offs).

November 8, 2026 — the site is live

- **GitHub Pages switched on:** the home page is at https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/ (checked: the page and its three projects load).
- The developer has a personal NASA API key (kept only in the browser and a local `.env`, never in the repository).
- The Claude project's instructions are in place.

November 8, 2026 — repository set up

- **The repository's structure:** a folder for each project (Asteroid Tracker, Mars Data, Image Explorer), `shared/` and `docs/`.
- **The shared look:** `shared/theme.css` (the Flight Deck Suite's navy, brass and cream, serif type) and an emblem.
- **A NASA API helper** (`shared/nasa.js`): one place for requests to api.nasa.gov, the visitor's key kept in their browser (the shared `DEMO_KEY` until they add one), answers cached for the visit, and plain-English errors.
- **A home page** (`index.html`) and a placeholder page for each project, ready for GitHub Pages.
- **Docs:** design guide, data sources, project notes, roadmap, history, and the Claude project's instructions.
- **Checks on GitHub:** the wording check (no personal details, neutral wording) and a key check (no API key in the repository) after every push.

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PART 5

Asteroid Tracker

Project 1: the near-Earth objects passing Earth, from NASA/JPL and ESA.

Asteroid Tracker

Source: `asteroid-tracker/README.md` · *last changed 2026-11-12*

Project 1 of NASA and ESA Data Projects. Status: **version 1 live** (Nov 8, 2026): https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/asteroid-tracker/

Every near-Earth object passing Earth, sorted by size and distance, with graphs to explore in an interactive dashboard.

What it does

- **This week:** every close approach from NASA NeoWs, live: the number passing Earth, the closest, the biggest, the fastest and those flagged potentially hazardous. Previous and next weeks.
- **Filter and sort:** potentially hazardous only, size, distance, day; sort by time, distance, size, speed or name.
- **Details for each object:** size, miss distance and speed, with comparisons people can picture.
- **Graphs:** size against miss distance, approaches per day, the week's biggest.
- **By year:** every approach within 1, 5 or 10 Moon distances on a timeline, any year from 2000 to next year, and how many were found each year (NASA/JPL data, refreshed daily).
- **Each asteroid's page:** its size and orbit, every close approach to Earth from 1900 to 2200, and the **orbit diagram**: the Sun, the planets and the asteroid to scale at any date from 1900 to 2050, turnable, with Play, a slider and a "Near Earth" view.
- **Solar System Map** (`map.html`): all 43,186 near-Earth objects, the planets and eleven spacecraft, turnable and zoomable at 60 frames a second, with a discovery timeline ("Found by the end of"), "Close to Earth in", Find and filters. A live preview runs on the home page.
- **Search:** results with every letter, a "Found in" filter, and a guide to how asteroids are named, with each code decoded (2024 YR4: found December 16–31, 2024, the 117th new small body given a code in that half-month).
- **Visible from here** (since Nov 12, 2026): from the visitor's place (device location or a town), the passes of the next 30 days that rise at least 10° in full darkness, with brightness, what is needed to see each (naked eye, binoculars, small or large telescope), a night chart and "Find it ›" into Sky Finder.
- **No key needed:** the weeks around today and their asteroids come from the daily copy of NeoWs; anything else uses NASA's shared demo key.

The numbers: which rule to use when

You have	You want	Rule	Example
km	Moon distances (LD)	LD = km ÷ 384,400	1,150,000 km ÷ 384,400 = **3.0 LD**
au (JPL)	km	km = au × 149,597,870.7	0.0000501 au = **7,495 km** = 0.0195 LD
km/s	km/h	km/h = km/s × 3,600	12.1 km/s = **43,560 km/h**

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You have	You want	Rule	Example
brightness H, no measured size	diameter range	D (km) = 1329 ÷ √p × 10^(−H ÷ 5), with p = 0.25 (smallest) and p = 0.05 (largest)	H = 22 → **106–237 m**
a measured diameter (JPL, rare)	size	use it as given	433 Eros: **about 16.8 km**
orbital elements and a date	position around the Sun	M = M0 + n·(days since epoch) → solve M = E − e·sin E → x′ = a(cos E − e), y′ = a√(1−e²)·sin E → rotate by ω, i, Ω	1998 VE31 on Nov 9, 2026: **0.1785 au** from Earth (NASA: 0.1785)
a provisional code, e.g. 2024 YR4	when it was found	year; 1st letter = half-month (A = Jan 1–15 … Y = Dec 16–31, no I); order = number × 25 + position of the 2nd letter (no I)	2024 YR4: Dec 16–31, 2024, **117th**
distance from the Sun r, orbit size a	speed around the Sun	v = √(GM☉ · (2/r − 1/a)), GM☉ = 1.327 × 10¹¹ km³/s²	r = 1.15 au, a = 1.05 au → **26.4 km/s**

Data

Source	Used for	Key	How
NASA NeoWs `feed` (`api.nasa.gov/neo/rest/v1/`)	This week	Visitors: none for the copied weeks; the data job uses the secret `NASA_API_KEY`	Daily copy `data/neows/week-*.json` (current week ± 2 weeks); other weeks live from the page (`DEMO_KEY`)
NASA NeoWs `neo/{id}`	Each asteroid's page	As above	Daily copy `data/neows/neo/<id>.json` for asteroids in the copied weeks; others live, 1 request per page
NASA/JPL Close-Approach Data (`ssd-api.jpl.nasa.gov/cad.api`)	By year	none	Daily files, `data/YYYY.json` and `data/years.json` (the API blocks calls from web pages)
JPL approximate planetary elements	The orbit diagram's and the map's planets	none	Copied into `orbit.js` (valid 1800–2050)
NASA/JPL Small-Body Database Query API	Map and search	none	Daily file `data/neos.json` (and `data/preview.json`)
NASA/JPL Horizons	The map's spacecraft and famous close passes	none	Daily file `data/spacecraft.json`; monthly `data/famous.json`
NASA/JPL Sentry (`ssd-api.jpl.nasa.gov/sentry.api`)	Impact risk (NASA's list)	none	Daily file `data/risk.json`
ESA NEO Coordination Centre (`neo.ssa.esa.int`)	Impact risk (ESA's list), ESA's close-approach distances	none	Daily files `data/risk.json` and `data/esa_close.json` (ESA's server blocks calls from web pages)

Limits: the `feed` covers at most 7 days per request; `DEMO_KEY` allows 30 requests an hour and 50 a day per connection, a personal key 1,000 an hour. Details: docs/DATA-SOURCES.md.

Files

```
asteroid-tracker/
├── index.html                  the tracker: This week, By year and Search tabs
├── object.html                 one asteroid: facts, the orbit diagram, every close approach
├── map.html                    the solar system map
├── tracker.css                 styles for all three pages (on top of shared/theme.css)
```

```
├── app.js                       the tracker's data, filters, sorting, tables and tabs
├── object.js                    the asteroid page: loading, the diagram's controls, the approaches
chart and list
├── map.js                       the map page: controls, selection, Find, year filters, full screen
├── skymap.js                    the map's renderer: 43,000 orbits in typed arrays, drawn into a pixel
buffer
├── kepler.js                    the map's orbit solving, shared by the page and the background thread
├── orbitworker.js               the background thread (Web Worker) that solves the orbits while the map
plays
├── back.js                      the "‹ Back" buttons and keeping a page's place when returning to it
├── preview.js                   the home page's live preview of the map
├── mapdata.js                   loads the map's and search's data files
├── search.js                    search over every name, ids, name decoding, result cards
├── searchview.js                the Search tab
├── charts.js                    the tracker's SVG charts and the shared tooltips
├── orbit.js                     orbit maths: Kepler's equation, planet and asteroid positions (JPL
elements)
├── orbitview.js                 the orbit diagram: projection, views, drawing, labels, dragging
├── format.js                    units, the size rule and number formatting
├── compare.js                   sizes next to familiar objects ("about the length of a school bus"),
with its rules
├── soon.js                      the home page's "closest pass, next 7 days"
├── risk.js                      the Impact risk tab: ESA's and NASA's lists side by side, filters,
sorting, details
├── neows.js                     reads the daily copy of NeoWs (data/neows/) for the week view and the
asteroid page
├── esa.js                       reads ESA's close approaches and both risk lists for the week's
details, the asteroid page and the map
├── visible.js                   the "Visible from here" tab: place, passes, night chart, the rules
├── share.js                     "Share this view" and "Share this week": the view as a link, opened by
the share sheet or copied
├── fetch_close_approaches.py    writes data/YYYY.json and data/years.json (daily, GitHub Actions)
├── fetch_map_data.py            writes data/neos.json, data/preview.json, data/spacecraft.json,
data/soon.json (daily), data/famous.json (monthly)
├── fetch_neows.py               writes data/neows/: NeoWs weeks around today and their asteroids' pages
(secret NASA_API_KEY)
├── fetch_sky.py                 writes data/sky.json: positions seen from Earth around the coming
passes (daily)
├── fetch_risk.py                writes data/risk.json (ESA NEOCC and NASA/JPL Sentry) and
data/esa_close.json (daily)
├── data/                        the files above
└── README.md                    this file
```

Run locally: `python3 -m http.server 8000` in the repository's top folder, then open `http://localhost:8000/asteroid-tracker/`. Refresh the data by hand: `python3 asteroid-tracker/fetch_close_approaches.py` (add `--all` for every year).

Asteroid Tracker · part of NASA and ESA Data Projects · Created by **Malcolm Syme** · mcsyme11@gmail.com

Sky Finder

Source: `sky-finder/README.md` · *last changed 2026-11-12*

An installable phone web app that points to tonight's passing asteroids, with the Moon, planets, bright stars and the space stations to check the aim. Part of NASA and ESA Data Projects, at `sky-finder/` on the site. Created by Malcolm Syme (mcsyme11@gmail.com), © 2026.

What it does

- **Tonight's list** (`index.html`, `app.js`): where you are (your device's location or a town), when it is dark, the asteroid passes worth looking for (tonight first), the Moon, the planets and the ISS and Tiangong passes you can see.

- **Camera view** (`view.js`): the camera picture with markers over it. The chosen asteroid is a diamond with its path hour by hour; the Moon, planets, bright stars and space stations are drawn for reference; an arrow points to anything off the screen and a line says which way to turn. Without motion sensors, the sky is dragged by hand.
- **Calibration:** with the Moon, a planet or a bright star in the brass circle, the difference between where the sensors say the phone points and where that object is (in azimuth and altitude) is taken off every marker. It is kept for 12 hours.
- **How-to page** (`how-to.html`): adding it to the home screen, the three permissions, finding an asteroid, what the marks mean, what is needed to see it.
- **Offline:** `sw.js` keeps the app's files and the last data in the browser's cache.

How it works

Step	Rule
Where the phone points	The W3C device-orientation angles α, β, γ give R = Rz(α)·Rx(β)·Ry(γ); the back camera looks along R·(0, 0, −1). Android's `deviceorientationabsolute` gives α from magnetic north; on an iPhone, `webkitCompassHeading` sets it (`orient.js`).
A direction to the screen	v·forward = f, v·right = x, v·up = y; X = W/2 + F·x/f, Y = H/2 − F·y/f, F = (L/2)/tan(θ/2), with θ the camera's view across the long side L (65° by default, adjustable). Behind the phone when f ≤ 0.
Where things are	`shared/sky.js`: precession, the observer's place on the WGS84 ellipsoid, refraction; asteroids from `asteroid-tracker/data/sky.json`; stations by SGP4 (`shared/sats.js`).
Seen tonight	Asteroid: at least 10° up with the Sun 12° down. Station: at least 10° up, in sunlight, the Sun 6° down.

Privacy and safety

Camera, location and motion readings stay on the device; nothing is recorded or sent. The place, the settings and the calibration are kept in the browser's local storage. A notice before first use and the site's Terms say never to look at the Sun and to look where you walk. See the site's Privacy and Terms pages.

Data

NASA/JPL (close approaches and Horizons), CelesTrak (orbital elements), d3-celestial (bright stars, BSD 3-Clause), GeoNames (towns, CC BY 4.0). Details in `docs/DATA-SOURCES.md`. Not endorsed by NASA or ESA.

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PART 6

Spacecraft map

A utility: the active spacecraft of NASA, ESA and others, from low Earth orbit to the Voyagers.

Source: `spacecraft/README.md` · *last changed 2026-11-12*

Where the active spacecraft are right now: crewed stations, telescopes, missions to the Sun, the Moon, Mars, Jupiter and beyond, and satellites watching Earth. A utility of NASA and ESA Data Projects at `spacecraft/`, previewed on the home page. Created by Malcolm Syme (mcsyme11@gmail.com), © 2026.

Files

File	What it is
`catalogue.json`	Every craft on the map, written by hand: name, agencies, kind, launch, description, timeline, links, and where its position comes from (`src`). Facts checked Nov 12, 2026.
`fetch_spacecraft.py`	The data job (workflow `spacecraft-data.yml`, twice a day): Horizons positions, CelesTrak orbital elements, the link check.
`data/positions.json`, `data/tle.json`, `data/links.json`	What the job writes.
`spacemap.js`	The map: a subclass of the asteroid map's `SolarMap`, with the craft, their icons, groups and paths.
`app.js`, `index.html`, `spacecraft.css`	The page: views, filters, list, card, time, share links.
`preview.js`	The home page's preview.

Positions

Each craft is drawn at its parent's position plus its own position relative to that parent:

Parent	Source	Samples
Sun	Horizons, `CENTER=500@10`	every 2–30 days (Parker Solar Probe every 6 hours)
Earth, near L1 and L2	Horizons, `CENTER=500@399`	every 6 hours, 60 days back to 120 ahead
Earth orbit	CelesTrak elements, SGP4 in the browser (satellite.js)	any moment
Moon, Mars	Horizons, `CENTER=500@301` and `500@499`	every 5 minutes (rovers 30), about a day and a half

Positions between samples follow a centripetal Catmull-Rom curve (as on the asteroid map). Satellite positions are turned from Earth's equator to the ecliptic by the obliquity, 23.4393°.

Adding or removing a craft

1. Check the mission's status and facts on the agency's own pages.
2. Add or remove its entry in `catalogue.json` (`src`: a Horizons id, Horizons names to look up, or CelesTrak names).

3. Push: the "Spacecraft data" workflow runs and its log says whether the craft was found.

Not endorsed by NASA, ESA or any other agency; no agency's logo is used.

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PART 7

Image Explorer

Project 2: the Earth from orbit, a globe inside the sky, and the picture libraries of NASA and ESA.

Source: `image-explorer/README.md` · *last changed 2026-11-11*

Project 2 of NASA and ESA Data Projects. Status: **live** (version 1, Nov 9, 2026).

NASA shares huge collections of scientific and satellite imagery. This project is a website that lets people search, filter and explore those images by details like date, location and mission.

Three tabs

Part	Source	What people do
Earth map	NASA GIBS (Global Imagery Browse Services)	Pick a date and a picture (true colour, night lights, sea temperature) with fires, snow, clouds, smoke and dust or sea ice on top; pan and zoom the whole planet; step or play through the days.
Globe	NASA Blue Marble (GIBS), NASA EONET, d3-celestial	Turn and zoom the Earth inside the real sky; pins for NASA′s natural events of the last 30 days; about 1,350 galaxies, nebulae and star clusters, each opening NASA′s pictures of it.
Picture library	NASA Image and Video Library (`images-api.nasa.gov`); ESA/Webb and ESA/Hubble picture archives (esawebb.org, esahubble.org)	Search NASA′s and ESA′s pictures together (a tag says whose each is; From picks one or both) by keyword, place, mission (Apollo, Hubble, Webb…), centre and year, with results and spelling suggestions as you type; space pictures first, launches and events under "Behind the scenes"; open a picture with its description and date, or find its galaxy or nebula on the globe.

Reason for using several sources: GIBS is satellite imagery **of Earth**, arranged by date and layer, with no "mission" to filter by. The Image Library has the mission photos. The globe ties places on Earth and objects in the sky to them.

Data notes

- **GIBS** needs no key. Tiles come from a web map tile service (WMTS), one URL per tile, for example: `https://gibs.earthdata.nasa.gov/wmts/epsg4326/best/MODIS_Terra_CorrectedReflectance_TrueColor/default/2026-06-01/250m/{level}/{row}/{column}.jpg`
- The map uses the `epsg4326` projection, the only one in which the fire layers load (checked Nov 9, 2026).
- Layers have their own dates and zoom limits; some lag a few days, others stop years ago. The page checks a layer′s dates before offering them.
- **GIBS acknowledgement** goes on the page (NASA asks for it; see docs/DATA-SOURCES.md).
- **Image Library** needs no key.

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Files

```
image-explorer/
├── index.html          the page: Earth map, Globe, Picture library
├── explorer.css        the page's styles (on top of shared/theme.css)
├── app.js              the Earth map (world, Arctic, Antarctic), place search, tabs and the picture
library
├── globe.js            the Globe tab (three.js)
├── fetch_astronaut_photos.py   daily astronaut photos (GitHub Actions; needs the EOL_API_KEY secret)
├── data/
│   ├── places.json     towns of 10,000 people or more and capitals (GeoNames, CC BY 4.0)
│   ├── features.json   mountains, ranges, lakes, rivers, deserts, islands, seas (Natural Earth, public
domain)
│   ├── astronaut.json  the newest astronaut photos of Earth with their positions (empty until the key
is set)
│   ├── stars.json      5,044 stars to magnitude 6        ┐
│   ├── sky.json        about 1,350 deep-sky objects      ├ d3-celestial (BSD licence,
│   ├── constellations.json  the 88 constellations        ┘ LICENSE-d3-celestial.txt)
│   └── LICENSE-d3-celestial.txt
├── tools/
│   ├── build_sky.py        builds stars.json, sky.json, constellations.json
│   ├── build_places.mjs    builds places.json
│   └── build_features.py   builds features.json
└── README.md           this file
```

Rebuilding the data:

```
npm install d3-celestial all-the-cities@3.1.0                 # in any scratch folder
python3 image-explorer/tools/build_sky.py node_modules/d3-celestial/data
node image-explorer/tools/build_places.mjs image-explorer/data/places.json   # run from the scratch
folder
python3 image-explorer/tools/build_features.py path/to/natural-earth-geojson   # the six files named in
the script
```

Image Explorer · part of NASA and ESA Data Projects · Created by **Malcolm Syme** · mcsyme11@gmail.com

PART 8

Mars Data

Project 3: weather measured on Mars, cleaned, organised, analysed and charted.

Mars Data

Source: `mars-data/README.md` · *last changed 2026-11-12*

Project 3 of NASA and ESA Data Projects. Status: **live**, with the whole released Curiosity record (sols 1–4843, August 2012 to March 2026). Automatic updates for new NASA releases are planned.

The weather on Mars as measured by NASA's Curiosity rover at Gale Crater since August 2012: the measurements cleaned, organised by season, time of day and the rover's height, summarised, and charted.

The page

`index.html`, in the project's look (`docs/DESIGN.md`), drawn as hand-made SVG charts from `data/summary.json` and, for one sol at a time, `data/hourly/MY<n>.json`:

- **The Martian year:** air pressure through the year, one line per Mars Year, with a correction for the rover's climb; how much the pressure swings in a day; humidity at night (preliminary, as NASA marks it); midday ultraviolet; average air temperature by season and local hour.
- **One sol:** air and ground temperature and pressure hour by hour, for any sol with data.
- **The whole mission:** daily highs and lows of air temperature; the rover's height above its landing site.

How the data is made

Step	What happens	Where
Download	One MODRDR and one ADR file per sol from NASA's Planetary Data System (Atmospheres Node), in chunks of about 90–130 sols; the corrected UV files, 100 sols each	`analysis/run_rems.py`, `analysis/run_uv.py`
Clean	The rules in `analysis/CLEANING.md`: NASA's quality flags read character by character, physical limits, missing-value codes; every removal counted	`analysis/rems.py`
Organise	Sol, Ls (season) and Mars Year (`analysis/mars_time.py`), local hour, rover height	`analysis/rems.py`
Summarise	Hourly and daily statistics; a sol is "full" with good data in 22 of 24 hours	`analysis/rems.py`, `analysis/run_uv.py`
Check	A cleaning report per chunk, checked before the next batch	`data/rems/reports/`
Publish	The small files the page reads	`analysis/make_page_data.py`

Raw downloads are never committed; `data/rems/manifest.csv` lists every file used, with its size and SHA-256.

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Files

```
mars-data/
├── index.html            the page
├── analysis/             Python: download, clean, organise, summarise (CLEANING.md: the rules)
├── data/summary.json     one row per sol and the season-by-hour grid, read by the page
├── data/hourly/          hourly means per sol, one file per Mars Year
├── data/rems/            sols.csv, hourly.csv, uv.csv, manifest.csv and the cleaning reports
└── README.md             this file
```

Data

NASA Planetary Data System, Atmospheres Node: MSL REMS MODRDR, ADR and UVRDR data sets. Citations and checks: docs/DATA-SOURCES.md. Planned next: Perseverance's weather station (MEDA) and ESA's Mars Express radio occultation profiles.

Created by **Malcolm Syme** · mcsyme11@gmail.com · part of NASA and ESA Data Projects.

Mars Data: cleaning rules

Source: `mars-data/analysis/CLEANING.md` · *last changed 2026-11-12*

Rules applied by `rems.py` to NASA's REMS MODRDR tables (Curiosity's weather station, Gale Crater). Each rule says what is done and why. A change to any rule raises the version number in `rems.py` (`RULES_VERSION`), and every row of `data/rems/sols.csv` records the version that produced it. Nothing is removed silently: every chunk of sols has a cleaning report in `data/rems/reports/` with a count for each reason.

Source: NASA Planetary Data System, Atmospheres Node, volume `mslrem_1001`; data set MSL-M-REMS-5-MODRDR-V1.0 (the REMS team's highest processing level: environmental values corrected for rover heat, shadows and dust). Column meanings are taken from the archive's own `LABEL/MODRDR6.FMT`; the column list is compared with the expected list on every run, and processing stops if it has changed.

Which rule for which situation

Situation	Rule	Action
A value is `-999.00`, `-9.990E+02`, `UNK`, `NULL` or blank	0	Treated as missing (NASA's codes for "cannot be determined", ERRATA.TXT, 2024-09-30)
Two rows share a timestamp	1	The first is kept, the other removed
A row's local time cannot be read	2	Removed
A row's local time is on a different sol from its file	2	Counted; not used in that sol's statistics
A quality flag says the reading is bad	3	The value is removed (the rest of the row stays)
A value is outside physical limits	4	The value is removed
A flag gives information, not quality	5	Kept, and counted in the report
A sol has good data in fewer than 22 of its 24 hours	6	Statistics are given but marked "partial"
Corrected UV is missing (Sun more than 55° from overhead)	9	No UV value for that minute

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Rule 0 — Missing values

NASA writes `-999` (as `-999.00` or `-9.990E+02`) where a value cannot be determined; files released before September 2024 used `UNK` or `NULL`. All are treated as missing. The wind columns are missing throughout: the wind sensor was damaged during landing, and the REMS team does not release processed wind data.

Rule 1 — One row per second

Rows are sorted by timestamp (seconds since noon, January 1, 2000). A repeated timestamp keeps its first row.

Rule 2 — Local time

Local Mean Solar Time (LMST) has the form `SSSSSMHH:MM:SS.sss`: sol number, `M`, then hours, minutes and seconds of the Mars day. The hour of the day (0–24 Mars hours, each 1/24 of a sol) comes from it.

Rule 3 — NASA's quality flags

Each flag is a string of characters, one per check, counted from the left; `1` = good, `0` = bad, `X` = unknown. Not every character is a quality check, so each is read for its own meaning (MODRDR6.FMT):

Measurement	Column	Characters required to be `1`	Meaning
Pressure	`PS_CONFIDENCE_LEVEL`	2	No shadow effect
Air temperature	`AMBIENT_TEMP_CONFIDENCE_LEVEL`	8 and 9	Error under 5 K; observation reliable
Ground temperature	`GTS_CONFIDENCE_LEVEL`	2, 3 and 4	High-quality recalibration; rover still; no shadow in the sensor's view
Relative humidity	`HS_CONFIDENCE_LEVEL`	2 must not be `0`	Relative humidity quality not "bad" (it is `X`, unknown, in the files examined)
Water vapour	`HS_CONFIDENCE_LEVEL`	2 not `0`, and 3	As above, and the amount of water vapour within the sensor's range (under 400 ppm)

An `X` in a character required to be `1` counts as not passing. For humidity, NASA marks the quality character `X` (unknown) throughout, so only readings marked `0` (bad) are removed.

Rule 4 — Physical limits

Values outside these limits cannot be real at Gale Crater and are removed:

Measurement	Limits	Reason
Pressure	400–1,200 Pa	Gale Crater's readings stay roughly within 700–950 Pa through the year
Air temperature	130–310 K (−143 to 37 °C)	Well below the frost point of carbon dioxide at this pressure (about 148 K) / well above any air temperature measured on Mars
Ground temperature	130–330 K (−143 to 57 °C)	As above, with room for warm sunlit ground
Relative humidity	0–100 %	Relative humidity cannot be outside this range
Water vapour	0–400 ppm	The sensor's stated range (character 3 of the flag)
Corrected UV, band ABC	0–30 W/m^2	Well above the largest flux possible at Mars's distance from the Sun

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The limits are deliberately wide: they catch impossible values, not unusual ones. Unusual values are listed for review instead (rule 8).

Rule 5 — Information kept with the values

- **Pressure oscillator:** the first character of `PS_CONFIDENCE_LEVEL` says which of the sensor's two oscillators was used (`0` = oscillator 1, `1` = oscillator 2). It is not a quality check; readings from both are kept, and the number from oscillator 1 is counted.
- **Ground temperature power supply:** the first character of `GTS_CONFIDENCE_LEVEL` is `0` when the sensor's power supply is outside 4.87–4.97 V. Late in the mission this is true of almost every reading, while NASA continues to release the values. They are kept and counted; ground temperature is shown as lower confidence than pressure and air temperature.
- **Humidity is preliminary:** the first character of `HS_CONFIDENCE_LEVEL` is `0` ("preliminary") on every reading in the files examined. Humidity is used (decided Nov 11, 2026) and labelled "preliminary" wherever it appears.

Rule 6 — Coverage of a sol

REMS measures for 5 minutes at the start of every hour, plus some full hours; the full hours move around the day from sol to sol. A local hour counts when it has at least 60 good one-second readings. A sol is **full** for a measurement when at least 22 of its 24 hours count, otherwise **partial**. Charts of daily averages use full sols only.

Rule 7 — Daily statistics

- **Daily mean:** the mean of the hourly means, so every hour of the sol counts once, however long REMS measured in it.
- **Daily minimum and maximum:** from one-minute medians (each minute with at least 30 good readings), so a single one-second gust or noise spike does not set the day's extreme. The local hour of each is recorded.
- **Season:** the solar longitude (Ls) and Mars Year at the middle of the sol's measurements, calculated from the date by the method of Allison and McEwen (2000) in `mars_time.py`, checked against the archive's stated values (sol 1: Ls 151.0, Mars Year 31; sol 4488: Ls 60.1, Mars Year 38). The REMS column `SOLAR_LONGITUDE_ANGLE` is not used: despite its name it is the Sun's direction relative to the rover.
- **Humidity:** the humidity sensor is accurate only when it is cold, so relative humidity is summarised as the day's highest value (from one-minute medians; it is reached at night) and its local hour, and water vapour as the median over local hours 2–6.
- **Rover height:** the median of `ROVER_POSITION_Z` from the ADR file (metres above the landing site). Curiosity has been climbing Mount Sharp; pressure falls with height, so height is kept beside every sol's pressure.

Rule 8 — Listed for review, not removed

Each report lists, for a person to check: a change of more than 15 Pa in daily mean pressure between consecutive full sols; daily air temperatures outside 150–300 K or ground temperatures outside 150–320 K; and a change of more than 50 m in rover height between sols with data.

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Rule 9 — Ultraviolet (UV)

UV comes from NASA's corrected UV data set (MSL-M-REMS-5-UVRDR-V1.0, folder `DATA_UV_CORRECTED`), the REMS team's highest UV processing level: fluxes corrected for dust settling on the sensor and for errors in the original calibration. It covers sols 1–3423, and only times when the Sun is less than 55° from overhead (other times are `-999`). The uncorrected UV in the MODRDR files is not used: dust on the sensor makes it misleading, and after sol 3423 there is no corrected UV, so UV charts end there.

- Band used: UV_ABC (the broad band, 220–370 nm), in W/m^2.
- Per sol: the midday peak (the highest one-minute median, each minute with at least 30 readings), its local hour, and the Sun's angle from overhead at that time (the peak depends on the season and on dust in the air).
- The peak counts as the sol's **midday** peak only if it falls between 10:30 and 13:30 local time and the sol has at least 30 good minutes of UV (`uv_midday`). A peak later in the afternoon, or a sol with only a few minutes, means there were no measurements near noon; charts of the midday peak leave such sols out.
- Each file holds 100 sols (about 1.2 GB); it is read as it downloads and not stored. Its row count is checked against its label.

Rule 10 — Values set aside for review

A value that passes every rule but is plainly not weather (for example one hour 80 °C colder than the hours either side) is listed in `exclusions.csv` with the sol, the measurement and the reason. It stays in the data files but is left out of what the page shows until it has been reviewed. The automatic rules are not changed for a single case.

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PART 9

Data Sources

Every NASA and ESA source the projects use: addresses, keys, limits, terms and credit lines.

Source: `docs/DATA-SOURCES.md` · *last changed 2026-11-12*

Every NASA source the projects use or may use, with how to reach it, what it needs, and its limits. Checked November 8, 2026 (Image Explorer sources: November 9, 2026). Each project's chat confirms the details again before building on them.

Keys at a glance

Source	Key needed?	Used by
api.nasa.gov (NeoWs and others)	Visitors: no (daily copy). The data job: the developer's key, repository secret `NASA_API_KEY`	Asteroid Tracker
JPL SSD APIs (close approaches, small-body database)	No	Asteroid Tracker
PDS Atmospheres Node (Mars weather)	No (file downloads)	Mars Data
GIBS (satellite imagery tiles)	No	Image Explorer
NASA Image and Video Library	No	Image Explorer

api.nasa.gov

- **Sign up:** api.nasa.gov, free, the key arrives by email. One key works for every api.nasa.gov service.
- **Passing it:** as the `api_key` query parameter.
- **Limits:**
  - `DEMO_KEY` (NASA's shared key for trying things out): **30 requests an hour, 50 a day**, per connection (IP).
  - A personal key: **1,000 requests an hour**.
  - Every answer carries `X-RateLimit-Limit` and `X-RateLimit-Remaining` headers; going over returns **429**.
- **In this repository:** no key is committed. Since Nov 12, 2026 the daily data job (`fetch_neows.py`) uses the developer's key from the repository secret `NASA_API_KEY` (GitHub › Settings › Secrets and variables › Actions) to copy the NeoWs data most visitors see into `asteroid-tracker/data/neows/`, so visitors need no key. A key cannot be shared through the pages instead: anything in a public page is readable by anyone, and all visitors would share one key's limit. For weeks and asteroids outside the copy, web pages use `shared/nasa.js` with `DEMO_KEY`, whose limit is counted per visitor's connection (the pages no longer offer to store a visitor's own key). Python scripts run by hand read `NASA_API_KEY` from `.env`.

Daily copy of NeoWs (`asteroid-tracker/data/neows/`, since Nov 12, 2026)

- **What:** the `feed` for the current week (Sunday to Saturday, UTC) and the two weeks before and after it (`week-YYYY-MM-DD.json`), and `neo/{id}` for every asteroid in those weeks (`neo/<id>.json`), in NeoWs's own shape, trimmed to the fields the pages use (Earth approaches only); `index.json` lists the weeks, the asteroids, when each was fetched and whether the developer's key was used.

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- **Requests:** a first run about 5 + 600; later runs only asteroids new to the window, plus copies older than 14 days, at most 750 a run; the script stops when NASA reports fewer than 25 requests left. Without the secret it uses `DEMO_KEY` for the weeks only.
- **If it fails:** the step is marked with a warning, the previous copy is kept, and the pages ask NASA directly.
- **Pages:** `asteroid-tracker/neows.js` reads the copy first; the week view and the asteroid page ask NASA only for what the copy does not hold.

Asteroid Tracker

NeoWs (Near Earth Object Web Service)

Base: `https://api.nasa.gov/neo/rest/v1/`

Endpoint	What it returns
`feed?start_date=YYYY-MM-DD&end_date=YYYY-MM-DD`	Every object with a close approach in the range, grouped by date. **At most 7 days per request.**
`neo/{id}`	One object: its orbit and all its known close approaches
`neo/browse?page=n&size=20`	The whole catalogue, a page at a time

Useful fields per object: `name`, `estimated_diameter` (min and max, in km, m, miles and feet), `is_potentially_hazardous_asteroid`, and in `close_approach_data`: `close_approach_date_full`, `relative_velocity.kilometers_per_hour`, `miss_distance.kilometers` and `miss_distance.lunar`.

Example: one week's feed

```
https://api.nasa.gov/neo/rest/v1/feed?start_date=2026-11-01&end_date=2026-11-07&api_key=DEMO_KEY
```

Checked Nov 8, 2026 (live requests from the published site):

- **Browsers may call it** (CORS allowed), so the Asteroid Tracker requests the week's feed from the page.
- **The rate-limit headers are not readable from a web page** (NASA does not expose them to browsers), so the page cannot show "requests left"; it shows which key is in use and that key's limits instead.
- `close_approach_data[].epoch_date_close_approach` is the close-approach time in milliseconds, UTC (1794183300000 = 2026-11-09 00:15 UTC, matching `close_approach_date_full`). Speed used: `relative_velocity.kilometers_per_second`.
- `estimated_diameter.meters` gives min and max; the range corresponds to albedos of about 0.25 and 0.05 (see the size rule below).
- A typical week: 42 close approaches, 5 flagged potentially hazardous (Nov 8–14, 2026).

`neo/{id}`, checked Nov 8, 2026 (used by the Asteroid Tracker's page for one asteroid, 1 request per page):

- Top level: `id`, `neo_reference_id`, `name`, `designation`, `absolute_magnitude_h`, `estimated_diameter`, `is_potentially_hazardous_asteroid`, `is_sentry_object`, `nasa_jpl_url`, `close_approach_data`, `orbital_data`.
- `close_approach_data`: every computed approach, 1900–2200 (for example 36 for 1998 VE31, all to Earth); same fields as in the feed.
- `orbital_data` (numbers given as strings): `semi_major_axis` (au), `eccentricity`, `inclination`, `ascending_node_longitude`, `perihelion_argument`, `mean_anomaly` (degrees, J2000 ecliptic), `mean_motion` (degrees per day), `epoch_osculation` (Julian date of these elements), `perihelion_distance`, `aphelion_distance`, `orbital_period` (days), `minimum_orbit_intersection`

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(MOID, au), `orbit_class` (type, description, range), `observations_used`, `first_observation_date`, `last_observation_date`, `data_arc_in_days`, `orbit_determination_date`, `orbit_uncertainty` (0 best to 9).

- **Ids:** an unnumbered object has a 7-digit id (3092201 for 1998 VE31); a numbered asteroid's id is 2,000,000 + its number (433 Eros = 2000433).

JPL Close-Approach Data API

`https://ssd-api.jpl.nasa.gov/cad.api` (no key). Close approaches over any date range, past and future, filtered by distance (`dist-max`, for example `10LD`) and date (`date-min`, `date-max`). Suited to "the closest approaches of the year" and to history beyond NeoWs's 7-day window.

Checked Nov 8, 2026 against the API's documentation (version 1.5) and live answers:

- **No CORS headers:** browsers block a web page from calling it. The Asteroid Tracker therefore reads files, `asteroid-tracker/data/YYYY.json` (one per year, 2000 to next year, approaches within 10 LD) and `asteroid-tracker/data/years.json` (the years and their counts within 1, 5 and 10 LD), written by `asteroid-tracker/fetch_close_approaches.py` on GitHub Actions (`.github/workflows/asteroid-data.yml`): daily for this year and next year and any missing year; every year when run by hand with "all".
- **Defaults:** `date-min=now`, `date-max=+60` (days), `dist-max=0.05` (au), NEOs only, Earth, sorted by date.
- **Useful options:** `diameter=true`, `fullname=true`, `sort` (`date`, `dist`, `-dist`, ...), `limit`, `total-only=true` (the count only).
- **Each row is an array** in the order given by `fields`: `des`, `orbit_id`, `jd`, `cd` (calendar time, TDB, about 69 s from UTC), `dist`, `dist_min`, `dist_max` (au), `v_rel`, `v_inf` (km/s), `t_sigma_f`, `h` (absolute magnitude), then `diameter` and `diameter_sigma` (km, usually null) and `fullname` when asked for.
- An empty answer is HTTP 200 with `count` 0. Errors: 400, 405, 500, 503.
- Counts for the whole of 2026, as known on Nov 8, 2026: 1,053 close approaches within 10 LD and 144 within 1 LD (about 50 KB as the daily file); 1,558 within 0.05 au.

JPL Sentry API (impact monitoring), checked Nov 9, 2026

`https://ssd-api.jpl.nasa.gov/sentry.api` (no key; documentation at `https://ssd-api.jpl.nasa.gov/doc/sentry.html`, API version 2.0). NASA/JPL's automatic impact-monitoring list: every object with a calculated, non-zero chance of hitting Earth within the next 100 years or so.

- **Summary (no parameters):** one row per object: `des`, `fullname`, `ip` (cumulative impact probability), `ps_cum`, `ps_max` (Palermo), `ts_max` (Torino), `n_imp` (number of possible impacts), `range` (years), `h`, `diameter` (km), `v_inf` (km/s), `last_obs`. Numbers are sent as strings.
- **`all=1`:** every possible impact of every object (`des`, `date`, `ip`, `ps`, `ts`); used to find each object's most likely impact date.
- **No CORS header** in the replies recorded by the data job, so pages read `asteroid-tracker/data/risk.json`, written by `asteroid-tracker/fetch_risk.py` on GitHub Actions (twice a day, with the close-approach job).
- A speed or other figure can arrive as `nan`; the job keeps only finite numbers and refuses to write NaN, because a browser cannot read JSON containing it.
- Count on Nov 9, 2026: 2,242 objects. Credit: "NASA/JPL (Sentry)".

Planet positions (JPL approximate elements)

`https://ssd.jpl.nasa.gov/planets/approx_pos.html`, Table 1 (valid 1800–2050, mean ecliptic and equinox J2000), checked Nov 8, 2026 and copied into `asteroid-tracker/orbit.js`: each planet's a, e, I, L,

ϖ, Ω and their rates per century, with the procedure (elements at the date → ω = ϖ − Ω and M = L − ϖ → Kepler's equation → in-plane coordinates → rotation into the ecliptic). The orbit diagram therefore covers 1900–2050.

Checks of `orbit.js` (Nov 8, 2026): Earth 0.98330 au from the Sun at the 2026 perihelion and 1.01670 au at aphelion; 1998 VE31 0.1785 au from Earth on 2026-11-09 00:15 UTC (NeoWs: 0.1784980 au), with the computed closest moment within half an hour of NASA's. A century from the elements' epoch the two-body model drifts (1927 approach: 0.190 au computed against 0.151 au), which the page states as "good", "approximate" or "rough" for the date shown.

Size from brightness (when no measured diameter is given)

```
D (km) = 1329 ÷ √p × 10^(−H ÷ 5)
```

- H: absolute magnitude (brightness; a larger H means a smaller object). p: albedo (how much light the surface reflects).
- **p = 0.25** (bright surface) gives the **smallest** likely size; **p = 0.05** (dark) the **largest**.
- Example: H = 22 → 106 m to 237 m. This is the range NeoWs reports; the Asteroid Tracker applies the same rule to JPL's data (`asteroid-tracker/format.js`).

Mars Data

Checked Nov 11, 2026, from GitHub's servers (the project's cloud workspace cannot reach these archives) and against each archive's own documentation. Decided Nov 11, 2026: three missions are followed (PROJECT-NOTES.md).

Curiosity's weather station (REMS), NASA PDS Atmospheres Node — in use

- **Address:** `https://atmos.nmsu.edu/PDS/data/mslrem_1001/` (volume `MSLREM_1001`, PDS3). No key, no account, free. Plain directory pages; folders `DATA/SOL_<first>_<last>/SOL<nnnnn>/`, one folder per sol, 44 chunk folders on Nov 11, 2026, the newest `SOL_04710_04843` (sol 4843, March 22, 2026). The REMS web page still gives sol 4488 as the newest; the folders are newer.
- **Files per sol:** `RME_<clock>RMD<sol>…TAB` (MODRDR: environmental values, corrected; the set used), … `RNV`… (ENVRDR), …`RTL`… (TELRDR), …`ADR`… (ancillary: rover position and Sun geometry), each with a PDS3 `.LBL` and a PDS4 `.xml` label. File sizes grow over the mission: sol 1, 0.3 MB (750 rows); sol 4843, 8.5 MB (22,793 rows). Sols 1–89 with their ADR files: 164 files, 1.0 GB.
- **Format:** comma-separated ASCII, fixed-length rows, 40 columns, one row per second of measurement; text fields in quotes. Column definitions: `LABEL/MODRDR6.FMT`; ADR: `LABEL/ADR_GEOM6.FMT`. Each ADR file holds two tables one after the other: corrections (4 columns), then geometry (16 columns, including `ROVER_POSITION_Z`, metres above the landing site, +Z up).
- **Missing values:** `-999.00` or `-9.990E+02` since the release of September 30, 2024 (`ERRATA.TXT`); earlier files used `UNK` or `NULL`.
- **Wind:** the processed wind columns are always missing (the sensor was damaged during landing).
- **Quality flags:** a string per sensor, one character per check (`1` good, `0` bad, `X` unknown); not every character is a quality check (for pressure, the first character only says which oscillator was used). The rules used are in `mars-data/analysis/CLEANING.md`.
- **Not the season:** the ADR column `SOLAR_LONGITUDE_ANGLE` is the Sun's direction relative to the rover, not Ls. Ls is calculated from the date (`mars-data/analysis/mars_time.py`, Allison and McEwen 2000) and matches the archive's stated values within 0.2° (sol 1: Ls 151.0, Mars Year 31; sol 4488: Ls 60.1, Mars Year 38).

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- **Index:** `INDEX/INDEX.TAB` lists products only up to sol 4133 (last updated September 2024), so the folders, not the index, are read to find new sols. Up to sol 4133, 140 sols have no data (for example sols 2–8 and 200–221).
- **Known events:** `DOCUMENT/DATA_EVENTS.TXT` lists sols with no activity, missing hours and calibrations.
- **Releases:** in batches every few months (`ERRATA.TXT` lists each release, its sols and any re-released files).
- **Citation (from the archive's catalog):** Gomez-Elvira, J., Mars Science Laboratory Rover Environmental Monitoring Station RDR Data V1.0, MSL-M-REMS-5-MODRDR-V1.0, NASA Planetary Data System, 2013.
- **Credit shown on the page:** "Data courtesy of NASA/JPL-Caltech and the REMS team (Centro de Astrobiología), NASA Planetary Data System."
- **Terms:** NASA data; generally not under copyright in the United States.

Perseverance's weather station (MEDA), NASA PDS Atmospheres Node — in use

Checked Nov 12, 2026, from GitHub's servers.

- **Address of the data files:** `https://atmos.nmsu.edu/PDS/data/PDS4/Mars2020/mars2020_meda/data_derived_env/` (PDS4), folders `sol_<first>_<last>/sol_<nnnn>/`, 16 chunk folders, sols 0–1859 on Nov 12, 2026 (May 13, 2026). No key, free. The release folders at `https://pds.nasa.gov/data/pds4/releases/atmos/mars2020_meda-<YYYYMMDD>/` hold only the labels and inventories.
- **Files per sol (derived level):** `WE__<sol>_____DER_<SENSOR>___…_P<nn>.CSV` with a PDS4 `.xml` label, for PS (pressure), RHS (humidity), TIRS (ground temperature and radiation) and ANCILLARY (rover position, Sun geometry). Air temperature (ATS) exists only at the calibrated level. One file spans one sol.
- **Format:** comma-separated ASCII with a header row; an empty value means not reliable for science (bundle `readme.txt`); `9999999999` marks an invalid value in some columns.
- **Coverage:** about 12 hours a sol at 1 Hz, even hours on even sols and odd hours on odd sols, with longer coverage on some sols. Humidity before sol 63 and wind before sol 36 are not reliable (readme).
- **Clock:** `SCLK`, seconds from noon on January 1, 2000; sol 1 falls on February 19, 2021 (Ls 6.2°, Mars Year 36), matching NASA's stated start. The ancillary column `SOLAR_LONGITUDE_ANGLE` is the Sun's azimuth, not Ls (as for REMS).
- **Rules:** `mars-data/analysis/CLEANING-MEDA.md`.
- **Citation:** J.A. Rodriguez-Manfredi et al. (2021), The Mars Environmental Dynamics Analyzer, MEDA, NASA Planetary Data System, DOI 10.17189/1522849.
- **Credit shown on the page:** "Data courtesy of NASA/JPL-Caltech and the MEDA team (Centro de Astrobiología), NASA Planetary Data System."

Mars Express radio occultation profiles (MaRS), ESA Planetary Science Archive — waiting for ESA

- **Address:** `https://archives.esac.esa.int/psa/ftp/MARS-EXPRESS/MRS/` (PDS3). No key, free. Level 5 (derived) data sets on Nov 11, 2026: `MEX-M-MRS-5-OCC-9101-V1.0` and `-V2.0`, `-9102-V1.0`, `-9103-V1.0`, `-9107-V1.0`, `-9124-V1.0` (one per occultation season; 9124 covers May 6 to June 21, 2022).
- **Files:** `DATA/<day>/LEVEL04/` with …`AIX`….`TAB` (atmosphere: temperature, pressure and number density against radius, latitude and longitude; 26 columns), …`IID`…/…`IIX`… (ionosphere), each with a `.LBL` and a `.TXT` description.

- **Citation (from the archive's catalog):** M. Paetzold, Mars Express MRS Level 5 data, Experimental Data acquired from Occultation , Version V1.0.
- **Terms:** data in ESA's Space Science Archives are distributed under the CC BY-NC 3.0 IGO licence (free use with the credit line; commercial use needs ESA's authorisation, `data.licences@esa.int`); ESA's Science Data Centre terms page, `cosmos.esa.int/web/esdc/terms-and-conditions`.
- **Access (Nov 12, 2026):** the simplified site `archives.esac.esa.int/psa/ftp/` answered "401 Unauthorized" while ESA's archive was down for maintenance (psa.esa.int: "The PSA is currently down for maintenance"). ESA's other documented routes, the web view `psaftp.esac.esa.int` (an interactive application) and anonymous SFTP (a password prompt), are not usable from a script without credentials, so the profiles wait until the simplified site returns. NASA's PDS Geosciences Node also lists Mars Express MaRS data (not yet checked). The reader (`run_mex.py`) and its rules (`CLEANING-MEX.md`) are ready.
- **Credit line, exactly as ESA's terms page gives it:** "Credit: ESA, M. Plätzold, K. Pieter". The archive itself names the principal investigator as Martin Pätzold (written "Paetzold"); the line is used as ESA gives it unless ESA confirms a corrected form.

Other candidates (not followed)

Source	Covers	Note
InSight weather (TWINS, pressure sensor)	Elysium Planitia, sols 1–1188 (2018–2022)	The mission has ended; a fixed set
Viking Lander meteorology	1976–1982	For a then-and-now comparison
ExoMars Trace Gas Orbiter (ACS, NOMAD), ESA	Since 2018	Archived mainly as calibrated spectra; gas amounts would need retrievals

SBDB Query API (every near-Earth object), checked Nov 8–9, 2026

`https://ssd-api.jpl.nasa.gov/sbdb_query.api` (no key; no CORS headers, so it is fetched by `asteroid-tracker/fetch_map_data.py` on GitHub Actions and saved as `asteroid-tracker/data/neos.json`).

- Query used: `sb-group=neo`, `full-prec=true`, `fields=spkid,full_name,pdes,a,e,i,om,w,ma,epoch,H,diameter,pha,class,first_obs`.
- Answer: `fields` and `data` (one array per object, numbers as strings) and `count`. 43,186 objects on Nov 8, 2026, 2,552 of them potentially hazardous; 209 are comets (classes JFc, JFC, HTC, ETc, …), the rest asteroids (APO 24,454, AMO 14,979, ATE 3,505, IEO 39). Nearly all share one epoch of osculation (2461200.5); a few thousand have their own.
- `first_obs` is the first observation found, which can be older than the discovery (precovery in archive images): Eros 1893, discovered 1898. The "year found" in `neos.json` is therefore the year in the provisional code (2004 MN4 → 2004, A898 PA → 1898), and `first_obs` only where there is no such code (survey codes such as 6344 P-L); comets have no year found.
- **Ids:** SBDB's `spkid` is 20,000,000 + number for numbered asteroids (Apophis 20099942) and an 8-digit id for the others (1998 VE31 = 50092224). NeoWs (`neo/{id}`) uses 2,000,000 + number for numbered asteroids (2099942) and accepts the 8-digit id for the others (checked: 50092224 returns 1998 VE31). NeoWs does not cover comets; the pages link comets to JPL's lookup page instead.
- Size: about 5.5 MB as JSON, about 1.8 MB as GitHub Pages sends it (compressed).

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Horizons API (spacecraft), checked Nov 8–9, 2026

`https://ssd.jpl.nasa.gov/api/horizons.api` (no key; no CORS headers; fetched by `fetch_map_data.py`).

- Query: `COMMAND='<id>'`, `EPHEM_TYPE=VECTORS`, `CENTER='500@10'` (the Sun), `OUT_UNITS=AU-D`, `REF_PLANE=ECLIPTIC`, `REF_SYSTEM=ICRF`, `VEC_TABLE=1` (position only), `CSV_FORMAT=YES`, `STEP_SIZE='<n> d'`. The table is between `$$SOE` and `$$EOE`; each line gives the Julian date, the calendar date, then x, y, z in au.
- Ids used: Voyager 1 −31, Voyager 2 −32, New Horizons −98, Parker Solar Probe −96, Lucy −49, Psyche −255, OSIRIS-REx/APEX −64, Hera −91, James Webb Space Telescope −170, Europa Clipper −159, Juno −61. On Nov 1, 2026 Voyager 1 was 172 au from the Sun, Voyager 2 145 au, New Horizons 66 au.
- When a range starts before a spacecraft's data, Horizons answers with the first date it covers; the script then retries from that date. Each spacecraft is sampled every 4 to 30 days from launch to a year ahead, and Parker Solar Probe every 6 hours (`STEP_SIZE='6 h'`; stored as a step of 0.25 days), since near the Sun it moves about 16 million km a day while passing about 7 million km from it (about 600 KB in all).
- The International Space Station and Hubble are not on the map: 400–540 km above Earth, they sit inside Earth's dot at any scale it shows.

Horizons API (famous close passes), checked Nov 9, 2026

- Same endpoint, written monthly (on the 1st, or with `--all`, or when the list of passes changes) to `data/famous.json` (about 780 KB, eight passes) by `fetch_map_data.py`. For each pass: the asteroid, Earth (`399`) and the Moon (`301`), with `VEC_TABLE=2` (position and velocity), `CENTER='500@10'`, ecliptic, au and au/day.
- Small bodies are named with a trailing semicolon: `'99942;'` for a numbered asteroid, `'DES=2020 QG;'` for one known only by its code.
- Samples: every 10 minutes within 12 hours of closest approach, hourly within 3 days, daily within 40 days (339 per body). The page joins them with cubic Hermite interpolation (positions and velocities).
- Check: the distance from Earth's centre at closest approach matches the JPL close-approach figures to within 16 km for all five passes; 2024 YR4's pass by the Moon gives 22,940 km against JPL's 22,938 km. Close-approach distances are measured from the centre of Earth (or the Moon), not the surface.
- Since Nov 9, 2026 each pass also stores `cad`: NASA/JPL close-approach data for that date (`cad.api`, `des`, `body=Earth` or `Moon`, `dist-max=0.2`, dates ±3 days): `cd` (UTC time), `km` (from the centre) and `v_rel` (km/s). The tour and its card use these. They differ slightly from earlier published figures because orbits are refined (for example 2020 QG: 9,317 km in 2026, 9,302 km as first published).

soon.json (home page), added Nov 9, 2026

- Written daily by `fetch_map_data.py` from the close-approach year files: every pass within 10 Moon distances from a day before the run to 15 days after it, as `[name, time_ms_utc, distance_ld, h, neows_id, designation]`. The NeoWs id is 2,000,000 + number for numbered asteroids, otherwise the SBDB `spkid`; comets get none. The home page shows the closest pass between the moment it is opened and 7 days later, in the visitor's time zone.

sky.json ("Visible from here" and Sky Finder), added Nov 12, 2026

- Written daily by `asteroid-tracker/fetch_sky.py` (a step of the Asteroid data workflow). Close-Approach Data (`cad.api`, `date-min` yesterday, `date-max` 30 days ahead, `dist-max=0.2`, `fullname=true`, `diameter=true`) gives the passes; each is kept when its estimated magnitude at closest approach, $V \approx H + 5\cdot\log10(\Delta) + 2.5$ (Δ in au, r taken as 1 au, +2.5 for a typical phase angle), is 16 or brighter; at most 40, brightest first.

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- For each, Horizons vectors as seen from Earth′s centre: `COMMAND='DES=<des>;'` (or `'<number>;'`), `CENTER='500@399'`, `REF_PLANE=FRAME`, `REF_SYSTEM=ICRF` (the J2000 equator), `OUT_UNITS=KM-S`, `VEC_TABLE=1`, from 6 days before to 6 days after closest approach (within yesterday to 31 days ahead), every hour when the pass is within 0.02 au and every 3 hours otherwise. Stored as whole kilometres.
- The pages (`shared/sky.js`, `shared/passes.js`) turn these into altitude and azimuth for the visitor′s own place: precession to the date (IAU 1976), the observer′s position on the WGS84 ellipsoid subtracted (parallax matters for close passes), refraction (Bennett), and the magnitude from the IAU H, G system (G = 0.15). Checked against Astropy at 40 random places and times on Nov 12, 2026: worst errors Sun 0.01°, Moon 0.018°, Jupiter 0.09°, bright stars 0.008°.
- The visitor′s place never leaves the browser: the file is the same for everyone.

Asteroid names (Minor Planet Center rules)

Used by `decodeCode()` in `asteroid-tracker/search.js`:

- **Provisional code** `YYYY XYn`: the year found; X the half-month (A = Jan 1–15, B = Jan 16–31, … Y = Dec 16–31; I and Z not used); Y and n the order in that half-month: 25 letters A–Z without I, n the number of times the alphabet was used up, so order = n × 25 + position of Y. Example: 2024 YR4 = Dec 16–31, 2024, 4 × 25 + 17 = 117th.
- Old-style codes before 1925 begin with A and the last three digits of the year (A898 PA = 1898).
- Survey codes `nnnn P-L` (Palomar–Leiden survey, 1960) and `nnnn T-1/T-2/T-3` (Trojan surveys, 1971, 1973, 1977).
- A permanent **number** follows once the orbit is secure, then possibly a **name** approved by the IAU; the full name keeps all three: 99942 Apophis (2004 MN4).
- Checked against known discovery dates: Eros Aug 13, 1898 (A898 PA); Apollo Apr 24, 1932 (1932 HA); Phaethon Oct 11, 1983 (1983 TB); Bennu Sep 11, 1999 (1999 RQ36); Apophis Jun 19, 2004 (2004 MN4); 2024 YR4 Dec 27, 2024.

Image Explorer

GIBS (Global Imagery Browse Services)

- **What:** over 1,000 satellite imagery layers of Earth, as map tiles, by date (1,332 layers on Nov 9, 2026).
- **No key.** Tile URL pattern (WMTS, REST):

```
https://gibs.earthdata.nasa.gov/wmts/{projection}/best/{Layer}/default/{YYYY-MM-DD}/{TileMatrixSet}/
{z}/{y}/{x}.{jpg|png|mvt}
```

- **Layer list and dates:** the GetCapabilities document of each projection, for example `https://gibs.earthdata.nasa.gov/wmts/epsg4326/best/wmts.cgi?SERVICE=WMTS&REQUEST=GetCapabilities`.
- **Required acknowledgement** (NASA asks every user of GIBS to show it):

> We acknowledge the use of imagery provided by services from NASA′s Global Imagery Browse Services (GIBS), part of NASA′s Earth Science Data and Information System (ESDIS).

Checked Nov 9, 2026 (live requests from the published site):

- **Browsers may call it** (CORS allowed): tiles and the GetCapabilities documents load from a web page.
- **Projections:** `epsg4326` (the world as a flat grid of latitude and longitude, 1,332 layers), `epsg3857` (web map, 1,332 layers), `epsg3413` (Arctic, 303 layers) and `epsg3031` (Antarctic, 263 layers).
- **The fire layers only work in `epsg4326`** (and the polar projections): they are vector tiles (`.mvt`, points), listed in `epsg3857` but every `epsg3857` request returned 404 (eight tiles at zoom 0–7, two dates),

while the same tiles in `epsg4326` returned data. **The Image Explorer therefore uses `epsg4326`**, as NASA Worldview does.

- **`epsg4326` tile sets:** 512 px tiles from the top-left corner (−180°, 90°); level 0 is 0.5625° per pixel (2 × 1 tiles), each level halves it. Deepest level: `250m` 8, `500m` 7, `1km` 6, `2km` 5. Tile path order is `{level}/{row}/{column}`.
- **Answers:** a tile outside a layer's dates returns 404 (true colour on 1990-01-01; the MUR layers on a day not yet processed); a level deeper than the layer's set returns 400. The page treats these as "no picture here", not as errors.
- **Size of the layer list:** the `epsg3857` GetCapabilities document is about 5.8 MB, too large to load on every visit; each layer's dates are also available on their own (the `…/all/--{TimeEnd}.xml` and `{TimeStart}--{TimeEnd}` templates under each layer's `ResourceURL`).
- **A layer's days (used since Nov 10, 2026):** `https://gibs.earthdata.nasa.gov/wmts/{projection}/best/1.0.0/{Layer}/default/{TileMatrixSet}/all/--{YYYY-MM-DD}.xml` returns, in about 0.1 s and well under 1 kB, the runs of days the layer has up to that date, in `<Domain>`, for example `2002-06-01/2021-02-19/P1D,2021-02-22/2022-11-08/P1D,…`; an unknown layer returns 404. Checked Nov 10, 2026: MUR sea temperature misses six short runs of days (newest day two days back), true colour misses nine runs in 2000–2022; MODIS layers end on the current, still incomplete UTC day. Browsers may read it.
- **Colour maps (used since Nov 10, 2026 for the colour keys):** each `ColorMap` in the v1.3 XML has a `Legend` (`continuous`, `discrete` or `classification`) with `LegendEntry` elements in order (`rgb`, `tooltip`, and on some `showTick`, `showLabel` and `label`), `minLabel` and `maxLabel`, and the map's `title` and `units` (for example °C, %, K). A "No Data" map lists the colour for missing values. Browsers may read them.
- **Colour keys:** each coloured layer lists an SVG legend (`https://gibs.earthdata.nasa.gov/legends/{name}_H.svg`, dark text on a light background) and a colour map (`https://gibs.earthdata.nasa.gov/colormaps/v1.3/{name}.xml`).

The eight layers chosen (Nov 9, 2026), with their tile sets in `epsg4326` and first dates:

On the page	GIBS layer	Tile set · format	From
True colour	`MODIS_Terra_CorrectedReflectance_TrueColor`	250m · jpg	2000-02-24
Night lights	`VIIRS_NOAA20_DayNightBand`	1km · png	2018-01-05
Sea temperature	`GHRSST_L4_MUR_Sea_Surface_Temperature`	1km · png	2002-06-01
Fires	`MODIS_Combined_Thermal_Anomalies_All`	1km · mvt (points)	2002-07-04
Snow	`MODIS_Terra_NDSI_Snow_Cover`	500m · png	2000-02-24
Clouds	`MODIS_Terra_Cloud_Top_Temp_Day`	2km · png	2000-02-24
Smoke and dust	`MODIS_Terra_Aerosol` (aerosol optical depth, land and ocean)	2km · png	2000-02-24
Sea ice	`GHRSST_L4_MUR_Sea_Ice_Concentration`	1km · png	2002-06-01

- Most layers have gaps in their dates (several ranges in the layer list); the page reads them rather than assuming every day exists.
- **Newest day:** on Nov 9, 2026 the MODIS and VIIRS layers reached Nov 8 or 9; the two MUR layers (sea temperature, sea ice) reached Nov 7.
- **Night lights:** `VIIRS_Black_Marble` (cloud-free city lights) exists only for 2012 and 2016, so the daily day/night band is used; it also shows moonlit cloud and fires.
- `Reference_Features_15m` and `Reference_Labels_15m` (coastlines, borders and names from OpenStreetMap) are available as overlays if wanted later; they require the OpenStreetMap credit.

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NASA Image and Video Library

- `https://images-api.nasa.gov/search?q=...&media_type=image` (no key).
- Filters: `q` (keywords), `center` (for example JPL, GSFC), `year_start`, `year_end`, `keywords`, `page`.
- Each result has a title, description, date, NASA centre and keywords; `asset/{nasa_id}` lists the image sizes.

Checked Nov 9, 2026 (live requests from the published site):

- **Browsers may call it** (CORS allowed), for searches, `asset/{nasa_id}` and the image files. One search request failed to connect and worked when repeated, so the page retries once before showing an error.
- **Answer:** `collection.items[]`, each with `data[0]` (`title`, `description`, `date_created`, `center`, `keywords`, `media_type`, `nasa_id`, and sometimes `photographer`, `secondary_creator`, `location`, `album`, `description_508`) and `links[]` (image sizes with width and height; `rel: "preview"` is the thumbnail); `collection.metadata.total_hits`; `collection.links` with `next` and `prev` pages.
- **Paging:** 100 results per page by default; `page_size` sets it (for example 24). **At most 10,000 results** per search: page 100 of 100 works, page 101 returns 400 ("Maximum number of search results have been displayed").
- `center` ignores letter case (`gsfc` = `GSFC`); `media_type=image,video` returns both. Examples: "hubble" 4,960 results, 598 from GSFC, 249 from 1990–1995; "apollo" 5,940 images and 222 videos.
- **`date_created` is the date in NASA's records,** sometimes not the date the picture was taken (an Apollo picture from 1967 carries 2008-07-11); the caption usually gives the real date.
- **`location` is where the picture was published or taken** (for example "Greenbelt, MD" for Goddard), not a map position: no result has coordinates. Searching by place therefore searches the words in titles, captions and keywords.
- **File names:** `https://images-assets.nasa.gov/image/{nasa_id}/{nasa_id}~{thumb|small|medium|large|orig}.jpg`. A listed size can fail to load for a while (`GSFC_20171208_Archive_e000775`: `thumb` and `medium` failed and `small` loaded on Nov 9, 2026; all three loaded an hour later), so the page tries the next size before showing "Picture unavailable". `asset/{nasa_id}` and older records give `http://` links; the page always uses `https://`.

Blue Marble (the globe's surface), checked Nov 9, 2026

- `BlueMarble_NextGeneration` (also `BlueMarble_ShadedRelief` and `BlueMarble_ShadedRelief_Bathymetry`): NASA's cloud-free composite of the whole Earth. Not listed in the GetCapabilities documents, but the tiles load: `https://gibs.earthdata.nasa.gov/wmts/epsg4326/best/BlueMarble_NextGeneration/default/default/500m/{level}/{row}/{column}.jpeg` (`default` in place of a date; levels 0–7 of the `500m` set).
- The globe joins level 2 (15 tiles, 2,560 × 1,280 pixels, about 16 km a pixel) and then level 3 (50 tiles, 5,120 × 2,560, about 8 km a pixel) where the graphics card accepts a picture that wide.

EONET (Earth Observatory Natural Event Tracker), checked Nov 9, 2026

- `https://eonet.gsfc.nasa.gov/api/v3/events` (no key; **browsers may call it**). Categories (`/api/v3/categories`): drought, dust and haze, earthquakes, floods, landslides, manmade, sea and lake ice, severe storms, snow, temperature extremes, volcanoes, water colour, wildfires.
- Each event: `id`, `title`, `categories[]`, `sources[]` (the reporting agency's page), `closed`, and `geometry[]` (dated points, sometimes polygons, with `magnitudeValue` and `magnitudeUnit`, for example acres for fires).

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- **Many events stay open long after their last report:** 7,236 open on Nov 9, 2026 (7,168 wildfires, 32 volcanoes, 35 sea and lake ice, 1 storm), but only 35 with a report in the last 30 days. The globe asks for `status=open&days=30`.

Sky: stars, galaxies, nebulae, clusters, constellations, checked Nov 9, 2026

- From the d3-celestial project (© 2015 Olaf Frohn, BSD 3-Clause licence, kept as `image-explorer/data/LICENSE-d3-celestial.txt`), npm package `d3-celestial`, folder `data/`.
- Positions are `[right ascension, declination]` in degrees, with right ascension from −180 to 180 (M42 the Orion Nebula: 83.85, −5.45; M57: −76.6, that is 283.4°); the repository's files store 0 to 360. Checked against known positions: Sirius 101.29, −16.72; M31 10.68, +41.27.
- Files used: `stars.6.json` (5,044 stars to magnitude 6, with colour index), `messier.json` (110 Messier objects with common names), `dsos.14.json` (15,809 deep-sky objects to magnitude 14, 10,336 of them galaxies), `dsonames.json` (724 common names), `constellations.json` and `constellations.lines.json` (88 constellations, plus Serpens drawn in two parts: 89 entries).
- **Chosen for the globe** (`image-explorer/tools/build_sky.py`): every Messier object, every object with a common name, and every galaxy of magnitude 12 or brighter (bright enough for a backyard telescope): 1,347 objects (1,001 galaxies, 165 nebulae, 181 star clusters) on Nov 9, 2026.
- Catalogue corrections made by the build: NGC 3372 named "Carina Nebula" (the catalogue says "η Car Nebula"), NGC 3132 "Southern Ring Nebula" (the catalogue gives the name "Eight-Burst Planetary Nebula" to NGC 3132 and "Southern Ring" to NGC 6563); objects split into several catalogue numbers (the Rosette Nebula, Stephan's Quintet) keep the name on the first part and the number on the others; dark nebulae lose their magnitude, which describes nothing visible.
- Four famous targets are not in the catalogue and are listed by hand (J2000, degrees): the Cartwheel Galaxy (9.421, −33.716), Westerlund 2 (155.992, −57.764), Cassiopeia A (350.85, +58.815) and the Rho Ophiuchi cloud complex (247.025, −24.542). **Checked Nov 9, 2026** against SIMBAD's values as published (Wikipedia's infoboxes, which cite SIMBAD, and NASA's Webb image page for the Cartwheel): the Cartwheel and Cassiopeia A were right; Westerlund 2 moved by 0.02°; Rho Ophiuchi had been placed on the star Rho Ophiuchi, 1.2° from the cloud complex named after it, and was corrected. SIMBAD itself did not answer requests from GitHub Actions within 3 minutes that night.

Place names (GeoNames), checked Nov 9, 2026

- GeoNames (geonames.org, CC BY 4.0) through the npm package `all-the-cities` 3.1.0 (MIT), which holds the 135,233 GeoNames places of 1,000 people or more with population and feature code.
- `image-explorer/data/places.json` (`tools/build_places.mjs`): every place of 10,000 people or more and every capital, 32,117 places (241 capitals), largest first, about 1.1 MB (about 0.5 MB as sent compressed). It loads the first time the place search or the map is used.
- Countries are found by their English names (the browser's `Intl.DisplayNames`) and shown around their capital.
- Mountains, lakes and rivers are not in this list.

Astronaut photos of Earth (Gateway to Astronaut Photography of Earth), checked Nov 9, 2026

- `https://eol.jsc.nasa.gov`, Earth Science and Remote Sensing Unit, NASA Johnson Space Center: astronaut photos of Earth from 1961 to today. **Web pages cannot call it** (no CORS headers); `robots.txt` allows everything.
- **Photos Database API** (documented at `https://eol.jsc.nasa.gov/SearchPhotos/PhotosDatabaseAPI/`): `https://eol.jsc.nasa.gov/SearchPhotos/PhotosDatabaseAPI/PhotosDatabaseAPI.pl?query=…`

`&return=…&key=…`, answering JSON (a list of records with `table.field` names). `query` is groups of `table|field|operator|value` (operators `gt ge lt le eq ne like`, joined with AND); `return` is groups of `table|field`. **A key is required**; keys are free and requested by email from jsc-earthweb@mail.nasa.gov.

- Tables: `frames` (cataloged photos: `mission`, `roll`, `frame`, `pdate` YYYYMMDD and `ptime` HHMMSS in UTC, `lat` and `lon` of the photo's centre to 0.1°, `nlat`/`nlon` of the point below the station, `cldp` cloud percentage, `tilt`, `geon` place name, `feat` features), `nadir` (photos not yet cataloged: only the point below the station), `images` (`directory`, `filename`, `width`, `height`, `filesize`; one record per image size).
- Images: `https://eol.jsc.nasa.gov/DatabaseImages/{directory}/{filename}`, for example `https://eol.jsc.nasa.gov/DatabaseImages/ESC/small/ISS073/ISS073-E-100000.JPG`; `ESC/small/…` and `ESC/large/….` Photo pages: `https://eol.jsc.nasa.gov/SearchPhotos/photo.pl?mission=ISS073&roll=E&frame=100000`.
- **Credit** (the site's conditions of use): "Image courtesy of the Earth Science and Remote Sensing Unit, NASA Johnson Space Center", with the photo number (mission-roll-frame) and the site's address.
- In this repository: `image-explorer/fetch_astronaut_photos.py`, run daily by `.github/workflows/astronaut-photos.yml` with the key from the repository secret `EOL_API_KEY`, writes `image-explorer/data/astronaut.json` (the newest 400 cataloged photos of the last 90 days, or the last year if fewer than 60, with 50% cloud or less). Without the secret the job changes nothing and the globe hides the astronaut photos.

Astronaut photos without a key: the "Latest ISS Imagery" set, checked Nov 11, 2026

- `https://eol.jsc.nasa.gov/SearchPhotos/PhotoIdSets/PhotoIdSets.pl?set=Latest_ISS_Imagery&view=1` sets a session cookie and forwards to `ShowQueryResults-Lightcycle.pl?results=Latest_ISS_Imagery`. With that cookie, `GetTSVFromQueryResults.pl?results=Latest_ISS_Imagery` returns the newest 1,000 Space Station photos as tab-separated text (ISO-8859-1), no key needed: `mission, roll, frame, Photo Date (YYYYMMDD), Photo Time GMT (HHMMSS), Center Point Lat/Lon, Nadir Lat/Lon, Spacecraft Altitude (nautical miles), Lens mm, geon, feat, Clouds Percentage, …, camera, RecordType`. Most are "NotCataloged": no centre point, place or cloud cover yet, only the point below the Station (nadir). `frame` has a leading space. `temp/Latest_ISS_Imagery.json` is the gallery's own listing.
- Used since Nov 11, 2026 by `fetch_astronaut_photos.py` when the repository has no `EOL_API_KEY`. Photos without a centre point are placed below the Station and say so on the globe.

Image and Video Library: videos, checked Nov 9, 2026

- A video's files are under `https://images-assets.nasa.gov/video/{nasa_id}/` (images are under `image/`): `~orig.mp4`, and usually `~large.mp4` or `~medium.mp4`, `~small.mp4`, `~preview.mp4`, `~mobile.mp4`, plus stills (`~thumb.jpg`, `~small.jpg`, `~medium.jpg`, numbered frames `~thumb_1.jpg` …) and subtitles (`{nasa_id}.vtt`). `asset/{nasa_id}` lists them (with `http://` links that also work as `https://`).
- Sizes for three Apollo 11 films: `~mobile.mp4` 118–177 MB, `~small.mp4` 534–799 MB, `~orig.mp4` 1.5–5.3 GB. The page plays the smallest version listed and links the largest.
- The video files send no CORS headers: they play in a page, but subtitles (`.vtt`) cannot be attached from another site, so the page shows none.
- "apollo 11": 71 videos.

Image and Video Library: what the results are, checked Nov 9, 2026

- NASA's own order puts launches, people and events first for most mission names: of the first results for "james webb", nearly all are testing, launch, people or illustrations. NASA's library holds only about 5

Webb science pictures (for example `carina_nebula`, `southern_ring_nebula`, `webb_first_deep_field`, `PIA26702`).

- Clues recorded with each result, used for the page's ranking: `nasa_id` starting `PIA` (Planetary Photojournal), `center` `STScI` or `STScI (Webb)` (Space Telescope Science Institute), `nasa_id` of mission frames (`AS11-`… Apollo, `iss0`…e… Space Station) point to pictures of space or of Earth from space; `nasa_id` starting `NHQ`, `KSC` or `jsc`, and photographers credited "NASA/…", point to event photography.
- NASA's APOD API answered 503 on Nov 9, 2026.

ESA/Webb and ESA/Hubble picture archives, checked Nov 9, 2026

Decided Nov 9, 2026: ESA's data is added alongside NASA's. The Image Explorer adds ESA's two telescope picture archives to the Picture library. Checked with requests from GitHub's servers and ESA's published terms.

- **Sites:** `https://esawebb.org/images/` (703 pictures) and `https://esahubble.org/images/` (5,511 pictures), the same archive software. No key and no JSON API.
- **Browsers may read them:** every archive page, picture page, feed and picture file answers with `access-control-allow-origin: *`.
- **Search:** `/images/?search={words}`, 50 results a page; further pages `/images/page/{n}/?search={words}`; newest first with `&sort=-release_date`. The page states the count ("Showing 1 to 50 of 166"); a search with no match has no count and no list. Examples: Webb "nebula" 166, "crab" 7, "hubble" 212; Hubble "andromeda" 116, "saturn" 131, "crab nebula" 47.
- **Results** are in the page's script as `var images = [ { id, title, width, height, src, url, potw }, … ]`: `src` is the 300 px-high thumbnail, `url` the picture's page, `potw` the date for a Picture of the Week or Month (empty otherwise). No date, credit or caption in the list.
- **A picture's page** (`/images/{id}/`) holds the title (`<h1>`), the caption, the credit line (`<div class="credit">`), and tables "About the Image" (Id, Type such as Observation, Release date, Size) and "About the Object" (name, type, constellation, category, distance), plus coordinates (RA, Dec, field of view). The markup differs slightly between the two sites (table classes, headings `h3` or `h4`).
- **Files:** `https://cdn.esawebb.org/archives/images/{size}/{id}.jpg` and `https://cdn.esahubble.org/…`, sizes `thumb300y`, `screen` (1280 px wide), `large`, `publicationjpg`, and others; originals are TIFF, up to 135 MB.
- **Licence:** Creative Commons Attribution 4.0 (CC BY 4.0) for pictures, videos and web texts. Each picture has its own credit line, to be shown in full, unchanged and beside the picture, with active links online. ESA/Webb and ESA/Hubble logos need written consent; the material must not suggest endorsement. Music, papers and code are not under CC BY 4.0.
- **Videos (used since Nov 10, 2026):** `/videos/?search={words}` and `/videos/page/{n}/?search={words}` on both sites, 50 a page, "Showing 1 to … of N" as for pictures, but no list in the page's script: each result is a thumbnail `https://cdn.{site}/archives/videos/videoframe/{id}.jpg` whose `alt` is the title (on esawebb.org prefixed "Video thumbnail for"). A video's page (`/videos/{id}/`) has the title, caption, credit line (`<div class="credit">`, often with a music credit) and a table with Id, Release date, Duration and Category. Files: `archives/videos/medium_podcast/{id}.mp4` (small), `hd_1080p25_screen/{id}.mp4` (full HD), `ultra_hd/{id}.mp4` (Webb); a Hubblecast in full HD is about 86 MB. The videos are under CC BY 4.0 except their music, which has its own terms. Checked Nov 10, 2026 from a web page: listing pages, video pages and files load from a browser. A first video search on esahubble.org took about 30 s (3 s when repeated), so the page waits at most 12 s for each ESA archive.
- **Used since Nov 9, 2026** by the Image Explorer's Picture library and the globe's selected card (live search, thumbnails `thumb300y`, viewer `screen`, "Open large size" `large`). A page past the last answers 404 and is treated as empty.

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- Example credit lines: weic2622c (Cat's Head Nebula) "NASA, ESA, CSA, STScI. Image Processing: J. Depasquale (STScI)"; heic1907a "NASA, ESA, and STScI".

GIBS polar views, checked Nov 9, 2026

- `epsg3413` (Arctic, 303 layers) and `epsg3031` (Antarctic, 263 layers). Tile grid: 512 px tiles from (−4,194,304 m, 4,194,304 m), 8,192 m a pixel at level 0, halving each level; deepest level `250m` 5, `500m` 4, `1km` 3. Map projections: polar stereographic, EPSG:3413 `+lat_ts=70 +lon_0=-45`, EPSG:3031 `+lat_ts=-71 +lon_0=0`.
- Of the eight layers: true colour (`250m`), night lights (`1km`), snow (`500m`) and clouds (`1km`, newest day a day behind) exist in both; fires (`MODIS_Combined_Thermal_Anomalies_All`, vector tiles) only in the Arctic view; sea ice comes from `MODIS_Terra_Sea_Ice` (`1km`, daily, from 2000) because the MUR sea-ice layer is world view only; sea temperature and smoke and dust are world view only. Test tiles for each answered 200 with CORS allowed.
- Blue Marble (`BlueMarble_NextGeneration`, `BlueMarble_ShadedRelief_Bathymetry`) is listed in both polar layer lists, unlike the world list.
- Checked with live imagery Nov 9, 2026: on Nov 8 the true-colour picture has a black area over the North Pole (polar night; MODIS is a camera); on Aug 15 the Arctic is covered. `MODIS_Terra_Sea_Ice` is a classification layer whose only coloured class is sea ice (rgb 255, 100, 100; colour map `MODIS_L2_Sea_Ice.xml`). GIBS's microwave sea-ice layers, which work in the dark, end in Aug 2021 (`SSMIS_Sea_Ice_Concentration`) and Sep 2025 (`AMSRU2_Sea_Ice_Concentration_12km`).
- OpenLayers note: `source.refresh()` did not fetch new tiles after the date changed; `setTileUrlFunction(fn, key)` with a new key does.

Place-name services not used

Natural features (Natural Earth), checked Nov 9, 2026

- Natural Earth 1:10m (naturalearthdata.com, **public domain**), GeoJSON files from the `nvkelso/natural-earth-vector` repository on GitHub: `ne_10m_lakes`, `ne_10m_rivers_lake_centerlines`, `ne_10m_geography_regions_points`, `ne_10m_geography_regions_elevation_points`, `ne_10m_geography_regions_polys`, `ne_10m_geography_marine_polys`.
- `image-explorer/data/features.json` (`tools/build_features.py`): 3,978 named features (1,063 rivers, 627 mountains with their heights, 453 lakes, 405 islands, 219 reservoirs, 214 mountain ranges, 158 island groups, 71 seas, 61 gulfs, 60 bays, 58 deserts and others), each with a label point, an importance (Natural Earth's `scalerank`) and, for areas and rivers, a bounding box so the map can show all of it. About 250 KB.
- Natural Earth gives some lakes a short English name ("Superior", "Nyanza" for Lake Victoria); the build uses the full local name when it starts with "Lake" or contains the short name.
- Small features (Lake Louise in Alberta, the Bow River) are below Natural Earth's 1:10m scale and are not included.

Place-name services not used

- **OpenStreetMap's Nominatim service** forbids search-as-you-type and allows at most 1 request a second (usage policy, checked Nov 9, 2026), so the place list is a file in the repository instead.

Earth map lines, names and countries (Natural Earth 1:50m), added Nov 11, 2026

- From the GeoJSON copies at `github.com/nvkelso/natural-earth-vector/tree/master/geojson`: `ne_50m_coastline`, `ne_50m_admin_0_boundary_lines_land`, `ne_50m_rivers_lake_centerlines`

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(rivers only), `ne_50m_admin_0_countries` (outlines and `LABEL_X`, `LABEL_Y`, `MIN_LABEL`; property names in capitals) and `ne_10m_populated_places_simple` (`scalerank` up to 7). Built by `tools/build_shapes.py` into `data/shapes.json` (1.2 MB, 0.37 MB compressed), `data/countries.json` (0.25 MB compressed) and `data/labels.json` (5,766 names). Public domain.

- The tap card finds the country by point-in-outline; outlines are simplified to about 2 km, so a spot within a few kilometres of a coast or border can be given the neighbouring country or the sea.

Earthquakes (US Geological Survey), checked Nov 11, 2026

- `https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/4.5_week.geojson`: earthquakes of magnitude 4.5 or more in the past 7 days (101 on Nov 11, 2026), GeoJSON; each with `mag`, `place`, `time` (ms), `url`, and coordinates [lon, lat, depth km]. `access-control-allow-origin: *`; no key. Public domain (US Government).

Blue Marble under the Earth map, checked Nov 11, 2026

- `BlueMarble_NextGeneration` (`default` time, `500m`, JPEG) exists in EPSG:4326, EPSG:3413 and EPSG:3031 and loads from a browser. It is drawn under the day's pictures so the gaps between a satellite's passes show the land.

Spacecraft map (`spacecraft/`), added Nov 12, 2026

The catalogue (`spacecraft/catalogue.json`)

- Written by hand: each craft's name, agencies, kind, launch date and rocket, a short description, a timeline and links to the agencies' own pages, with where its position comes from. Facts checked Nov 12, 2026 against the agencies' pages and news. Missions that ended before then are not included (for example MAVEN, lost in 2026; Swift, which re-entered on Nov 4, 2026; Gaia, retired in 2025; INTEGRAL, whose science ended in 2025).
- Military satellites are included only where the orbit is published by CelesTrak (for example WSF-M1); communication and navigation constellations (Starlink, GPS and similar) are left out, decided Nov 11, 2026.

Horizons API (spacecraft map), checked Nov 12, 2026

- `spacecraft/fetch_spacecraft.py`, twice a day (workflow `spacecraft-data.yml`). Same query as above, but `OUT_UNITS=KM-S`, whole kilometres, and the centre chosen by where the craft is: `500@10` (the Sun; every 2–30 days, Parker Solar Probe every 6 hours; from launch or a few years back to a year ahead), `500@399` (Earth; craft near L1 and L2 every 6 hours, 60 days back to 120 ahead), `500@301` (the Moon) and `500@499` (Mars; orbiters every 5 minutes, rovers every 30, half a day back to a day and a half ahead).
- Ids: as listed in the catalogue (`src.horizons`); craft whose id is not fixed there are looked up by name (`src.horizons_name`): the script asks Horizons for the name and takes the id from "Target body name … (−nnn)" or, when several match, the line naming the craft. The run log lists every id found and every craft not found.
- The page adds each craft's position to its parent's: the Sun at the origin, Earth and the planets from JPL's approximate elements, the Moon from Meeus (chapter 47) moved to the J2000 ecliptic.
- First run (Nov 12, 2026): every Horizons craft found except Tianwen-1 and Tianwen-2 (no Horizons target by those names), which stay off the map. Roman resolved to −211, Aditya-L1 to −156, IMAP to −43 and ESCAPADE to −9 (Blue). Perseverance's Horizons trajectory ends on Feb 19, 2026, so the map places it at a fixed point in Jezero crater (18.44° N, 77.45° E) and turns it with Mars (IAU rotation model: pole α_0 = 317.269°, δ_0 = 54.433°, prime meridian W = 176.050° + 350.8919824° a day).

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CelesTrak orbital elements, checked Nov 12, 2026

- `https://celestrak.org/NORAD/elements/gp.php?GROUP=<group>&FORMAT=tle` (no key), the groups `stations`, `science`, `weather`, `resource`, `noaa`, `goes`, `military`, `geodetic`, `active`, read in turn until every craft is found by its exact CelesTrak name (`src.tle`, several spellings allowed); otherwise `?NAME=<name>&FORMAT=tle`.
- A name can also be a catalogue number (`#43435` asks `?CATNR=43435`): CelesTrak does not list TESS by name. First run (Nov 12, 2026): all found except SPHEREx and WSF-M1 (names not matched), which stay off the map until their catalogue numbers are added.
- Usage policy (celestrak.org/usage-policy.php): data are updated about every 2 hours, and the same request repeated within that time is refused (HTTP 403). The job asks twice a day. Credit: "CelesTrak".
- Written to `spacecraft/data/tle.json`: name, the two lines, NORAD catalogue number and the elements' epoch. Positions are worked out in the browser with SGP4 (satellite.js 7.2.0); low-orbit positions drift by kilometres a day as the elements age.

Link check

- On the morning run, every catalogue link is requested (HEAD, then GET); links answering 404 or 410 are written to `spacecraft/data/links.json` and left off the page. Other answers (403, 429, timeouts) keep the link.

Sky Finder (`sky-finder/`), added Nov 12, 2026

- Asteroids: `asteroid-tracker/data/sky.json` (above). Space stations: `spacecraft/data/tle.json`, the ISS and Tiangong; a pass is shown when the station is at least 10° up, in sunlight (outside Earth's shadow, taken as a cylinder) and the Sun at least 6° below the observer's horizon.
- Bright stars: `sky-finder/data/stars.json`, the 921 stars of magnitude 4.5 or brighter from d3-celestial's Hipparcos star file (`stars.6.json`), with the proper names of those of magnitude 2.6 or brighter (d3-celestial's `starnames.json`); BSD 3-Clause, © 2015 Olaf Frohn.
- Moon, planets and Sun: computed in the browser (`shared/sky.js`).
- Nothing is requested from any other service; the camera, location and motion readings stay on the device.

Using NASA material

- NASA data and imagery are generally free to use and not under copyright in the United States, but NASA's insignia (the "meatball" logo) and logotype **may not be used** without permission. These projects use their own emblem.
- Pages say "Data courtesy of NASA" (and ESA where its data is used) and that the project isn't endorsed by NASA or ESA.

ESA (the European Space Agency)

Decided Nov 9, 2026: ESA's data is added alongside NASA's where it helps. Each ESA source gets its own entry here (address, key if any, limits, terms and the exact credit line), checked against ESA's current documentation before anything is built on it. ESA sources in use: the ESA/Webb and ESA/Hubble picture archives (Image Explorer, entry above) and the NEO Coordination Centre's risk lists and close approaches (Asteroid Tracker, entry below).

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Using ESA material, checked Nov 9, 2026

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ESA NEO Coordination Centre (NEOCC): risk lists and close approaches, checked Nov 9, 2026

Used by the Asteroid Tracker (Impact risk tab, the asteroid page, the week's details and the map's "On a risk list" filter), next to NASA's figures, never in place of them. Checked against ESA's "Automated data access" page (`https://neo.ssa.esa.int/computer-access`) and live answers from GitHub's servers.

- **Address:** `https://neo.ssa.esa.int/PSDB-portlet/download?file=<name>`; no key, no account, free.
  - `esa_risk_list`: every object with a non-zero impact chance (Nov 9, 2026: 2,209 objects).
  - `esa_special_risk_list`: objects monitored separately because their impact dates are beyond the usual window (Nov 9, 2026: 101955 Bennu and 29075 (1950 DA)).
  - `esa_recent_close_app` and `esa_upcoming_close_app`: close approaches to Earth within 0.05 au over the past 30 days and the coming year (Nov 9, 2026: 364 together).
- **Format:** plain text. A `Last Update: YYYY-MM-DD HH:MM UTC` line, two heading lines, a format line starting `AAAA`, then one row per object with `|` between columns. The risk list gives designation and name, diameter (m, * when estimated), the most likely impact (date and time, IP max, PS max, TS), speed (km/s), the years of possible impacts, IP cum and PS cum. The close-approach lists give the date only (no time), the miss distance in km, au and LD, diameter, H, brightest magnitude, speed and ESA's CAI index.
- **Designations** have no space: `2008JL3` is "2008 JL3"; survey designations such as `6344PL` are "6344 P-L"; a numbered object is given by its number.
- **Status:** ESA describes this access as experimental and liable to change without notice. The job checks each table's headings and, if they change, keeps the previous file and stops with an error instead of writing wrong figures; the NASA part still runs.
- **Update times:** the risk lists change after each orbit update (the Nov 9, 2026 list was stamped 18:48 UTC); the job reads them twice a day (06:17 and 14:47 UTC).
- **CORS:** no `Access-Control-Allow-Origin` header (recorded in `risk.json`, `cors`), so visitors' browsers never contact ESA; pages read `asteroid-tracker/data/risk.json` and `asteroid-tracker/data/esa_close.json`. No change to the Privacy page is needed.
- **Terms:** the NEOCC portal has no licence of its own and refers to ESA's general terms: educational and informational use is allowed with ESA's credit. No ESA logo is used, and pages say the project isn't endorsed by NASA or ESA.

- **Credit line, shown wherever the figures appear:** "ESA NEO Coordination Centre (©ESA)", with the list's date. If ESA asks for a different form, it is changed in `index.html`, `object.html`, `map.html`, `risk.js`, `object.js`, `app.js`, `CREDITS.md` and the Credits page. Contact for questions: the NEOCC (`neocc@esa.int`).

PART 10

Design Language

The navy, brass and cream look every page follows.

Design guide — "old money", under a night sky

Source: `docs/DESIGN.md` · *last changed 2026-11-12*

NASA and ESA Data Projects share the look of the developer's other projects (the Flight Deck Suite and Hot Mic): navy, brass and cream, serif type and thin brass rules; calm and precise, like a well-kept paper logbook. The one addition is a quiet night sky: a faint field of brass and cream stars behind the home page.

The values live in `shared/theme.css`. When in doubt, copy an existing component rather than inventing a new one.

1. Principles

1. **Calm and readable at a glance.** Big numbers, small labels, short notes. No paragraph where a number will do.
2. **Numbers people can picture.** "3.2 Moon distances", "as wide as two football fields", next to the exact value.
3. **Brass is for meaning, not decoration.** Labels, rules, the main button, selected states, the first chart series.
4. **Plain English.** Explain every science word the first time it appears (sol, Ls, LD), behind an ☒ where possible.
5. **Honest data.** Estimates look like estimates (ranges, "about"), gaps in the data stay visible, and every chart says where its numbers came from.

2. Colour

Role	Token	Value
Background (navy)	`--bg`	`#141B26`
Bars	`--bg2`	`#0F1520`
Card	`--card`	`#1A2333`
Charts, maps, image frames	`--deep`	`#0C1119`
Hairline / stronger border	`--line` / `--line2`	`#2A3446` / `#3A465C`
Brass (labels, accents, main button)	`--brass`	`#C0A062`; bright `#E2C88E`; dim `#7E6A45`
Cream text / secondary / muted	`--ink` / `--ink2` / `--muted`	`#EDE6D3` / `#B4AFA2` / `#98A0AB`
Faint (disabled)	`--faint`	`#5E6775`

Status: good `#8FB996`, problem `#C9675A`, caution `#D1A956`, information `#8AA9C9`.

Page background: navy with a faint brass glow at the top, `radial-gradient(800px 420px at 50% -160px, rgba(192,160,98,.10), transparent 70%), #141B26`.

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3. Type

- **Everything is serif:** `Georgia, "Palatino", "Palatino Linotype", "Book Antiqua", serif`.
- **Numbers:** `lining-nums tabular-nums`, so digits line up and don't jump as they change.
- **Code-like text** (IDs, coordinates, URLs): monospace, 12.5px, muted.

Use	Style
Page title	34px (44px on the home page), weight 400
Eyebrow / section label	11px, uppercase, brass, letter-spacing .32–.42em
Tile label	10.5px, uppercase, brass, letter-spacing .26em
Big number	26px
Body	16px/1.5
Note	12.5–13px, italic, muted
Buttons	14px, uppercase, letter-spacing .2em

4. Spacing and shape

- Page gutter 16px; content at most 1440px wide.
- Card padding 22px; gaps of 22px between cards; tiles joined by 1px hairlines.
- Corners: 3px on buttons, chips and fields; 6px on cards, tiles and frames.
- Cards have a dim-brass top rule. No drop shadows.

5. Components

The classes in `shared/theme.css`:

Component	Class	Notes
Top bar	`.topbar`, `.brand`, `.nav`	Sticky, 64px; the open page brass and underlined
Page head	`.phead`, `.eyebrow`, `h1`, `.lede`	A dim-brass rule under it
Section header	`.shead` with `h2` and `.aside`	Brass caps left, italic muted aside right
Card	`.card`	
Tiles (KPIs)	`.tiles` › `.tile` › `.label`, `.value`, `.note`	Label → big number → italic note
Buttons	`.btn`, `.btn.ghost`	Main = brass gradient with navy text; ghost = dim-brass outline
Filter chips	`.chip[aria-pressed]`	36px; the chosen one brass
Fields	`input`, `select`	44px, deep navy, brass border on focus
Status pill	`.pill.good/.bad/.warn/.info`	Caps, coloured outline
Tag	`.tag`	Italic, outlined (for example "Potentially hazardous")
Tabs	`.tabs` (Asteroid Tracker)	Caps, brass underline on the open tab; scroll sideways on narrow screens
How to read this	`details.howto`	Closed by default; brass caps summary with a › that turns
Notice	`.notice`, `.notice.warn`	A bordered box for messages

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Component	Class	Notes
Chart / map / image frame	`.frame`	Deep navy with a hairline
Footer	`.footer`, `.ftop`, `.copy`, `.fnote`	Decided Nov 12, 2026: one row with the site links (added by `shared/footer.js`) and "© 2026 Malcolm Syme · email", then one short line: "Data courtesy of NASA and ESA. An independent project, not endorsed by NASA or ESA." Nothing else goes in the footer
Site links	`.sitelinks`	About · Legal · Contact · Source code ↗️; every page loads `shared/footer.js`; Legal is marked on the Legal, Privacy, Terms and Credits pages
Data credits	`.srcline`	A page's detailed credits and acknowledgements (sources, ©ESA lines, the GIBS acknowledgement), in small grey italic lines just above the footer, next to the content they cover
Legal page	`.legalgrid`, `.legalcard`	`legal/`: one card each for Privacy, Terms of use, Credits and the MIT licence, with a one-line summary
Reading page	`.prose`, `.summary`, `.updated`, `table.credits`	About, Privacy, Terms and Credits (Privacy, Terms and Credits go back to Legal when opened directly): the full content width (decided Nov 9, 2026), a brass-ruled summary, the date of the last update
Back button	`.backlink#back`	"‹ Back to ..." above the heading, named for the previous page and using the browser's Back; "‹ Home" when the page was opened directly (wired by `shared/footer.js`)

6. Charts

- **Background** `--deep`, gridlines `--line`, axis text 11px muted serif, axis titles in brass caps.
- **Series colours, in order:** brass `#C0A062`, blue `#8AA9C9`, green `#8FB996`, copper `#C9784A`, lilac `#B79AC8`, amber `#D1A956` (`--s1` to `--s6`). Never more than six; group the rest as "Other".
- **Highlight** with brass-bright `#E2C88E`; everything else dims.
- **Hazard / warning** uses `--red` only for things that are actually flagged (for example NASA's "potentially hazardous" flag), never for decoration.
- **Every chart** has a title, units on its axes, a one-line source note under it (italic, muted), and a tooltip with exact values. Colour is never the only way to tell series apart (labels or shapes too).
- **Estimates** (asteroid diameters) are drawn as a range (a bar from min to max), not one point.
- **Agencies:** every source note names the agency the numbers come from (NASA, NASA/JPL, ESA). There are no agency logos or agency colours; NASA and ESA data share the same series colours.

7. Screen sizes

Everything must look right at 375, 390, 430 (phones; 320 must not break), 768, 1024 (tablets), 1280 and 1440+ (computers). Below 760px the nav collapses and buttons go full width.

8. Accessibility

- Text contrast at least 4.5 : 1 (cream and brass on navy both pass).
- Every control reachable and usable from the keyboard, with a visible focus ring.
- Images have descriptions (NASA's own captions where there are).
- Motion is never needed to understand anything, and stops with "reduce motion".

8a. Feel (decided Nov 12, 2026)

- **Feedback:** everything pressable (buttons, chips, tabs, links, cards that open something) changes colour or border over 0.15 s on hover, and buttons and chips move down 1 px while pressed. The rules are in `shared/theme.css`; "reduce motion" turns them off.
- **Tap targets:** at least 40 px tall on phones (44 px for the main buttons and fields).
- **Rows that don't fit:** tabs and chip rows scroll sideways on phones, with a soft fade on the side where there is more, and the selected tab scrolled into view (`shared/footer.js`: `.tabs`, `.popular`, `.gbar .seg`, `[data-hscroll]`). Rows never run off the screen with nothing to show they continue.
- **Labels on maps and diagrams:** a dark outline (canvas `strokeText`, SVG `paint-order: stroke`) so they stay readable over orbits, dots and imagery; fewer names on a phone-sized map.
- **Fields:** a brass glow on focus; date pickers, dropdowns and scrollbars drawn dark (`color-scheme: dark`); a search box's Clear button appears only when there is something to clear; placeholders short enough for a 375 px phone.
- **The header** is solid, with a soft shadow, so nothing shows through it while scrolling.
- **Labels on a row of controls** use the small brass capitals (`.barlbl`), never italics; no two controls on one page share a label.

9. Quality checklist (every change)

1. Look at the page at all the widths above.
2. Check the empty, loading and error states (no results, NASA out of reach, the demo key used up).
3. No console errors.
4. Rows of tiles have equal heights; spacing is even.
5. At 375 px: nothing runs off the screen, tap targets are 40 px or taller, placeholders aren't cut off.
6. Hover, press and focus look right on every control.

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PART 11

Newly added documents

Documents added to the repository that don't have their own place in this record yet. Each is here in full until it is moved into its Part.

Cleaning rules: Perseverance MEDA (version 1.0)

Source: `mars-data/analysis/CLEANING-MEDA.md` · *last changed 2026-11-12*

Rules applied by `run_meda.py` to NASA's Mars 2020 MEDA data (Perseverance's weather station, Jezero Crater). Every chunk of sols has a cleaning report in `data/meda/reports/`.

Source: NASA Planetary Data System, Atmospheres Node, `PDS4/Mars2020/mars2020_meda/data_derived_env/`, the derived level, which NASA recommends for most uses (bundle `readme.txt`). One folder per sol; each sensor in its own CSV table with a header row and a PDS4 XML label.

Situation	Rule	Action
A value is empty or `9999999999`	M1	Treated as missing (NASA: an empty derived value is not reliable for science)
A required column is missing from a table's header	M1	The sol is not processed and is listed in the report
Two rows share a clock time	M2	The first is kept
A row's local time is on another sol	M2	Not used for that sol
Pressure outside 400–1,200 Pa	M3	Removed
Humidity before sol 63	M4	Not used (NASA: not reliable before the sensor's first maintenance)
Relative humidity outside 0–100 %, water vapour outside 0–400 ppm	M4	Removed
A sol has pressure in fewer than 10 local hours	M5	Statistics given but marked "partial"

M1 — Missing values and columns

NASA leaves a derived value empty where it is not reliable, and uses `9999999999` as an invalid value in some columns. Both are missing. The columns used are found by name in each table's header row: pressure `PRESSURE`, `TRANSDUCER`; humidity `LOCAL_RELATIVE_HUMIDITY`, `VOLUME_MIXING_RATIO`; ancillary `ROVER_POSITION_Z`; and `SCLK` and `LMST` in each.

M2 — Time

Local Mean Solar Time (`LMST`, `SSSSSMHH:MM:SS.sss`) gives the sol and the hour. The date, season (Ls) and Mars Year come from the clock time `SCLK`, taken as seconds from noon on January 1, 2000 (sol 1859 gives May 13, 2026, Ls 281°, Mars Year 38), by the same method as REMS (`mars_time.py`).

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M3 — Pressure

The derived pressure (from transducer 1 or 2; the transducer is recorded) within 400–1,200 Pa is kept. NASA's readme notes that the transducer used for the derived level can show jumps; the report lists any day-to-day change of more than 15 Pa between consecutive full sols for review.

M4 — Humidity

Humidity before sol 63 is not used (NASA's readme). Relative humidity is summarised as the day's highest value (from one-minute medians with at least 30 readings), water vapour as the median over local hours 2–6, as for REMS.

M5 — Coverage and daily statistics

MEDA measures about 12 hours a sol, the even hours on even sols and the odd hours on odd sols. An hour counts with at least 60 readings; a sol is **full** with pressure in at least 10 hours. The daily mean is the mean of the hourly means: because the measured hours are spread evenly through the day, every second hour, this follows the daily average closely. Charts of daily averages use full sols only.

Created by **Malcolm Syme** · mcsyme11@gmail.com · part of NASA and ESA Data Projects.

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PART 12

Decisions and Working Rules

Working rules

How every change to the projects is made, as agreed with the developer (from the repository's CLAUDE.md).

- When the developer asks a question ("why…", "what do you think?"), answer it only. Build or change nothing until he explicitly says go.
- **After every change that's finished and checked,** commit and push it to `main` with a short plain-English commit message, and update `docs/ROADMAP.md` (remove what's done), `CHANGELOG.md` and, for milestones, `docs/HISTORY.md`.
- **Public repository: no secrets, ever.** No API key, password or token in any file. Visitors need no key: data that needs one is copied by the GitHub data jobs with repository secrets (`NASA_API_KEY`, `EOL_API_KEY`); web pages call NASA only through `shared/nasa.js` with `DEMO_KEY`; Python run by hand reads `NASA_API_KEY` from `.env` (ignored by Git).
- **Public-ready by default:** no personal details in code, docs or tests (no PC or user names, personal paths, callsigns or family details); use generic wording and made-up test data. **Exception: always keep "Malcolm Syme" and mcsyme11@gmail.com as the developer, copyright and contact credit.**
- **Look:** the Flight Deck Suite's "old money" look (navy, brass, cream, serif type), as set out in `docs/DESIGN.md` and `shared/theme.css`. New pages and big visual changes start as mock-ups the developer approves.
- **Free to run:** static pages on GitHub Pages and free NASA APIs; no paid services.
- **Site pages (since Nov 9, 2026):** every page loads `shared/footer.js` (the links to About, Legal, Contact and the source code; since Nov 12, 2026 the footer is one row of those links with the copyright, then one short credit line, and Privacy, Terms, Credits and the licence are reached through the Legal page, `legal/`). Each page's detailed data credits sit above the footer in `<p class="srcline">` lines, never in the footer. A new data source or library is added to `CREDITS.md` and the Credits page (`credits/index.html`) in the same change; anything new the site stores or sends is added to the Privacy page first.
- **Version tags on style files (since Nov 9, 2026; project style sheets since Nov 12):** pages link `shared/theme.css`, `shared/footer.js`, `asteroid-tracker/tracker.css` and `image-explorer/explorer.css` with a version tag (`?v=YYYYMMDDNN`). After changing any of them, raise the tag in every page (one search-and-replace), so browsers fetch the new copy instead of an old saved one.
- **Feel (since Nov 12, 2026):** everything pressable gives quick feedback (the shared hover and press transitions in `shared/theme.css`); tap targets are at least 40 px tall on phones; rows of tabs or chips that don't fit a phone scroll sideways with a soft edge fade (`shared/footer.js` adds it to `.tabs`, `.popular`, `.gbar .seg` and anything marked `data-hscroll`); labels drawn over maps and diagrams get a dark outline; placeholders fit a 375 px phone.
- **Project home tab (decided Nov 12, 2026):** each project's main page keeps its title and its row of tabs at the top, and the **first tab is "Home"**, the view a visitor lands on: what the project is and why it exists; the details (each data source, which agency it comes from, NASA or ESA, and how often it updates; what you can do; how to use it, with short rules for the words and numbers); a few live headline

numbers; and cards linking to every other tab and related page. The project's numbers start on the second tab. Old links to the other tabs (`#name`) keep working. Same look and shared parts as every page: the theme, the Back button, the data credits in `srcline` lines and the short footer. Each project's Home tab starts as a mock-up the developer approves.

- **ESA data (decided Nov 9, 2026):** ESA's data is added alongside NASA's where it helps; NASA stays the main source, and pages say which agency each number or picture comes from. For every ESA source:
  - check it against ESA's current documentation before building on it (address, format, limits, terms, credit), and record it in `docs/DATA-SOURCES.md` (ESA section) with its exact credit line;
  - show ESA's credit wherever its data or pictures appear, in ESA's own form (for example "ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO", with a link to the licence where one applies), and add the source to `CREDITS.md` and the Credits page in the same change;
  - use only material ESA allows: works released under CC BY-SA 3.0 IGO, or ESA's terms for educational and informational use; never ESA's logo (it needs ESA's written permission);
  - keep it free to run: no paid ESA services; a free account or key follows the same no-secrets rule as NASA's;
  - if visitors' browsers contact an ESA service directly, add it to the Privacy page first.
- **Respect NASA's and ESA's terms:** show "Data courtesy of NASA" (and ESA where its data is used) and that the project isn't endorsed by NASA or ESA; show the GIBS acknowledgement wherever GIBS imagery appears; never use NASA's insignia or logotype, or ESA's logo.
- **Check facts about the data:** confirm endpoints, fields and limits against NASA's current documentation before building on them, and record anything new in `docs/DATA-SOURCES.md`.
- Quality bar: correct, professional, tested. Check every page at the widths in DESIGN.md, and its empty, loading and error states (no results, NASA out of reach, the demo key used up).
- Keep every pending to-do in `docs/ROADMAP.md`; the developer tests at his own pace.
- Explain things with examples and clear, concise, correct notes. When teaching a method, give a clear set of rules saying which formula or step to use in which situation.

Project Record (since Nov 11, 2026)

`docs/NASA and ESA Data Projects - Project Record.pdf` is the whole project as one PDF, built from the repository by `docs/record/` (see its README). **GitHub rebuilds it automatically** (`.github/workflows/project-record.yml`) after every push to main and commits it as "Project Record rebuilt (automatic)". Pushes that change only data files (`asteroid-tracker/data/`, `image-explorer/data/`, `mars-data/data/`: the daily data updates) don't start a rebuild. So: **never build or commit the PDF yourself** (two builds at once clash). Pull and rebase before every push, since the automatic commit lands on main. After pushing, check a few minutes later that the automatic commit arrived (`git fetch` and `git log -1 origin/main`); only if the run failed, fix the cause. **Everything goes in:** every Markdown or text document in the repository is in the record; one not yet listed in `PARTS` (`docs/record/build_record.js`) lands in a "Newly added documents" Part by itself. When you add a document, add it to its proper Part in `PARTS` in the same change. Keep "Where things stand" in `docs/record/front.md` current.

Protected main branch (since Nov 10, 2026)

`main` has a ruleset that blocks force pushes and its deletion. Pull and rebase before pushing; never force-push. To undo a change, make a new commit (`git revert <commit>`).

Writing style

Rule (decided Nov 8, 2026): all documentation is written in a professional tone, as project documentation, never in the tone of an AI assistant talking to a person. No greetings, no "I'll", "we'll", "let's", "happy to", "great question" or exclamation marks; no addressing the developer; no narrating what a chat did. State facts, decisions, steps and limits plainly.

Documents, commit messages and code comments are neutral, professional project documentation, not conversation: "the developer" for Malcolm Syme (his full name and email stay in every credit line), "the user" or "you" in text for people using the pages. Record decisions as dated facts ("Decided Nov 8, 2026: ..."), and describe work by what was done, not by who in a chat did it. `python3 dev/check_wording.py` checks this, and GitHub runs it after every push.

Project notes and decisions

Source: `docs/PROJECT-NOTES.md` · *last changed 2026-11-12*

How NASA and ESA Data Projects is built, the decisions made so far, and how to run and publish it.

Decisions

- **Decided Nov 12, 2026 (project home tab):** every project's main page opens on a "Home" tab, the first in its tab row: the project's mission, its data sources and agencies, what can be done and how, headline numbers and links to every other tab. The title and tabs stay at the top of the page; the numbers start on the second tab. The rule is in CLAUDE.md. A separate landing page above the tabs was considered and not chosen.
- **Decided Nov 12, 2026 (Sky Finder):** an installable phone web app in its own folder, `sky-finder/`, on the same site (not a separate repository). It marks where an asteroid is with a marker and its path: most passing asteroids are far too faint for a phone camera, so the asteroid itself is not shown. The Moon, planets, bright stars and the space stations are drawn to check the aim, and calibration on one of them removes compass error and magnetic declination together. Camera, location and motion readings stay on the device; a safety notice (never the Sun, look where you walk) is shown before first use and the Terms say the same.
- **Decided Nov 12, 2026 (spacecraft map):** a utility, not a fourth project, at `spacecraft/`, replacing the asteroid map on the home page's large panel (the asteroid map is a link under it). It shows every active craft with a public position that the hand-checked catalogue lists; military craft only where CelesTrak publishes the orbit; no communication or navigation constellations (Starlink, GPS). Each craft is drawn relative to its parent (the Sun, Earth, the Moon or Mars), so orbiters stay on their planets at any zoom. Ended missions come off the list.
- **Decided Nov 12, 2026 ("Visible from here"):** a pass counts as visible when the asteroid is at least 10° up while the Sun is at least 12° below the horizon; what is needed follows magnitude (naked eye ≤ 6, binoculars 6–9, small telescope 9–12, large telescope 12–15, too faint above 15). The place is kept in the browser and shared with Sky Finder (same storage key).
- **Decided Nov 11, 2026:** the project has a **Project Record**, `docs/NASA and ESA Data Projects - Project Record.pdf`: the whole project as one PDF (front matter, roadmap, history, change log, each project, data sources, design, decisions and working rules, the site's pages, licence and credits, repository history, glossary), built from the repository by `docs/record/` in the same way as the Flight Deck Suite's and Hot Mic's. GitHub rebuilds it after every push to main and commits it; chats never build it themselves. Pushes that change only data files (the daily data updates, in the `data/` folders) don't start a rebuild, since none of those files are in the record; the Repository History counts them instead of listing them. The record is public, like the repository; its build stops if a personal detail would end up in it.

- **Decided Nov 12, 2026 (Mars Data):** the developer approved the "Two craters" chart: daily average pressure at Gale (Curiosity, solid lines) and Jezero (Perseverance, dashed), one colour per Mars Year, for the years both rovers measured.
- **Decided Nov 11, 2026 (Mars Data):** the developer approved the charts page mock-up (`mars-data/mockup.html`): pressure through the year with a correction for the rover's climb, daily pressure range, humidity at night (preliminary), midday UV, season × time of day, one sol hour by hour, daily highs and lows, and the rover's climb.
- **Decided Nov 11, 2026 (Mars Data):** the whole Curiosity record (sol 1 to the newest released sol) is used from the start, not a single Mars year. It is downloaded and cleaned in chunks (a range of sols at a time), each chunk checked and its cleaning results recorded before the next is processed: rows kept and removed, the reason for each removal, and the gaps found. Cleaning is done slowly and carefully; correctness comes before speed.
- **Decided Nov 11, 2026 (whole site):** the projects are numbered in build order: Project 1 Asteroid Tracker, Project 2 Image Explorer, Project 3 Mars Data (Mars was previously labelled Project 2). The home page lists them in that order.
- **Decided Nov 11, 2026 (Mars Data):** the project follows a fixed set of three missions and updates itself as new data is published: NASA's Curiosity (REMS, Gale Crater) and Perseverance (MEDA, Jezero Crater), and ESA's Mars Express (MaRS temperature and pressure profiles, CC BY-NC 3.0 IGO). InSight is not followed (the mission ended in 2022). Mission data reaches the public archives in batches, months after it is measured, so "up to date" means each new batch is picked up soon after release, not live readings. A GitHub Actions job checks the archives twice a week to start (weekly once stable), downloads only new or replaced files, processes them with the same code as the rest, and commits the summaries only if every check passes; otherwise the previous summaries stay. New data and failed runs open a GitHub issue, which reaches the developer by email. Raw downloads are never committed.
- **Decided Nov 10, 2026:** the repository stays a single public repository (no separate private copy: only the developer and the chats he works with can push; others can only fork, comment or propose changes, which need his approval). `main` is protected by a branch ruleset, "Protect main": deletion of `main` and force pushes to it are blocked; ordinary pushes from the chats and the daily data jobs are unaffected. History on `main` is therefore never rewritten: a mistake is undone with a new commit (`git revert`), not by force-pushing over it.
- **Decided Nov 9, 2026:** ESA (the European Space Agency) data is added **alongside** NASA's across the projects; NASA stays the main source, and ESA is added where it helps. Each project's chat changes its own project; the project-wide files (name, CLAUDE.md, credits, terms, footers) are changed in the general chat. The rules for ESA sources are in CLAUDE.md: check against ESA's documentation first, record in DATA-SOURCES, show ESA's credit wherever its data appears, list it in CREDITS.md and on the Credits page, never use ESA's logo, keep it free, and add any ESA service the browser contacts to the Privacy page first. Every page says the project isn't endorsed by NASA or ESA.
- **Decided Nov 9, 2026:** the project is renamed **NASA and ESA Data Projects**, and the repository **mcsyme11-byte/NASA-ESA-Data-and-API-Projects** (formerly NASA-Data-and-API-Projects). The site moved with it to https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/ (GitHub does not redirect the old site address; it redirects the old repository address for links and Git). Saved browser data keeps its old `nasa-data-projects.` names, so visitors' keys and places are not lost.
- **Decided Nov 9, 2026:** the site has information pages (About, Privacy, Terms of use, Credits, Contact) at `about/`, `privacy/`, `terms/`, `credits/` and `contact/`, linked from a row of site links in every footer (`shared/footer.js`, one file for every page). The footer also links to the licence and the source code on GitHub. **Decided Nov 12, 2026:** the footer was shortened to one row (About, Legal, Contact, Source code ↗ and the copyright) and one short credit line. A Legal page (`legal/`) gathers Privacy, Terms of use, Credits and the MIT licence; their own addresses are unchanged. Each page's detailed data credits (including the GIBS acknowledgement and ESA's credit lines) moved out of the footer to `.srcline` lines

just above it. The Privacy page states what the site actually does (nothing collected; the API key in local storage; the place in a page in session storage; GitHub Pages and NASA's and ESA's services contacted directly; the code libraries are served from the site itself since Nov 10, 2026) and is updated before the site stores or sends anything new.

- **Decided Nov 9, 2026:** the project's own code and documents are released under the **MIT License** (replacing "all rights reserved"); third-party data and libraries keep their own terms, listed in `CREDITS.md`. A new data source or library is added to `CREDITS.md` (and the site's Credits page) in the same change.
- **Decided Nov 8, 2026:** three projects in one repository, each in its own folder (a "stream"): `asteroid-tracker/`, `mars-data/`, `image-explorer/`. Shared code and the look live in `shared/`.
- **Decided Nov 8, 2026:** the repository is **public**, so the pages can be hosted free on GitHub Pages and shown as a portfolio. Rule that follows: no API key, password or personal detail is ever committed.
- **Decided Nov 8, 2026:** the look follows the Flight Deck Suite and Hot Mic ("old money": navy, brass, cream, serif type), see DESIGN.md.
- **Suggested order (Nov 8, 2026):** Asteroid Tracker, then Image Explorer, then Mars Data.
- **Each project is built in its own chat** inside the Claude project "NASA and ESA Data Projects"; a general chat is for questions and exploring.
- **Decided Nov 8, 2026 (Asteroid Tracker):** charts are hand-drawn SVG (no library), the "This year" view (JPL close-approach data) is part of version 1, and the page was built from an approved mock-up.
- **Decided Nov 8, 2026 (Asteroid Tracker):** the JPL close-approach API blocks calls from web pages, so GitHub Actions fetches the year's data once a day and commits `asteroid-tracker/data/close-approaches.json` when it has changed (one automatic commit a day at most, "Asteroid data: daily close-approach update").
- **Decided Nov 8, 2026 (Asteroid Tracker):** after version 1, three additions: a page for each asteroid (every approach, from `neo/{id}`), past years in the year view (2000 to next year) and an orbit diagram, with the orbit diagram to be done with particular care. The developer allowed building these without separate mock-ups.
- **Decided Nov 8, 2026 (Asteroid Tracker):** a solar system map on its own page (`map.html`) with every near-Earth object, the planets and spacecraft; a live preview of it on the home page; a Search tab. Mock-ups approved Nov 8.
- **Decided Nov 9, 2026 (Asteroid Tracker):** a guide to how asteroids are named, with each code decoded on the pages; "Found by the end of" (a discovery timeline) and "Close to Earth in" (a year) on the map; "Found in" in search; comets kept on the map as their own type. Built without a separate mock-up, at the developer's request.
- **Map performance (Nov 9, 2026):** positions are worked out only when the date changes, in typed arrays, starting each object from its previous frame's solution; dots are written straight into a pixel buffer at most 1.5× the screen's resolution; nothing is redrawn unless something changed. Measured with all 43,186 objects: dragging at 60 frames a second on every screen tested; playing at 60 on standard and phone screens and 50–55 on 2× desktop and tablet screens, on a test machine without a graphics processor.
- **Map performance (Nov 9, 2026):** the Kepler solving moved to a background thread (`orbitworker.js`, sharing `kepler.js` with the page, which falls back to solving itself if a worker cannot start); lines are cut to the visible area before drawing. Measured: playing at a median of 60 frames a second at every width, zoomed in or out.
- **Decided Nov 9, 2026 (Asteroid Tracker):** spacecraft paths are smooth curves through every sample; Parker Solar Probe and Juno show only their last year unless selected, because their repeated loops piled up into a tangle.

- **Decided Nov 9, 2026 (Asteroid Tracker):** famous close passes are placed from Horizons positions (`data/famous.json`), not two-body orbits, because Earth's pull bends the path during a close pass. Five passes: Apophis 2029, Duende 2013, 2020 QG, 2019 OK and 2024 YR4 (2032).
- **Decided Nov 9, 2026 (Asteroid Tracker):** "‹ Back" uses the browser's own Back when the previous page is part of the site; the tracker and the map keep their place in the browser tab (`back.js`), and NASA's answers are kept for an hour in the tab (`shared/nasa.js`), so returning costs no requests.
- **Decided Nov 8, 2026 (Asteroid Tracker):** the orbit diagram computes positions itself (two-body Kepler orbits from NASA's elements; planets from JPL's approximate elements), so it works offline from the one NeoWs answer. Its limits are shown on the page; NASA's official close-approach figures are shown beside it.
- **Decided Nov 9, 2026 (Image Explorer):** one page with two tabs, "Earth map" and "Picture library". The map uses OpenLayers in the `epsg4326` projection (the fire layers work only there; see DATA-SOURCES.md), with Arctic and Antarctic views. Eight layers to start: true colour, night lights and sea temperature as the picture underneath; fires, snow, clouds, smoke and dust, and sea ice on top (fires with anything, the coloured layers one at a time). A date picker, a day-by-day slider and Play. Place search with suggestions after every letter (names and coordinates); tapping the map offers NASA's pictures of that place. The picture library is images first, with videos available as a filter. A mock-up comes first (`image-explorer/mockup.html`).
- **Decided Nov 9, 2026 (Image Explorer):** after the first mock-up the developer asked for a globe of the Earth that turns and zooms like the Asteroid Tracker's solar system map, with NASA pictures pinned to places on the Earth and to galaxies and nebulae in the sky around it. The daily Earth map stays. The page now has three tabs: **Earth map · Globe · Picture library**. The globe uses three.js (MIT licence) with NASA's Blue Marble, turned to the current sidereal time inside the real sky. Pins on the Earth: NASA EONET events (live) and, later, astronaut photos of Earth. Pins in the sky: all 110 Messier objects and famous Hubble and Webb targets, each opening NASA's pictures of it. Several rounds of mock-ups are expected.
- **Decided Nov 9, 2026 (Image Explorer):** the developer approved the globe mock-up for publishing and asked to see it with all the galaxies. Version 1 went live the same night with about 1,350 sky objects (every Messier object, every named object and every galaxy to magnitude 12) and the constellations, the real place list (GeoNames) and the live picture library. Further rounds of changes are expected.
- **Decided Nov 9, 2026 (Image Explorer):** the developer asked for five additions at once, built the same night: the four hand-listed sky positions checked, playable videos, natural features in place search (Natural Earth), Arctic and Antarctic views (proj4js with OpenLayers), and astronaut photos on the globe (a daily GitHub Actions job, active once the free API key is stored as the secret `EOL_API_KEY`).
- **Decided Nov 9, 2026 (Image Explorer):** the picture library puts pictures taken in space or of space first and moves launches, people, events and artists' illustrations under "Behind the scenes"; results appear as you type from 3 letters; suggestions and corrections allow for small spelling mistakes; mission buttons search several phrasings; a picture naming a sky object links to it on the globe. Whether to add ESA's Webb and Hubble archives is left to the developer.
- **Decided Nov 9, 2026 (Image Explorer):** the developer approved the mock-up of ESA's pictures in the library. Each search also asks ESA/Webb and ESA/Hubble's archives live (their pages allow browsers to read them); results are merged and sorted by the same score (ESA's archive pictures start 2 higher, as telescope science pictures), taking turns between libraries. A tag on each picture names its library; "From" picks NASA, ESA or both. The James Webb and Hubble buttons list ESA's whole archive for that telescope; the other mission buttons ask NASA only. ESA's archives have no videos; the NASA centre filter applies to NASA's pictures; the years apply to ESA's pictures by the year in their ID. An ESA picture's credit line, caption and details are read from its page when it opens.

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- **Decided Nov 9, 2026 (Asteroid Tracker):** ESA's NEO Coordination Centre is added beside NASA's data. A new "Impact risk" tab lists every object on ESA's risk list (including its special risk list) or NASA/JPL's Sentry list, side by side, with the Torino and Palermo scales explained as rules; an asteroid's page and the week's details show ESA's and NASA's impact chances and ESA's close-approach distance; the map gains an "On a risk list" filter and on-screen arrow buttons for panning on phones. Neither centre's server allows calls from web pages, so the data job writes `data/risk.json` and `data/esa_close.json`. NASA stays the main source; every figure says which agency it comes from. The developer waived the mock-up step for this addition ("just do as much as you can").
- **Decided Nov 12, 2026 (Asteroid Tracker, keys):** the site must work for every visitor without a key. NeoWs (the "This week" list and each asteroid's page) needs an api.nasa.gov key, and a key placed in a public page could be read and used up by anyone, so the daily data job copies the NeoWs data most visitors see (the current week, two weeks either side, and their asteroids) with the developer's key from the repository secret `NASA_API_KEY`. Pages read the copy first; other weeks and asteroids are asked of NASA directly with the visitor's own key or the demo key (limit counted per connection). The Key settings box was removed the same day: with the copy it served no purpose, and a key saved by the earlier version is deleted from the visitor's browser.
- **Decided Nov 10, 2026 (Asteroid Tracker):** asteroids on an impact-risk list are drawn in rose on the map, a colour no orbit type uses, at 2 × 2 pixels; diamond outlines are added only when zoomed in, because at full view about 2,500 outlines covered the map. Share links carry the view in plain, readable parameters (`view=1` marks a shared link) and are removed from the address once applied, so Back and reload return to where the user was; nothing is stored or sent, so the Privacy page is unchanged.
- **Decided Nov 9, 2026 (licence position):** ESA's NEOCC portal has no licence of its own and refers to ESA's general terms, which allow educational and informational use with ESA's credit. The figures are shown with the credit "ESA NEO Coordination Centre (©ESA)" and no ESA logo; if a question arises, the NEOCC is asked by email (`neocc@esa.int`).
- **Decided Nov 11, 2026 (Image Explorer):** the developer asked for fixes to the map's tap card ("Open water" in the Sahara), more detail on the Earth map, more points on the globe, and astronaut photos without waiting for the key. Done: country, feature and distance on the tap card; Blue Marble under the day's pictures; coastlines, borders, rivers, names and a grid (Natural Earth); earthquakes (USGS), volcanoes, launch sites and natural wonders on the globe; the public Latest ISS Imagery set for astronaut photos.
- **Not possible (Nov 9, 2026):** seeing individual houses. NASA's sharpest free imagery is about 30 m a pixel (Landsat) and the daily layers 250 m; houses need commercial imagery, which is neither NASA's nor free.

How it's built (starting plan; each project's chat confirms it)

Part	Built with	Why
Web pages	Plain HTML, CSS and JavaScript modules, no build step	Free hosting on GitHub Pages; easy to read and learn from
Charts	Hand-drawn SVG, styled to DESIGN.md (Asteroid Tracker); chosen per project	No library to load; matches the look exactly
Globe (Image Explorer)	three.js 0.187 (MIT licence, from a CDN), WebGL	A textured Earth, 5,000 stars and markers drawn smoothly; the markers and names are drawn on a 2D layer above
Maps (Image Explorer)	OpenLayers 10 (BSD licence) and proj4js 2.22 (MIT) from a CDN; `epsg4326`, `epsg3413` (Arctic), `epsg3031` (Antarctic)	Handles GIBS's tile grids, vector fire tiles and polar views; Leaflet does not without add-ons

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Part	Built with	Why
Sky positions	`shared/sky.js`, `shared/passes.js`, `shared/sats.js` (satellite.js 7.2.0, MIT, in `shared/vendor/`)	Sun, Moon, planets, stars, asteroids and satellites for any place and time, in the browser; checked against Astropy
Spacecraft map	`spacecraft/spacemap.js`, a subclass of the asteroid map's `SolarMap`	The same controls and projection; positions relative to each craft's parent
Mars analysis	Python 3 (pandas, matplotlib) in `mars-data/analysis/`	Cleaning and statistics are easiest in Python; it writes small summary files the web page reads

API keys

- **Web pages:** `shared/nasa.js` calls api.nasa.gov with NASA's `DEMO_KEY` (limit counted per visitor's connection). Since Nov 12, 2026 the pages have no key box: the data job copies what needs a key, with the developer's key kept as a repository secret. A key placed in a public page's code would be visible to anyone, which is why it isn't done.
- **Python:** reads `NASA_API_KEY` from `.env` (copy `.env.example`). `.env` is in `.gitignore`.
- **Check on GitHub:** `dev/check_wording.py` also fails if anything that looks like a NASA key is committed.

Running it locally

```
python3 -m http.server 8000      # from the repository's top folder
```

Then open `http://localhost:8000`. (Opening the HTML files directly doesn't work, because browsers block JavaScript modules on `file://`.)

Publishing (GitHub Pages)

Live since Nov 8, 2026: https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/

One-time setting, done on github.com: the repository's **Settings › Pages › Build and deployment**, Source "Deploy from a branch", Branch **main**, folder **/ (root)**, Save. The site then appears at `https://mcsyme11-byte.github.io/NASA-ESA-Data-and-API-Projects/` a minute or two after each push.

Checks on GitHub

`.github/workflows/checks.yml` runs `dev/check_wording.py` after every push to main:

- no first name on its own (the credit "Malcolm Syme" is fine), no chat narrating its own work;
- no personal paths or user names;
- no NASA API key.

Run it by hand: `python3 dev/check_wording.py`.

`.github/workflows/astronaut-photos.yml` runs once a day (07:41 UTC) and on demand: it fetches the newest astronaut photos of Earth into `image-explorer/data/astronaut.json` if the secret `EOL_API_KEY` is set, and commits the file if it changed.

`.github/workflows/asteroid-data.yml` runs twice a day (06:17 UTC, and 14:47 UTC as a backup) and on demand (Actions › Asteroid data › Run workflow, with "all" to refresh every year): it refreshes the Asteroid Tracker's close-approach files, every near-Earth object's orbit, the spacecraft positions and the coming passes

(`soon.json`) from JPL, and commits them if they changed. If other work has been pushed meanwhile, it rebases onto it and retries the push (up to five times).

Decided Nov 9, 2026 (whole site): "‹ Back" buttons (`<a class="backlink" id="back">`) are named and wired by `shared/footer.js` alone. Page code must not add its own click handler to them, or one press goes back two pages; the Asteroid Tracker's `back.js` only un-hides the map's button.

Decided Nov 9, 2026 (Asteroid Tracker): the map pans (right, middle or Shift drag, two fingers, or the ✥ button), and the wheel and a pinch zoom towards the pointer; a plain drag still turns the view.

Decided Nov 9, 2026 (Asteroid Tracker): the orbit diagram marks a close approach at both bodies' positions at that moment, and Play stops there; its normal speed is 5 days a second (60 was too fast to follow).

Decided Nov 9, 2026 (Asteroid Tracker): a backup daily run, after GitHub did not start the 06:17 run that day. The size comparisons (`compare.js`) replace the earlier "school buses / football fields" phrase everywhere, with one rule set; the famous passes take their times and distances from NASA/JPL close-approach data in `famous.json` rather than figures typed into the page; Chelyabinsk (2013) is not a famous pass, because no orbit exists for it (it was not seen before it entered the atmosphere).

Writing style

Decided Nov 8, 2026: all documentation is written in a professional tone, as project documentation, never in the tone of an AI assistant talking to a person (no greetings, first-person phrases, sign-offs or exclamation marks; no addressing the developer; no narrating what a chat did). `dev/check_wording.py` checks for the common phrases.

Documents, commit messages and code comments are neutral, professional project documentation. The developer is "the developer" (his name and email stay in every credit line); people using the pages are "the user" or "you" in user-facing text. Decisions are recorded as dated facts ("Decided Nov 8, 2026: …").

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PART 13

The Site's Pages

The About, Privacy, Terms of use, Credits and Contact pages, as they appear on the site.

About

NASA and ESA Data Projects is a free website for learning about space and Earth with real data, straight from NASA and ESA (the European Space Agency). It turns their open data and APIs into things you can explore: tables, maps, charts, a globe and pictures.

The projects

- **Asteroid Tracker** (live): the near-Earth objects passing Earth this week and every year since 2000, a page and orbit diagram for each, a search, and a map of all of them with the planets and spacecraft, with impact-risk lists from both NASA/JPL and ESA side by side. Distances are also given in Moon distances, so they are easy to picture.
- **Image Explorer** (live): the whole Earth from orbit, day by day, from NASA's satellite imagery; a globe with natural events, astronaut photos and the sky; and the picture libraries of NASA and of ESA/Webb and ESA/Hubble, searchable by subject, mission and date.
- **Mars Data** (planned): weather measured on the surface of Mars, cleaned and organised by time and place, with statistics and charts that show the patterns of a Martian year.

How it works

- **Real data.** The numbers and pictures come from NASA and NASA/JPL services, with ESA's data added where it helps, either live as the page loads or from data files refreshed automatically every day. Every page says where its data comes from.
- **Runs in your browser.** There is no account and nothing to install. The site is plain HTML, CSS and JavaScript, hosted on GitHub Pages.
- **Open to learn from.** All of the code is public on GitHub under the MIT License, so anyone can read how it works and build on it.

Who made it

NASA and ESA Data Projects is designed and created by **Malcolm Syme**, and built with Claude. It is an independent project and is not endorsed by NASA or ESA.

More

- Credits: every data source and library the site uses.
- Source code on GitHub and the MIT License.
- Contact: questions, ideas and bug reports.

Privacy

Last updated November 12, 2026

NASA and ESA Data Projects collects nothing about you. There are no accounts, no cookies, no analytics, no advertising and no tracking. Nothing you do on the site is sent to the developer.

What is kept in your browser

A few things are saved in your own browser, on your own device, to make the site work. They are never sent to the developer.

- **No key.** The site does not ask for or keep a NASA API key. A key saved by an earlier version of the Asteroid Tracker is deleted from your browser the next time you open it.
- **Where you were,** such as the open tab, your filters and your scroll position, so the "Back" buttons return you to the same place. This is kept in your browser's session storage and is deleted when you close the tab.
- **The place you choose** in the Asteroid Tracker's "Visible from here" tab or in Sky Finder (a town's name and its latitude and longitude, or your device's location rounded to about 1 km), so it is there next time. It is kept in your browser's local storage and used only on your device to work out what is in your sky; "Forget this place" deletes it.
- **Sky Finder's settings:** that you have read its safety notice, whether red light is on, the camera view width, and its compass calibration (two angles, kept for 12 hours). Local storage, on your device only.
- **Sky Finder's offline copy:** Sky Finder saves its own pages and data files in your browser's cache (a service worker) so it still works with a weak connection outdoors. Nothing about you is in it; your browser's "clear site data" removes it.

Sky Finder: camera, location and motion sensors

Sky Finder asks for three permissions, each only when you use the feature that needs it, and you can refuse any of them:

- **Camera:** the picture is shown on your screen behind the markers and nothing else. It is never recorded, saved, analysed or sent anywhere. The camera is turned off when you leave the camera view.
- **Location:** used on your device to work out which way to look. It is never sent anywhere; the data files Sky Finder reads are the same for every visitor. You can type a town instead.
- **Motion and orientation:** used on your device, while the camera view is open, to know where the phone points. Readings are not kept or sent.

Browsers show their own prompt for each permission, and you can withdraw it at any time in the browser's site settings.

Services your browser contacts

The pages load data and code straight from the services below. Like any website, each one receives the basic technical information every browser sends (your IP address, browser type and the address requested), under its own privacy policy.

- **GitHub Pages,** which hosts the site. GitHub logs each visitor's IP address for security (GitHub Pages data collection, GitHub Privacy Statement).
- **Pages that read only this site's own files:** the spacecraft map, the "Visible from here" tab and Sky Finder contact no other service; their data (from NASA/JPL Horizons and CelesTrak) is copied to the site twice a day.
- **NASA and NASA/JPL services** that supply the data and imagery: api.nasa.gov, GIBS, the NASA Image and Video Library, EONET and JPL's Solar System Dynamics (NASA Web Privacy Policy).

- **The US Geological Survey** (earthquake.usgs.gov): the Image Explorer's globe reads the past week's earthquakes from it (USGS privacy policies).
- **ESA/Webb and ESA/Hubble** (esawebb.org, esahubble.org and their picture servers cdn.esawebb.org and cdn.esahubble.org): the Image Explorer's Picture library and globe search their picture archives and show their pictures, under those sites' own privacy policies (ESA/Hubble privacy policy; ESA/Webb's is linked from its site).

Links to other sites

Some pages link to NASA, ESA (including its NEO Coordination Centre), the Minor Planet Center, other space agencies' mission pages, CelesTrak and other sites. Those sites have their own privacy policies.

Changes

If the site ever starts storing or sharing anything new, this page is updated first, with a new date at the top.

Contact

Questions about privacy: mcsyme11@gmail.com.

Terms of use

Last updated November 12, 2026

NASA and ESA Data Projects is free and educational. It is offered as it is, without any guarantee, and it is not a source for safety decisions. By using the site, you agree to these terms.

Using the site

The site is free to use for learning, teaching and personal interest. Please use it fairly: it reads data from NASA's public services, so automated or heavy use should go to NASA's APIs directly, with your own key.

Not for safety decisions

The data comes from NASA, NASA/JPL and ESA, but it can be estimated, delayed, simplified or wrong, and the site can make mistakes in showing it. Asteroid sizes, for example, are often estimated from brightness. Nothing on the site is an official warning or forecast. For official information about asteroid close approaches and impact risk, see NASA/JPL's Center for Near Earth Object Studies.

Sky Finder and looking at the sky

Sky Finder and the "Visible from here" tab help you find things in the night sky. Using them, you agree that:

- **You never point your eyes, binoculars, a telescope or a camera at the Sun.** Looking at the Sun through any optical aid causes instant and permanent eye damage. Sky Finder marks the Sun in red so it can be avoided; it does not make looking at it safe.
- You look after your own safety outdoors at night: watch where you walk, stay away from roads, water, edges and private land, and never use Sky Finder while driving or cycling.
- Directions, times and brightness are estimates for learning and enjoyment. A phone's compass and motion sensors can be several degrees off, positions are worked out from published data that can be out of date, and an object may not be visible at all from your place, weather and sky.

The spacecraft map

Positions on the spacecraft map are worked out from NASA/JPL Horizons and from CelesTrak's published orbital elements; satellite positions grow less accurate as the elements age, and are not for navigation or any operational use. Mission descriptions are summaries; each agency's own pages, linked from each craft, are the authority. Names of missions and agencies belong to their owners; the map uses no agency's logo or insignia.

NASA and ESA

This is an independent project. It is not run by, connected with or endorsed by NASA or ESA (the European Space Agency), and it does not use NASA's insignia or logotype or ESA's logo. NASA material is generally not under copyright in the United States, but some pictures in the NASA Image and Video Library include work by others and carry their own credit or rights. Check an item's credit before reusing it, and see NASA's media usage guidelines.

ESA material is shown with ESA's credit. Some ESA images and videos are released under the CC BY-SA 3.0 IGO licence: they may be reused with the same credit, and changed versions must be shared under the same licence. Other ESA material may be used for educational and informational purposes; see ESA's terms and conditions. Pictures from ESA/Webb and ESA/Hubble are released under CC BY 4.0, each with its own credit line, which is shown with the picture and must be kept when reusing it.

The code and other material

The site's own code and documents are released under the MIT License. Data, imagery and code libraries from others keep their own terms, listed on the Credits page.

No warranty

The site is provided "as is", without warranty of any kind. To the extent the law allows, the developer is not responsible for any loss or damage arising from using the site or relying on its information.

Availability

The site depends on NASA's and ESA's services, CelesTrak and GitHub. Parts of it may change, stop working for a while or be removed, without notice.

Changes

These terms may be updated. The date at the top shows the latest version.

Contact

Questions about these terms: mcsyme11@gmail.com.

Credits

Last updated November 12, 2026

NASA and ESA Data Projects is created by **Malcolm Syme**. It is built on the work of NASA, NASA/JPL, ESA (the European Space Agency) and the people and projects below. Each keeps its own terms.

Data and imagery

Source	Used for	Terms
NASA NeoWs (api.nasa.gov)	Asteroid Tracker: close approaches, sizes, the hazard flag	NASA data
NASA/JPL Solar System Dynamics: Close-Approach Data, Small-Body Database, Horizons, approximate planetary elements, Sentry	Asteroid Tracker: approaches, orbits, spacecraft and planet positions, NASA's impact-risk list, the positions behind "Visible from here"; the spacecraft map's deep-space, Lagrange-point, Moon and Mars positions; Sky Finder's asteroid positions	NASA/JPL data
CelesTrak (celestrak.org), general perturbations orbital elements	Spacecraft map: satellites in Earth orbit; Sky Finder: the space stations' passes	Published orbital data, credited "CelesTrak"
NASA GIBS (Global Imagery Browse Services), including Blue Marble	Image Explorer: the Earth map, polar views, the globe's surface	NASA imagery; acknowledgement below
NASA Image and Video Library	Image Explorer: pictures and videos	NASA material; some items carry their own credit, shown with each item
NASA EONET (Earth Observatory Natural Event Tracker)	Image Explorer: natural events on the globe	NASA data
Gateway to Astronaut Photography of Earth, Earth Science and Remote Sensing Unit, NASA Johnson Space Center (eol.jsc.nasa.gov)	Image Explorer: astronaut photos	NASA material
d3-celestial, © 2015 Olaf Frohn	Image Explorer: stars, galaxies, nebulae, clusters, constellations; Sky Finder: bright stars and their names (from the Hipparcos catalogue)	BSD 3-Clause (licence)
GeoNames (geonames.org), through the all-the-cities package (MIT) and GeoNames' own downloads	Image Explorer, the Asteroid Tracker's "Visible from here" and Sky Finder: place search (towns, cities, capitals; smaller natural features with an English Wikipedia article; the globe's volcanoes and natural wonders)	CC BY 4.0
NASA Planetary Data System, Atmospheres Node: Perseverance's Mars Environmental Dynamics Analyzer (MEDA)	Mars Data: pressure, humidity and the rover's height at Jezero Crater	NASA data; credit "NASA/JPL-Caltech and the MEDA team (Centro de Astrobiología), NASA Planetary Data System"
NASA Planetary Data System, Atmospheres Node: Curiosity's Rover Environmental Monitoring Station (REMS)	Mars Data: pressure, air and ground temperature, humidity, ultraviolet and the rover's height	NASA data; credit "NASA/JPL-Caltech and the REMS team (Centro de Astrobiología), NASA Planetary Data System"
Natural Earth (naturalearthdata.com)	Image Explorer: mountains, lakes and rivers; the Earth map's coastlines, borders, rivers and names; the country of a tapped spot	Public domain
US Geological Survey earthquake feed (earthquake.usgs.gov)	Image Explorer: the globe's earthquakes of the past week	Public domain (US Government)

We acknowledge the use of imagery provided by services from NASA's Global Imagery Browse Services (GIBS), part of NASA's Earth Science Data and Information System (ESDIS).

ESA (the European Space Agency)

ESA data and imagery are added alongside NASA's where they help. Each ESA source is listed here, with its exact credit line, from the day it appears on the site; ESA's credit is also shown wherever its data or pictures appear. Some ESA material is released under the CC BY-SA 3.0 IGO licence (credit in ESA's own form, for example "ESA/DLR/FU Berlin, CC BY-SA 3.0 IGO"); other ESA material follows ESA's terms and conditions. ESA's logo is not used.

Source	Used for	Terms
ESA NEO Coordination Centre (NEOCC, neo.ssa.esa.int): risk list, special risk list, close approaches	Asteroid Tracker: ESA's impact-risk list beside NASA's, and ESA's close-approach distances	ESA's terms for educational and informational use; credit "ESA NEO Coordination Centre (©ESA)"
ESA/Webb picture and video archives (esawebb.org)	Image Explorer: Picture library and the globe's pictures, beside NASA's	CC BY 4.0 (ESA/Webb terms); each picture's own credit line, shown in full with it, for example "NASA, ESA, CSA, STScI. Image Processing: J. Depasquale (STScI)"; captions credited to ESA/Webb
ESA/Hubble picture and video archives (esahubble.org)	Image Explorer: Picture library and the globe's pictures, beside NASA's	CC BY 4.0 (ESA/Hubble terms); each picture's own credit line, shown in full with it, for example "NASA, ESA, and STScI"; captions credited to ESA/Hubble

Code libraries

Copies are kept in this repository (image-explorer/vendor/ and shared/vendor/), each with its licence file, so the pages load them from this site.

Source	Used for	Terms
OpenLayers 10.11.0	Image Explorer: the Earth map	BSD 2-Clause
proj4js 2.22.0	Image Explorer: polar map projections	MIT
three.js 0.187.0	Image Explorer: the globe	MIT
satellite.js 7.2.0, © 2013 Shashwat Kandadai (github.com/shashwatak/satellite-js)	Spacecraft map and Sky Finder: satellite positions from orbital elements (SGP4)	MIT (licence)
pandas (analysis tool, run on GitHub's servers; not sent to visitors)	Mars Data: cleaning and statistics	BSD 3-Clause

This project's licence

The site's own code and documents are released under the MIT License: anyone may use, copy and change them, as long as the copyright notice (© 2026 Malcolm Syme) stays with them. The same list of credits is kept in the repository as CREDITS.md.

NASA and ESA

Data courtesy of NASA and ESA. This is an independent project and isn't endorsed by NASA or ESA. NASA's insignia, logotype and seal and ESA's logo are not used. The spacecraft map names missions of other agencies (JAXA, ISRO, CNSA, CMSA, CSA, KASA, the UAE Space Agency and others) and links to their pages; no agency's logo is used and none endorses the project.

Contact

Questions about the data, ideas for something new, or something that isn't working? Send a message.

mcsyme11@gmail.com

Write an email

Start with a subject

opens your email app, subject filled in

- Bug report · Something isn't working · Say which page, what you did, and your device and browser.
- The data · A question about a number · An asteroid, a picture, a layer: name it and what looks odd.
- Ideas · Something new to explore · A feature, a dataset, or a way to make it clearer.

Your message goes to the developer's inbox only. The site itself stores nothing (see Privacy).

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PART 14

Licence and Credits

The project's licence, and every outside data source and library with its terms.

Licence

Source: `LICENSE.md` · *last changed 2026-11-09*

MIT License

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This licence covers the code and documents written for this project. Data, imagery, libraries and other material from third parties (NASA, JPL and ESA data and imagery, the d3-celestial star catalogue, GeoNames place names, Natural Earth, OpenLayers, proj4js, three.js) keep their own terms, listed in CREDITS.md. NASA's and ESA's names, NASA's insignia and logotype and ESA's logo are not covered by this licence, and the project is not endorsed by NASA or ESA.

Credits

Source: `CREDITS.md` · *last changed 2026-11-12*

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The same list appears on the site's Credits page. When a new data source or library is added, it is added here and there in the same change.

Data and imagery

Source	Used for	Terms
NASA NeoWs (api.nasa.gov)	Asteroid Tracker: close approaches, sizes, the hazard flag	NASA data; generally not under copyright in the United States
NASA/JPL Solar System Dynamics: Close-Approach Data, Small-Body Database, Horizons, approximate planetary elements, Sentry	Asteroid Tracker: approaches, orbits, spacecraft and planet positions, NASA's impact-risk list, the positions behind "Visible from here"; spacecraft map: deep-space, Lagrange-point, Moon and Mars positions; Sky Finder: asteroid positions	NASA/JPL data
CelesTrak (celestrak.org), general perturbations orbital elements	Spacecraft map: satellites in Earth orbit; Sky Finder: the space stations' passes	Published orbital data; credit "CelesTrak"
NASA GIBS (Global Imagery Browse Services), including Blue Marble	Image Explorer: the Earth map, the polar views, the globe's surface	NASA imagery; acknowledgement required (below)
NASA Image and Video Library	Image Explorer: the picture library	NASA material; some items carry their own credit or rights, shown with each item
NASA EONET (Earth Observatory Natural Event Tracker)	Image Explorer: events on the globe	NASA data
Gateway to Astronaut Photography of Earth, Earth Science and Remote Sensing Unit, NASA Johnson Space Center	Image Explorer: astronaut photos on the globe	NASA material; credit the unit
d3-celestial, © 2015 Olaf Frohn	Image Explorer: stars, galaxies, nebulae, clusters and constellations; Sky Finder: bright stars and their names (Hipparcos catalogue), `sky-finder/data/stars.json`	BSD 3-Clause, kept as `image-explorer/data/LICENSE-d3-celestial.txt`
GeoNames (geonames.org), through the `all-the-cities` package (MIT) and GeoNames' own downloads	Image Explorer, the Asteroid Tracker's "Visible from here" and Sky Finder: place search (towns, cities, capitals; smaller natural features with an English Wikipedia article; the globe's volcanoes and natural wonders)	GeoNames: Creative Commons Attribution 4.0
Natural Earth (naturalearthdata.com)	Image Explorer: mountains, lakes and rivers in place search; the Earth map's coastlines, borders, rivers and names; the country of a tapped spot	Public domain
US Geological Survey earthquake feed (earthquake.usgs.gov)	Image Explorer: the globe's earthquakes of the past week	Public domain (US Government)
NASA Planetary Data System, Atmospheres Node: Perseverance's Mars Environmental Dynamics Analyzer (MEDA), Mars 2020 MEDA derived data (DOI 10.17189/1522849)	Mars Data: pressure, humidity and the rover's height at Jezero Crater (the two-craters comparison)	NASA data; credit "NASA/JPL-Caltech and the MEDA team (Centro de Astrobiología), NASA Planetary Data System"

Source	Used for	Terms
NASA Planetary Data System, Atmospheres Node: Curiosity's Rover Environmental Monitoring Station (REMS), data sets MSL-M-REMS-5-MODRDR-V1.0, MSL-M-REMS-6-ADR-V1.0 and MSL-M-REMS-5-UVRDR-V1.0	Mars Data: pressure, air and ground temperature, humidity, ultraviolet and the rover's height, sol by sol	NASA data; credit "NASA/JPL-Caltech and the REMS team (Centro de Astrobiología), NASA Planetary Data System"; citations in `docs/DATA-SOURCES.md`

GIBS acknowledgement (NASA asks every user of GIBS to show it): We acknowledge the use of imagery provided by services from NASA's Global Imagery Browse Services (GIBS), part of NASA's Earth Science Data and Information System (ESDIS).

ESA (the European Space Agency)

Decided Nov 9, 2026: ESA data and imagery are added alongside NASA's where they help. Each ESA source is listed here, with its exact credit line, in the same change that adds it, and ESA's credit is shown wherever its data appears.

Source	Used for	Terms
ESA/Webb picture and video archives (esawebb.org)	Image Explorer: Picture library and the globe's pictures, beside NASA's	CC BY 4.0 (esawebb.org/copyright); each picture's own credit line, shown in full with it, for example "NASA, ESA, CSA, STScI. Image Processing: J. Depasquale (STScI)"; captions credited to ESA/Webb
ESA/Hubble picture and video archives (esahubble.org)	Image Explorer: Picture library and the globe's pictures, beside NASA's	CC BY 4.0 (esahubble.org/copyright); each picture's own credit line, shown in full with it, for example "NASA, ESA, and STScI"; captions credited to ESA/Hubble
ESA NEO Coordination Centre (NEOCC, neo.ssa.esa.int): risk list, special risk list, close approaches	Asteroid Tracker: ESA's impact-risk list beside NASA's, and ESA's close-approach distances	ESA's terms for educational and informational use; credit "ESA NEO Coordination Centre (©ESA)"

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Code libraries (copies kept in the repository, `image-explorer/vendor/` and `shared/vendor/`, each with its licence file)

Library	Version	Used for	Licence
OpenLayers (`ol`)	10.11.0	Image Explorer: the Earth map	BSD 2-Clause
proj4js (`proj4`)	2.22.0	Image Explorer: the polar map projections	MIT
three.js (`three`)	0.187.0	Image Explorer: the globe	MIT

Library	Version	Used for	Licence
satellite.js, © 2013 Shashwat Kandadai	7.2.0	Spacecraft map and Sky Finder: satellite positions from orbital elements (SGP4); bundled as an ES module, `shared/vendor/satellite.min.js`, licence `shared/vendor/LICENSE-satellite.js.md`	MIT

Analysis tools (run on GitHub's servers to prepare data; not sent to visitors): pandas (BSD 3-Clause), for the Mars Data cleaning and statistics in `mars-data/analysis/`.

NASA and ESA

Data courtesy of NASA and ESA. This is an independent project and is not endorsed by NASA or ESA. ESA's logo is not used. The spacecraft map names missions of other agencies (JAXA, ISRO, CNSA, CMSA, CSA, KASA, the UAE Space Agency and others) and links to their pages; no agency's logo is used and none endorses the project. NASA's insignia ("meatball"), logotype ("worm") and seal are not used.

NASA and ESA Data Projects · Created by **Malcolm Syme** · mcsyme11@gmail.com

PART 15

Repository History

Every change saved to the repository, newest first, except the daily data updates (counted, not listed).

142 changes since the repository was created, the latest on 2026-11-12 (f98086c). Not listed: 20 daily data updates (asteroid data, astronaut photos and other automatic data refreshes) and the record's own rebuilds.

Date	Commit	Change
2026-11-12	`f98086c`	Rule: each project's main page opens on a Home tab (mission, data and agencies, how to use, links to every tab)
2026-11-12	`b8e908c`	Image Explorer home mock-up: no back button, as on the Mars Data home page
2026-11-12	`8a9bf70`	Mars Data: home page mock-up for approval
2026-11-12	`acbcd3e`	Image Explorer: mock-up of the project's own home page, for approval
2026-11-12	`ffa0c2d`	Mars Data: no back button on the project's home page
2026-11-12	`30a4ca2`	Asteroid Tracker tabs wrap at every width instead of scrolling off the edge
2026-11-12	`f879fa6`	Mars Data: the approved Two craters chart (Gale and Jezero) on the page
2026-11-12	`7011690`	Shorter footer on every page and a Legal page for Privacy, Terms, Credits and the licence; detailed data credits moved above the footer
2026-11-12	`dccb2d2`	Mars Data: Perseverance MEDA data (sols 0-1859) and a two-craters mock-up for approval
2026-11-12	`0d29781`	Map page: remove the Back button; the tracker's tabs are right below the title
2026-11-12	`abb918f`	Layout fixes: spacecraft filters, card close button, Sky Finder place panel, tracker tabs
2026-11-12	`6f594a2`	Mars Data: a distinct colour for each of the eight Mars Years
2026-11-12	`b89a3d2`	Mars Data: the whole Curiosity record on the page (sols 1-4843); evening glitch set aside on 26 sols
2026-11-12	`dfff9d2`	Mars Data: sols 3645-4257 added; evening air-temperature glitch set aside on 12 sols
2026-11-12	`828f316`	Spacecraft data: positions and orbital elements update
2026-11-12	`1c699b5`	Mars Data: sols 3069-3644 added; corrected UV complete
2026-11-12	`197e091`	Mars Data: sols 2483-3068 added; sol 2844 air temperature set aside for review
2026-11-12	`44296a5`	Mars Data: sols 1870-2482 added to the page
2026-11-12	`c8b4649`	Mars Data: sols 1294-1869 added to the page
2026-11-12	`91042b5`	Visible from here: distances under 10 Moon distances to one decimal; spacecraft map fits far Mars and Moon orbits
2026-11-12	`ff70097`	Spacecraft data: positions and orbital elements update
2026-11-12	`7a9caa9`	Spacecraft map and sky data: fixes from the first data runs

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Date	Commit	Change
2026-11-12	68a1805	Spacecraft data: positions and orbital elements update
2026-11-12	b274803	Sky Finder, "Visible from here" and the spacecraft map
2026-11-12	659df70	Mars Data page live: built from the approved mock-up, with Curiosity's weather for sols 1-1293
2026-11-12	e85eced	Mars Data: mock-up approved
2026-11-12	80ff4a7	Mars Data: mock-up with humidity and corrected UV (sols 1-1293); midday UV rule
2026-11-12	45f90a2	Roadmap: Sky Finder and Spacecraft map, mock-ups awaiting approval
2026-11-12	27f237e	Number tiles shrink their figures on narrow tiles so no value is cut off; style version tag raised
2026-11-12	8951a9b	Site-wide polish: hover and press feedback, phone-friendly tab and chip rows, larger tap targets, readable map labels, consistent tabs, tags and filter labels
2026-11-12	edc30bb	Asteroid Tracker: key box removed; no visitor needs a key, and a key saved earlier is deleted from the browser
2026-11-12	02fb3c6	Roadmap: NASA_API_KEY secret added; the daily NeoWs copy is running
2026-11-12	c32419c	Asteroid Tracker: no key needed for visitors; the daily data job copies NASA's NeoWs weeks and asteroid pages with the key kept as a repository secret
2026-11-12	e326814	Image Explorer: the astronaut photo job checks that the photos' image files exist
2026-11-12	1f13529	Image Explorer: black gaps in the true-colour pictures made see-through, so the Blue Marble picture shows
2026-11-12	dd940f8	Image Explorer: the astronaut photo job waits up to 5 minutes for NASA's photo site, three tries
2026-11-12	8f8fde0	Image Explorer: map tap card names the country, feature and distance; map lines and names; satellite gaps filled; earthquakes, volcanoes, launch sites and natural wonders on the globe; astronaut photos without a key
2026-11-12	b31f205	Project Record builds again: NASA's and ESA's own e-mail addresses allowed by its check; Mars cleaning rules in their Part
2026-11-12	f44c733	One-off source check: the astronaut photo list download (removed after use)
2026-11-12	101b117	One-off source check: astronaut photo listings and the Space Station's position (removed after use)
2026-11-12	61cb811	One-off check of data sources for the Image Explorer (removed after use)
2026-11-12	01c5a37	Mars Data: sources checked, cleaning rules 1.1 and analysis code, credits for Curiosity REMS, and a page mock-up for approval
2026-11-11	bcb5630	Roadmap: developer testing of the Image Explorer's Nov 10 additions
2026-11-11	099c705	Image Explorer: the map's place card offers pictures from both libraries

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Date	Commit	Change
2026-11-11	266ff5d	Image Explorer: place search help and credits name the smaller natural features
2026-11-11	8cf596b	Image Explorer: smaller natural features rebuilt from GeoNames
2026-11-11	15b5173	Image Explorer: smaller places ranked by how widely they are named; Antarctica's last
2026-11-11	9aec1e6	Image Explorer: each ESA archive gets at most 12 seconds before results are shown without it
2026-11-11	c918bf6	Image Explorer: smaller natural features rebuilt from GeoNames
2026-11-11	4c00b3f	Project Record: the whole project as one PDF, rebuilt by GitHub after every push except the daily data updates
2026-11-11	230aad9	Image Explorer: ESA's videos in the Picture library; smaller places ranked by Wikipedia languages
2026-11-11	f986b09	Asteroid Tracker: risk-list asteroids in rose on the map, ESA and NASA compared on each asteroid page, share links, orbit diagram tilts all the way round
2026-11-11	1006e93	Image Explorer: smaller natural features rebuilt from GeoNames
2026-11-11	a4577c1	Image Explorer: the smaller-places job checks its list and commits it; place search reads it
2026-11-11	df4a7fe	Image Explorer: smaller places use their English names from GeoNames
2026-11-11	ba3c368	Image Explorer: the globe's finer pictures start sooner, from screen resolution, and cover a wider patch
2026-11-11	312a5a7	Projects numbered in build order (Image Explorer 2, Mars Data 3); Mars Data plan uses the whole Curiosity record, cleaned in chunks
2026-11-11	da229a8	Mars Data: plan for automatic updates from three missions, checked twice a week, with an email notice
2026-11-11	02fa2f1	Image Explorer: sharper globe close up, and day and night shading
2026-11-11	0bc8c7b	Image Explorer: smaller places kept when they have an English Wikipedia article
2026-11-11	0fef496	Image Explorer: a manual job that builds a list of smaller natural features from GeoNames
2026-11-11	77f5e5c	Image Explorer: colour keys drawn in the project's look from GIBS's colour maps
2026-11-11	794e46b	Image Explorer: each Earth map layer's days read from GIBS, with the nearest earlier day where a layer has none
2026-11-11	6749e15	Image Explorer: OpenLayers, proj4js and three.js served from the site instead of a CDN
2026-11-10	ce4f203	Main branch protected: force pushes and deletion blocked; changes are undone with git revert
2026-11-10	b5bdd9f	Asteroid Tracker map: tilt turns all the way round, including the view from underneath
2026-11-09	b97daac	Image Explorer: the searching message names the libraries being asked

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Date	Commit	Change
2026-11-09	`0a83c9b`	README: link to the live site, projects' status, NASA and ESA sources, the GitHub workflows
2026-11-09	`15e74a3`	ESA check: Privacy, Terms, About, home page, README and CLAUDE.md cover the ESA sources now on the site
2026-11-09	`8cd7c9a`	Image Explorer: ESA's Webb and Hubble pictures beside NASA's in the Picture library and on the globe, with each picture's ESA credit line
2026-11-09	`849dd28`	Asteroid Tracker: ESA's impact-risk list and close approaches beside NASA's, and arrow buttons for panning the map
2026-11-09	`a0324cf`	Asteroid data job: impact-risk file keeps only finite numbers (a speed given as nan made it invalid JSON)
2026-11-09	`f6a535e`	Asteroid data job: ESA's special risk list (Bennu, 1950 DA) included
2026-11-09	`da5dbd3`	Asteroid data job: impact-risk lists from ESA NEOCC and NASA/JPL Sentry, and ESA's close approaches; runs when its scripts change
2026-11-09	`8b3517a`	Image Explorer: ESA picture archives checked and recorded; mock-up of ESA pictures in the library
2026-11-09	`bb1dada`	ESA alongside NASA: rules for ESA sources, ESA credits and terms, and 'not endorsed by NASA or ESA' on every page
2026-11-09	`cb24d31`	Renamed NASA and ESA Data Projects: the new name and repository address on every page and document
2026-11-09	`df7608d`	Asteroid Tracker: pan the map (right, middle or Shift drag, two fingers, or the pan button); zoom towards the pointer; Back buttons go back one page
2026-11-09	`5f8f8ad`	Asteroid Tracker: orbit diagram marks where Earth and the asteroid meet at the close approach; Play stops there; slower speeds; famous passes can be turned off
2026-11-09	`029939c`	Version tags on the shared style sheet and footer script, so updates reach browsers at once
2026-11-09	`d0daf01`	Site pages: full-width text and a Back button on every information page
2026-11-09	`d631b13`	Home page: the coming pass's time says "your time" instead of a zone code
2026-11-09	`14365d0`	Asteroid Tracker: closest pass of the week on the home page, size comparisons, three more famous passes
2026-11-09	`f8785aa`	Site pages: About, Privacy, Terms, Credits and Contact, with site links in every footer
2026-11-09	`87fbd89`	Asteroid data job: soon.json, the close passes of the next 15 days with page ids, for the home page
2026-11-09	`9a3405b`	Image Explorer: space pictures first, results and spelling suggestions as you type, viewer fix
2026-11-09	`b9d413d`	MIT License for the project's own code and documents; CREDITS.md lists every outside source and library
2026-11-09	`cda487b`	Asteroid data job: three more famous passes with NASA's close-approach figures; backup daily time; push retries
2026-11-09	`91cc964`	Image Explorer: date changes reload every picture on the map; polar sea ice key and polar-night note

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Date	Commit	Change
2026-11-09	b460445	Asteroid Tracker: data check recorded; Parker path confirmed smooth with the new data
2026-11-09	1aab454	Image Explorer: Arctic and Antarctic views, mountains and lakes in place search, playable videos, astronaut photos on the globe (daily job), two sky positions corrected
2026-11-09	cfdee6e	Image Explorer version 1: live page with the Earth map, the globe (about 1,350 galaxies, nebulae and clusters, constellations) and the picture library
2026-11-09	c72e05f	Asteroid Tracker: smooth spacecraft paths; Parker and Juno show their last year unless selected
2026-11-09	957e46b	Image Explorer: globe works on phones; globe sources, decisions and to-dos recorded
2026-11-09	d026971	Image Explorer mock-up: the globe opens facing Sagittarius, beside the Earth
2026-11-09	35c7a24	Image Explorer mock-up: globe script versioned
2026-11-09	aa757b0	Image Explorer mock-up: globe shows events reported in the last 30 days; softer atmosphere edge
2026-11-09	52cac94	Image Explorer mock-up: Globe tab (Blue Marble Earth in the real sky, NASA events, galaxies, nebulae and clusters) and live picture search
2026-11-09	6883095	Asteroid Tracker: famous close passes, back buttons that return to the same spot, smoother zoom
2026-11-09	5561a7c	Image Explorer mock-up: a clearer Close button on the picture viewer; data note on picture files corrected
2026-11-09	b2cb397	Image Explorer mock-up: the map comes first on tablets and phones
2026-11-09	7362f7d	Image Explorer mock-up: the place card stays inside the map near its edges
2026-11-09	413f0b4	Image Explorer mock-up: the map stays inside the world's edges
2026-11-09	66ae7d0	Image Explorer: data checks recorded, plan and roadmap updated, picture viewer fallback
2026-11-09	531f682	Image Explorer mock-up: world view centred; a missing picture size falls back to the next size
2026-11-09	315c8f8	Image Explorer mock-up: the map fills its frame on load; fire dots scale with zoom
2026-11-09	d8b9d51	Image Explorer mock-up: each layer's newest day, map fills its frame, readable colour keys, one colour layer at a time
2026-11-09	3f40d7e	Image Explorer: mock-up for review (live GIBS map, sample picture results)
2026-11-09	96f1a3a	Data job: fetch the famous passes' paths (the previous commit carried only the description)
2026-11-09	c993651	Data job: precise paths for the map's famous close passes (NASA/JPL Horizons)
2026-11-09	25c9563	Roadmap: delete the test branch left from the map's data job test
2026-11-09	ef82d2c	Asteroid Tracker: solar system map, search tab, home page preview, asteroid names explained

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Date	Commit	Change
2026-11-08	`b84902c`	Asteroid Tracker: notes closed by default, layout fixes at small widths, smoother dragging of the orbit diagram
2026-11-08	`beb1d9c`	Orbit diagram: time ticks that follow the asteroid's speed near Earth; clearer figures for very close passes
2026-11-08	`354f66c`	Asteroid Tracker: a page for each asteroid with an orbit diagram, and past years in the year view
2026-11-08	`6ac3827`	Asteroid Tracker: a 'How to read this' note under each chart and plain labels on the size chart
2026-11-08	`84e9842`	Asteroid Tracker: note that distances are measured from Earth's centre; correct the 2026 counts in the data notes
2026-11-08	`b0596f9`	Asteroid Tracker version 1: this week's close approaches, filters, three graphs, the year view and key settings
2026-11-08	`fa7a20e`	Documentation tone rule: professional project documentation, never chat; documents reworded and the check extended
2026-11-08	`3691e3d`	The site is live on GitHub Pages; roadmap updated
2026-11-08	`2e3517e`	Set up the repository: three project folders, shared look and NASA API helper, home page, docs and checks
2026-11-08	`15c69c3`	Initial commit

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PART 16

Glossary

Term	Meaning
API	Application programming interface: a web address a program asks for data, answered in a fixed format (usually JSON).
API key	A code that identifies who is asking an API. NASA's are free; a key is never written into this project's code.
astronomical unit (au)	The average distance from the Earth to the Sun, about 149.6 million km.
CC BY 4.0, CC BY-SA 3.0 IGO	Creative Commons licences: reuse is allowed with credit; "SA" (share-alike) means changed versions must use the same licence. ESA/Webb and ESA/Hubble pictures use CC BY 4.0; some other ESA material uses CC BY-SA 3.0 IGO.
CDN	Content delivery network: servers that hand out files (such as code libraries) quickly from near the visitor.
close approach	The moment an object passes nearest the Earth on its orbit, with its miss distance and speed.
DEMO_KEY	NASA's shared key for trying the APIs: 30 requests an hour and 50 a day per connection.
EONET	NASA's Earth Observatory Natural Event Tracker: wildfires, storms, volcanoes and other events as they happen.
ESA	The European Space Agency.
force push	Overwriting a branch's history on GitHub; blocked on this repository's `main`.
GIBS	NASA's Global Imagery Browse Services: daily satellite pictures of the whole Earth, delivered as map tiles.
GitHub Actions	Jobs GitHub runs by itself: the checks after every push, the daily data updates, and this record's rebuild.
GitHub Pages	GitHub's free web hosting, which publishes the site from the repository.
Horizons	NASA/JPL's service for the positions of planets, moons, asteroids and spacecraft at any time.
LD (lunar distance)	The average distance from the Earth to the Moon, about 384,400 km; used to make miss distances easy to picture.
Ls (solar longitude)	Where Mars is on its orbit, as an angle: the Martian season (0° is the start of northern spring).
MIT License	The licence of this project's own code and documents: anyone may use and change them, keeping the copyright notice.
NEO	Near-Earth object: an asteroid or comet whose orbit brings it within 1.3 au of the Sun at its closest, and so near the Earth's orbit.
NEOCC	ESA's NEO Coordination Centre, which keeps ESA's impact-risk list.
NeoWs	NASA's Near Earth Object Web Service, the API behind the Asteroid Tracker's weekly list.
PDS	NASA's Planetary Data System, the archive of data from planetary missions.

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Term	Meaning
potentially hazardous asteroid	NASA′s label for an asteroid larger than about 140 m whose orbit comes within 0.05 au of the Earth′s. It describes a possibility, not a predicted impact.
REMS	Rover Environmental Monitoring Station: Curiosity′s weather station on Mars.
repository secret	A value kept encrypted by GitHub for its jobs (such as a service key); never visible in the code or the site.
Sentry	NASA/JPL′s impact-monitoring system and its list of objects with a possible future impact.
sol	A Martian day: 24 hours 39 minutes.

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